Measuring Crime Reporting and Incidence: Method and Application to #MeToo

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Abstract

A long-standing issue with crime statistics is that they reflect both variations in the propensity to report crimes and actual crime incidence. I propose a two-way fixed effects model that exploits the delay between an incident and its police report to separate these two margins. I then study the Me Too movement's impact on sex crimes in New York City. While reporting was already rising before October 2017, #MeToo led to higher victim reporting and arrest probabilities. Using non-sexual crimes as a control group, difference-in-differences estimates suggest the movement also deterred sex crimes.

Keywords: crime reporting, crime deterrence, sex crimes, #MeToo
JEL Classification: C18, C24, C41, J16, K14, K42

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Replication files: https://github.com/PinchOfData/measuring-crime-replication

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"I do not fear to say that all we possess of statistics of crime and misdemeanors would have no utility at all if we did not tacitly assume that there is a nearly invariable relationship between offenses known and adjudicated and the total unknown sum of offenses committed." – Quêtelet (1831)

1 Introduction

In October 2017, the hashtag #MeToo went viral on social media and triggered a large-scale public discussion of sexual violence. Enthusiastic commentators argued that the movement lowered the social costs of speaking out and increased victims' willingness to report. Others worried about backlash, false accusations, or uneven reach across socioeconomic and racial groups.[1] This paper focuses on three substantive questions: whether #MeToo changed victim reporting, whether it changed underlying sex crime incidence, and whether these effects differed across socio-demographic groups.

As with most interventions against crime, answering these questions is difficult because many victims do not come forward. Between 1995 and 2010, U.S. national surveys estimated over 6 million rape and sexual assault victims, of which 60 to 70% did not report the incident to the police (Planty et al., 2013). In turn, a long-standing empirical challenge has been interpreting variations in reported crimes as changes in the number of offenses committed or victims' propensity to report (Quêtelet, 1831). To address this issue, I develop a simple econometric framework to disentangle victim reporting and crime incidence from police data in the presence of reporting *delays*.

The empirical strategy exploits crimes that occurred before an intervention but were reported after it. Because the intervention cannot affect the number of crimes that had already occurred, changes in the rate at which these past incidents are reported reveal changes in victims' propensity to come forward. I formalize this intuition with a Poisson two-way fixed effects model for the count of crimes committed at date t and reported at date τ. Incident-date fixed effects capture crime incidence, while delay fixed effects capture the baseline probability of reporting after $k = \tau - t$ periods. A treatment indicator then captures the intervention's effect on reporting.

Under familiar no-anticipation and parallel-trends assumptions, identification follows from a ratio-of-ratios argument. First, within a given pre-intervention incident cohort, I compare reports filed at a longer delay after the intervention with reports filed at a shorter delay before it. Because both counts correspond to crimes committed on the same date, their ratio cancels out the unobserved number of crimes in that cohort. Second, I compare this ratio with the corresponding ratio for an earlier incident cohort, for which both the short- and long-delay reports occurred before the intervention. This second comparison cancels out baseline differences in reporting probabilities across delays. This non-linear difference-in-differences logic point identifies the proportional

[1] For contrasting perspectives, see AP News (2017); Psychology Today (2017); Berkeley Law (2019) and Forbes (2020); Harvard Business Review (2019); New York Post (2020); New York Times (2017a); AP News (2021); New York Times (2017b).

change in reporting for "switcher" cohorts: those whose incident date generates both pre- and post-intervention reports.

Recovering the full post-intervention path of incidence requires an additional step. Cells whose incident date falls after the intervention are never observed untreated, so the reporting response must be extrapolated from the identified cells to them. Rather than fix the functional form of this extrapolation a priori, I let the data guide it. An imputation estimator first recovers an unrestricted reporting effect for each identified cell (Borusyak et al., 2024; Moreau, 2025). I then examine how these effects vary across event time and delay and use the resulting pattern to determine the restriction under which the reporting response can be extrapolated to post-intervention cohorts. A key feature of the proposed model is that it relies only on the internal delay structure of police records: because it identifies *variations* in crime reporting and incidence rather than their levels, it requires no external information on victims' reporting rates at any point in time.

I apply this method to incident-level police records from New York City between 2006 and 2019.[2] Delayed reporting is pervasive in the data: over half of sex-crime reports are filed after the incident date, with delays ranging from days to decades. Among sex crimes, the mean incident-to-report delay is 204 days, and the distribution has a long right tail. Consistent with the idea that reporting increased over time, average reporting delays measured by report date rise dramatically over the decade, more than tripling relative to 2006 levels, and jump after October 2017, indicating that more victims came forward to report past incidents. A complementary descriptive pattern is visible in the aggregate hazard of reporting for pre-#MeToo incidents, which increases sharply after the movement's surge in public salience.

Turning to my main estimation, I aggregate both incident and report dates at the quarterly level and estimate a Poisson model with incident-quarter fixed effects, reporting-delay fixed effects, and quarterly event-time indicators. The event-time indicators allow reporting to vary freely across quarters starting in 2010. The specification nevertheless rules out heterogeneity by reporting delay: within a given event-time quarter, it assumes that the movement changes reporting by the same proportion for recent and older incidents. This restriction is justified by the imputation estimator, which recovers a separate effect for every identified combination of event time and reporting delay. After #MeToo, the unrestricted estimates show substantial variation across event time: the effect builds over the quarters following the movement's viral spread, then gradually recedes. By contrast, within the ten-year delay window of the baseline sample, the effect is essentially unrelated to reporting delay. Victims reporting incidents dating back several years respond about as strongly as those reporting recent incidents.

Given this specification, the first main result is that the propensity to report sex crimes increased substantially throughout the 2010s and accelerated after #MeToo. Reporting rates rise by roughly 8% per year between 2010 and 2017, then display a clear break in 2017 Q4. In a simple before-

[2]Common U.S. datasets such as UCR and NIBRS typically do not include both incident and report dates at the incident level. City-level open data provide this key structure, though the quality of data sets may vary. The NYPD dataset is very high quality and extends far back in time relative to other cities.

after comparison, the pooled Poisson Pseudo Maximum Likelihood (PPML) estimate implies that the movement raised the probability of reporting a sex crime by approximately 47%; the heterogeneity-robust imputation estimator yields a nearly identical 51%. These estimates do not net out the pre-existing trend, so I also assess the movement's contribution against a trend-based counterfactual. Between 2010 and 2019, the propensity to report a sex crime to the police increased 3.7-fold. Extrapolating the 2010–2017 trend implies it would have risen only about 2.1-fold by 2019 absent #MeToo. By 2019, the movement had raised reporting about 81% above this no-#MeToo counterfactual, accounting for roughly 61% of the decade-long increase. The increase in reporting appears broad-based across victim and suspect groups and across crime severity levels. It is robust to various alternative sample restrictions (i.e., using the full sample, and samples excluding left-truncated cells, first-of-month placeholder dates, and long reporting delays; see Online Appendix Table D.1).

The second main result concerns crime incidence. Although the number of reported sex crimes increased by roughly 30% over the study period, the two-way fixed effects decomposition implies that underlying incidence declined by roughly two-thirds. The rise in observed reports therefore masks two opposing forces: a large increase in victims' propensity to report and a substantial decline in offending. Using estimated crime incidence, I then isolate the effect of #MeToo with a (generalized) difference-in-differences design that uses non-sexual crimes as a control group. In my preferred specification, which uses matrix completion (Athey et al., 2021), I estimate that #MeToo reduced sex crime incidence by about 38% relative to the control-based counterfactual. Cumulated over 2010–2019, the movement prevented roughly 6% of the sex crimes that would otherwise have been committed over that period, despite its deterrent effect operating only over the final two years. A joint test of the pre-treatment coefficients is consistent with no differential trends, and the results are robust to alternative counterfactual models, including an interactive fixed effects model (Xu, 2017), a two-way fixed effects model, and a specification with crime-specific linear trends and seasonality.

For robustness, I replicate my main results with an alternative model of crime reporting. I model each victim's time to report as a duration model, working with individual reports rather than aggregate counts and explicitly allowing for victims who never come forward. This model rests on assumptions closely related to those behind my main estimates but adopts a different functional form and requires an external estimate of the share of victims who never report. It nonetheless corroborates each of the main findings: the victim reporting rate more than doubled, while sex crime incidence fell by roughly 57% over the decade. #MeToo reduced incidence by about 25%. That two models with such different functional forms reach the same conclusions suggests that the findings are not artifacts of any particular modeling choice.

These findings markedly differ from what one would conclude using more standard approaches that treat reported crimes as the outcome. In my setting, aggregating reports by incident date or by report date yields materially different time series and treatment effects, and both obscure the joint movement of reporting and incidence. Concretely, a difference-in-differences in the spirit of

Levy and Mattsson (2025) that takes *reported* sex crimes as the outcome – using non-sexual crimes as controls, with crime-specific linear time trends and crime-specific seasonality – attributes to #MeToo a 17% increase in reported sex crimes when reports are aggregated by incident date, and a 25% increase when they are aggregated by report date. This jump is naturally read as a rise in the reporting rate rather than in incidence; but it is far smaller than the reporting response my decomposition recovers, because the decline in incidence offsets part of the increase in reporting within the reported-crime series.

The results also clearly contrast with the survey and emergency-department measures most often used to sidestep police records and obtain direct estimates of variations in crime incidence: the National Crime Victimization Survey suggests sexual-violence incidence increased by roughly 145% through 2018 (Worthen and Schleifer, 2024), and emergency-department visits for sexual assault rose by approximately 28% over the study period. These measures, however, still depend on victims coming forward. In fact, the one proxy that does not, sex-related homicides, instead falls by 70 to 80% between 2010 and 2018, consistent with the decline I estimate. All three are national series, not directly comparable to my New York City estimates. The homicide counts are also very small (fewer than 20 sexual homicides in the United States in 2017) and are thus suggestive rather than conclusive evidence. That such alternatives are confined to national aggregates also highlights a strength of my approach: relying on rich incident-level police records, it can be applied to fine-grained geographical units such as individual cities.

I then consider several channels that may explain these results. Unfounded allegations, changes in the legal definitions of sex crimes, and changes in statutes of limitations are unlikely to account for these patterns. One plausible explanation is a shift in social norms: the social cost of reporting a sex crime may have fallen, while the social cost of committing such a crime may have risen. This interpretation is consistent with increased awareness of sexual violence, as reflected in Google searches and Twitter data. A second plausible explanation operates through deterrence and incapacitation. Higher reporting increased the probability that sex offenders were arrested, which roughly quadrupled between 2006 and 2019. Using #MeToo as the source of variation, the data imply an elasticity of sex crime incidence with respect to the probability of arrest of about -1. This estimate should be interpreted as an upper bound on the deterrent effect of arrest. #MeToo increased the probability of arrest, but as noted above, it also increased the social and reputational cost of committing a sex crime. Because these two margins moved together, the calculation attributes to the threat of arrest a decline in incidence that may also be explained by changing gender norms.

This paper speaks to four strands of the literature: the measurement of crime, crime deterrence, gender-based violence, and the Me Too movement.

First and foremost, the paper contributes to the measurement of crime. Scholars have long recognized that observed crime statistics reflect both true incidence and reporting behavior – the so-called *dark figure of crime* (Quêtelet, 1831; Coleman and Moynihan, 1996). Existing solutions rely on victimization surveys (Fogliato et al., 2024), homicides (Levitt, 1998), proxy variables (Stephens-Davidowitz, 2013; Bellégo and Drouard, 2024), instruments (Bellert et al., 2026), or informative Bayesian priors (Bradshaw and Blei, 2024). All implicitly assume that crimes are reported quickly or never. This assumption fails for sex crimes, domestic violence, harassment, and corruption, among other crimes where reporting delays of months or years are common. I show how to exploit these delays for identification, providing a new tool for credible impact evaluation in settings where underreporting is potentially correlated with treatment.

Second, by recovering incidence rather than reported crime, the paper speaks to the literature on criminal deterrence. A robust finding of this literature is that offenders respond more to the *certainty* of apprehension than to the *severity* of punishment (Becker, 1968; Chalfin and McCrary, 2017). The margin I study operates through certainty rather than severity: by raising victims' propensity to report, #MeToo mechanically raised the probability that a sex offender is arrested, and my estimates imply an arc elasticity of sex crime incidence with respect to this probability of about -1. This magnitude is in line with the literature. It is larger than standard probability-of-punishment elasticities, whose median sits near -0.7, and the police-manpower elasticities of roughly -0.4 for violent crime and -0.2 for property crime (Chalfin and McCrary, 2017, 2018), as expected for an elasticity defined on the probability of arrest, the most salient certainty margin. It is smaller than the responses elicited by salient, offender-specific increases in detection risk, such as the rollout of DNA databases (Doleac, 2017; Anker et al., 2021).

Third, the paper contributes to the economics of gender-based violence, a field where underreporting is a central policy concern. Gender-based violence has drastic consequences in the workplace (Folke et al., 2020; Folke and Rickne, 2022, 2025; Adams-Prassl et al., 2024) and responds sharply to victims' economic circumstances (Bhalotra et al., 2025). The monitoring and punishment of such crimes depends critically on victims' willingness to come forward (Acemoglu and Jackson, 2017; Lee and Suen, 2020; Cheng and Hsiaw, 2022). Yet reporting is costly: retaliation threats and deteriorating outside options silence harassment victims (Dahl and Knepper, 2026), and reports of sexual harassment rise dramatically when disclosure is made safer (Boudreau et al., 2023). Studies using administrative data show that female political representation (Iyer et al., 2012), female police officers (Miller and Segal, 2019), specialized justice centers (Sviatschi and Trako, 2024), and highly publicized cases (Bottan and Perez-Truglia, 2015; Mathur et al., 2019; McDougal et al., 2021) can increase reporting, while the criminal-justice consequences triggered by a report shape victims' subsequent willingness to call the police (Amaral et al., 2023). My results add that social movements can shift reporting and that such shifts may also deter offending.

Fourth, the paper adds to the rapidly growing body of work on the Me Too movement itself, arguably the largest public awareness campaign against sexual violence in history. Previous studies document impacts on financial markets (Borelli-Kjaer et al., 2021), labor markets in entertainment, venture capital, mutual funds, and academia (Luo and Zhang, 2022; Sophie Calder-Wang and Sweeney, 2021; Cici et al., 2021; Gertsberg, 2025; Batut et al., 2021), and the treatment of harassment and sex crime cases in courts (Marchenko and Pakzad-Hurson, 2025; Cai et al., 2024; Langen, 2024). Most relevant to this paper, Levy and Mattsson (2025) provide a comprehensive analysis of the movement's effects on reported crimes across the United States and other OECD countries, and Chen and Long (2024) shows these effects concentrated in US regions with historically lower sexism. Like most of the literature, these studies take reported crimes as their outcome. I go one step further by decomposing observed changes into victim reporting and crime incidence. The two respond to #MeToo in opposite directions: I estimate average treatment effects of approximately $+51\%$ on the victim reporting rate and approximately -38% on sex crime incidence. Because these responses partly offset within reported-crime counts, the reduced-form increases in reported sex crimes that prior studies recover (approximately 11% in the United States and 8% across the OECD) likely underestimate the movement's effects.

Finally, although this paper focuses on reported crime statistics, the econometric framework applies more broadly whenever records contain both an occurrence date and a reporting or disclosure date, and some events enter the data only after a delay (and potentially never at all). Divorce provides a leading example. An increase in filings following liberalizing reforms may reflect not only additional marital dissolutions, but also the accelerated formalization of marriages that had already ended; separation dates recorded in divorce filings provide the delay structure needed to distinguish these two margins (Wolfers, 2006). Similar issues arise in voluntary disclosures of offshore wealth following enforcement shocks (Alstadsæter et al., 2019) and in adverse drug event reporting, where safety scandals can prompt the reporting of events that occurred well before the scandal (Pariente et al., 2007). The same logic may also inform the study of sexual orientation disclosure: greater social acceptance may lead individuals to identify publicly as LGBT sooner, so increases in observed self-identification need not reflect changes in the underlying prevalence of sexual orientation within the population (Coffman et al., 2017).

The remainder of the paper proceeds as follows. Section 2 describes the data and the institutional context. Section 3 documents the prevalence of delayed reporting and the raw trends surrounding #MeToo. Section 4 develops the econometric framework. Section 5 presents the main results. Section 6 discusses mechanisms and alternative explanations. Section 7 concludes.

2 Data and Context

2.1 Data Sources

My empirical strategy requires incident-level datasets that distinguish the date of the incident and the date of its report to the police. Commonly used police datasets in the United States do not provide this information. The Uniform Crime Reporting (UCR) datasets provide yearly aggregate reports per agency but no incident-level observations. The National Incident-Based Reporting System (NIBRS) provides incident-level observations, recording either the incident date or the report date to the police, but not both. To circumvent this problem, I rely on city-level police datasets (Police Data Intiative, 2024). I focus on New York City, which (i) provides incident-level data for the period 2006–2019, (ii) includes sex crime reports, and (iii) distinguishes the date of the incident and the date of its report to the police. With a population of approximately 8.3 million, New York City is the largest city in the United States. The records are official administrative data; their collection is meant to be rigorous and systematic, and the data are later sent to the FBI for consolidation. I also downloaded offender-level arrest datasets for New York City over the same period, which allow me to compute arrest rates per crime category. Finally, I restrict the sample to crimes reported until October 2019 because lockdowns and other restrictions to fight the COVID pandemic may have affected sex crime incidence and reporting from 2020 onwards.

I process the data in the following way. I manually classify offenses as *sexual* or *non-sexual*. I exclude sexual offenses related to *pornography*, *indecency*, *loitering*, *prostitution*, and *consensual acts*. For non-sexual offenses, I focus on four broad categories: *burglary*, *robbery*, *assault*, and *murder*. I exclude all other non-sexual offenses from the analysis (see Online Appendix Table A.1 for the full classification). I can further distinguish sex offenses between *misdemeanors* and *felonies*, and I observe the socio-demographic characteristics of complainants and alleged offenders, such as their self-declared race, age, and sex. The sex variable has three groups: *male*, *female*, and *unknown*. For the self-declared race, I form four categories: *black*, *hispanic*, *white*, and *other/unknown*. For the age of complainants and offenders, I create a dummy variable *juvenile* that takes values zero for adults (above 18 years old) and one for children (below 18 years old). The age recorded is the complainant's (offender's) age upon filing the complaint. Finally, I exclude from the sample all complaints with incoherent dates for the incident or its report to the police.[3]

2.2 The Me Too Movement

The Me Too movement is a social movement against all forms of sexual misconduct in which people share and publicize allegations of sex crimes. Its explicit goal is to raise awareness of the pervasiveness of sexual violence in society. Social activist Tarana Burke launched the movement on MySpace in 2006. For over ten years, the campaign focused on female minorities (mainly black women) and received limited media coverage. On the 15^{th} of October 2017, it was popularized by actress Alyssa Milano in reaction to the Harvey Weinstein scandal. She tweeted: *"If you've been sexually harassed or assaulted write 'me too' as a reply to this tweet."* In the following days, the hashtag #MeToo spread virally on social media and was posted millions of times on Twitter, Facebook, and other platforms worldwide.

Following its mass mediatization, mentions of the hashtag #MeToo have dwarfed past references to the Me Too movement (see Online Appendix Figure F.4). However, this large discontinuity in the time series should not obscure the social and historical context in which the movement emerged. Several pieces of anecdotal evidence suggest that attitudes toward sex crimes had been changing for over a decade. The movement appeared 11 years before it became popular. The traditional media also brought many affairs to the spotlight during the 2000s: examples include sex crime allegations in the Catholic Church, the Bill Cosby sexual assault cases, and allegations con-

[3] In some cases, the incident's date is later than the date of its report, or one of the two dates is missing. These represent 1.5% of the raw data.

cerning Harvey Weinstein before 2017. On social media, several hashtags denouncing violence against women preceded #MeToo but were less viral (e.g., #YesAllWomen, #IAmNotAfraidToSpeak, #myHarveyWeinstein and #BeBrave). Given these early signs of changing attitudes towards women, one would expect positive time trends in crime reporting before October 2017. I find empirical support for this hypothesis in Section 5.

Debated on television and in the newspapers, the Me Too movement also drew criticism as it gained momentum. Some pointed to the risk of false allegations (Forbes, 2020; New York Post, 2020). Others claimed the movement failed to recognize the heightened vulnerability that women of color frequently face (New York Times, 2017a; AP News, 2021). Finally, proponents warned against the potential backlash (Harvard Business Review, 2019). I address these concerns in my empirical assessment of the movement's impact on victims and offenders. Overall, in the scope of this study, false allegations are unlikely to be the drivers behind the large increase in sex crime reports, the reporting rate increased across demographic groups, and there was no backlash effect in sex crime incidence (see Section 5).

3 Stylized Facts

3.1 The Prevalence of Delayed Reports

Between 2006 and 2019, over 1.5 million crime incidents were reported. Among those, approximately 100,000 were sexual crimes (see Online Appendix Table A.2).[4] Sexual criminality is largely gender-specific, with 85% of reports filed by women in the sample. In terms of declared race, Black and Hispanic complainants account for the bulk of sex crime reports. Furthermore, children and teenagers are particularly exposed to sexual criminality, with around 50% of complainants being below 18 when filing the complaint. Alleged offenders, on the other hand, are mainly adult males belonging to a racial minority.

A striking feature in the data is the prevalence of delayed reporting: approximately 56% of sex crimes are not reported on the day of the incident (I refer to these complaints as *delayed* reports as opposed to *direct* ones). This figure hides substantial heterogeneity across offenses. 44% of sex crimes are direct reports as opposed to 81% for non-sexual assaults and 84% for robberies. Burglaries also display a sizable share of delayed reports (45%), but their delays are overwhelmingly short: the median burglary is reported within a day, typically once the break-in is discovered. The contrast is starker for average reporting delays. The average time-to-report a sex crime is 204 days, as opposed to three to five days for non-sexual assaults, robberies, and burglaries.[5] The standard deviation in reporting delays is roughly thirteen to sixteen times as large for sex crimes relative to non-sexual assaults, robberies, or burglaries. Approximately 44% of complainants for sex crimes report on the day of the incident, 79% within the first month, and 91% within the first

[4] For comparison, note that the National Crime Victimization Survey (NCVS) estimates rates of sexual violence based on less than a hundred self-declared cases per year.

[5] Murders recorded with long delays correspond to cold cases.

year (see Online Appendix Figure A.1). Delayed reporting also varies along socio-demographic lines. Longer reporting delays are typically expected for subgroups experiencing higher costs to reporting. Juveniles report over more extended periods than adults. Hispanic and Black complainants also display longer reporting delays than other racial categories (see Online Appendix Figure A.2).

3.2 Trends around #MeToo

I now turn to the study of trends in reported crimes surrounding the Me Too movement's mass mediatization (in October 2017). Sex crime reports increased by approximately a third between 2006 and 2019, whereas non-sexual crime reports fell by about 19% over the same period (see Panel A of Figure 1). The surge starts before #MeToo. This can be rationalized as an increase in sex crime reporting or an increase in sex crime incidence, or variations in both margins.

Several pieces of evidence point to an increase in the reporting rate of victims. I split reported sex crimes into those filed within a year of the incident, and those filed a year or more later (see Panel B of Figure 1). Reports of these older incidents increased more than four times as much as reports of recent incidents between 2006 and 2019, and drove a sizable share of the increase in reported sex crimes. This stylized fact is consistent with the depletion of a large stock of unreported sex crimes being progressively reported to the police.

To better understand this increase in delayed reports, I examine average reporting delays and probabilities of reporting a sex crime over time (see Figure 2). If victims become increasingly likely to report sex crimes to the police over time, then we should observe an increase in average reporting delays for a given report date. Panel A clearly shows that average reporting delays more than tripled during this period. The increase is particularly large after #MeToo went viral on social media.

A complementary view comes from the probability of reporting crimes committed at time t in subsequent periods. Panel B plots, for several incident cohorts – sex crimes committed before 2006, 2008, 2010, 2012, and 2014 – the number of complaints filed after these cutoff dates. For each cohort, the number of reports progressively decreases over time as the stock of unreported crimes depletes. However, every cohort displays an unusual increase in reports after #MeToo. Given that #MeToo cannot impact the number of crimes committed before each cohort's cutoff, this suggests the victim reporting rate increased after #MeToo.

To summarize, we have strong reasons to believe that the reporting rate increased, particularly after #MeToo. This raw empirical evidence alone, however, cannot tell us *by how much* sex crime reporting and incidence evolved over time, hence the empirical strategy that I outline below.

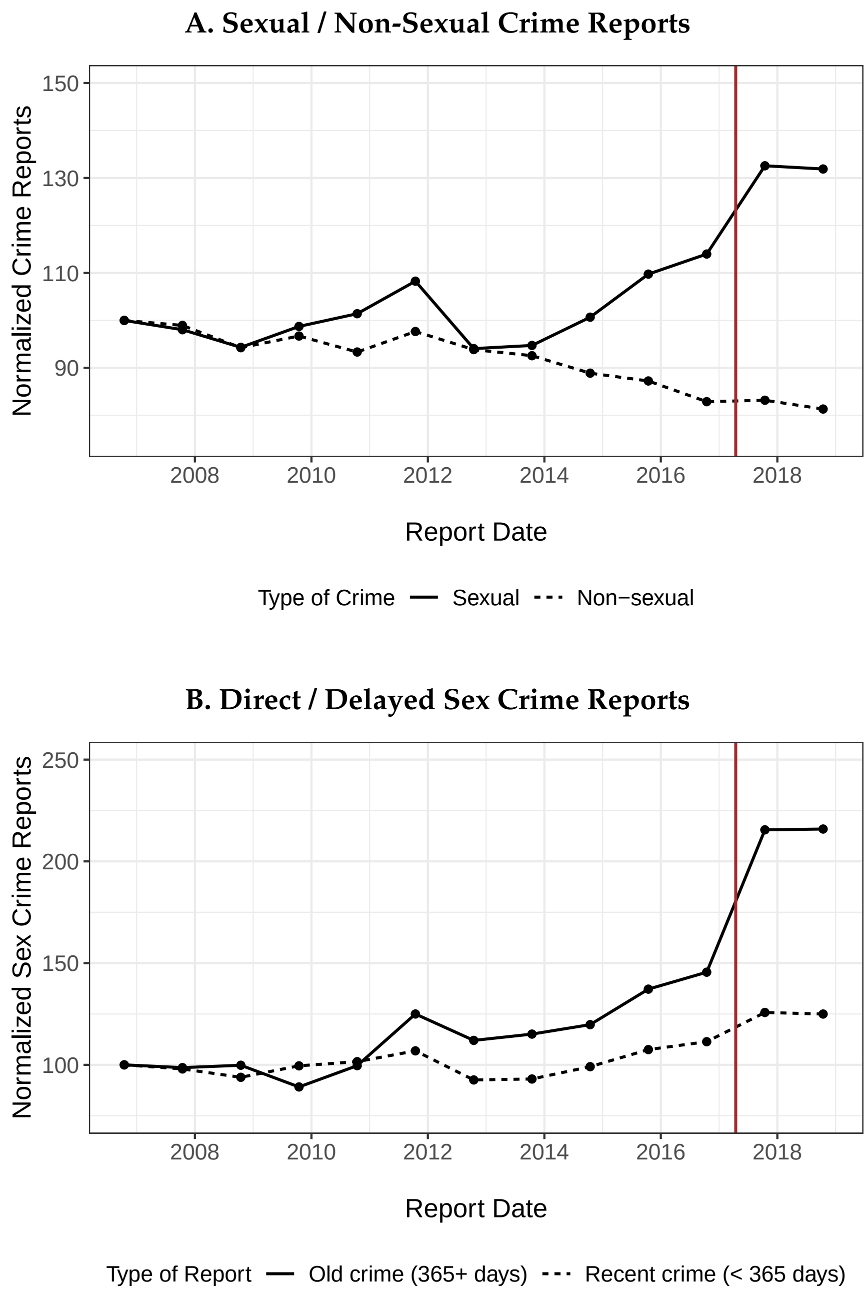

Figure 1: Trends in Reported Crimes

Notes: This figure presents trends in reported crimes. Crime reports are aggregated on the report date. The vertical solid line corresponds to #MeToo (Oct 2017). Panel A compares sexual crime reports to non-sexual crime reports. Panel B compares reports of sex crimes filed within a year of the incident ("recent") to those filed a year or more after the incident ("old"); both series are normalized to 100 at the start of the sample.

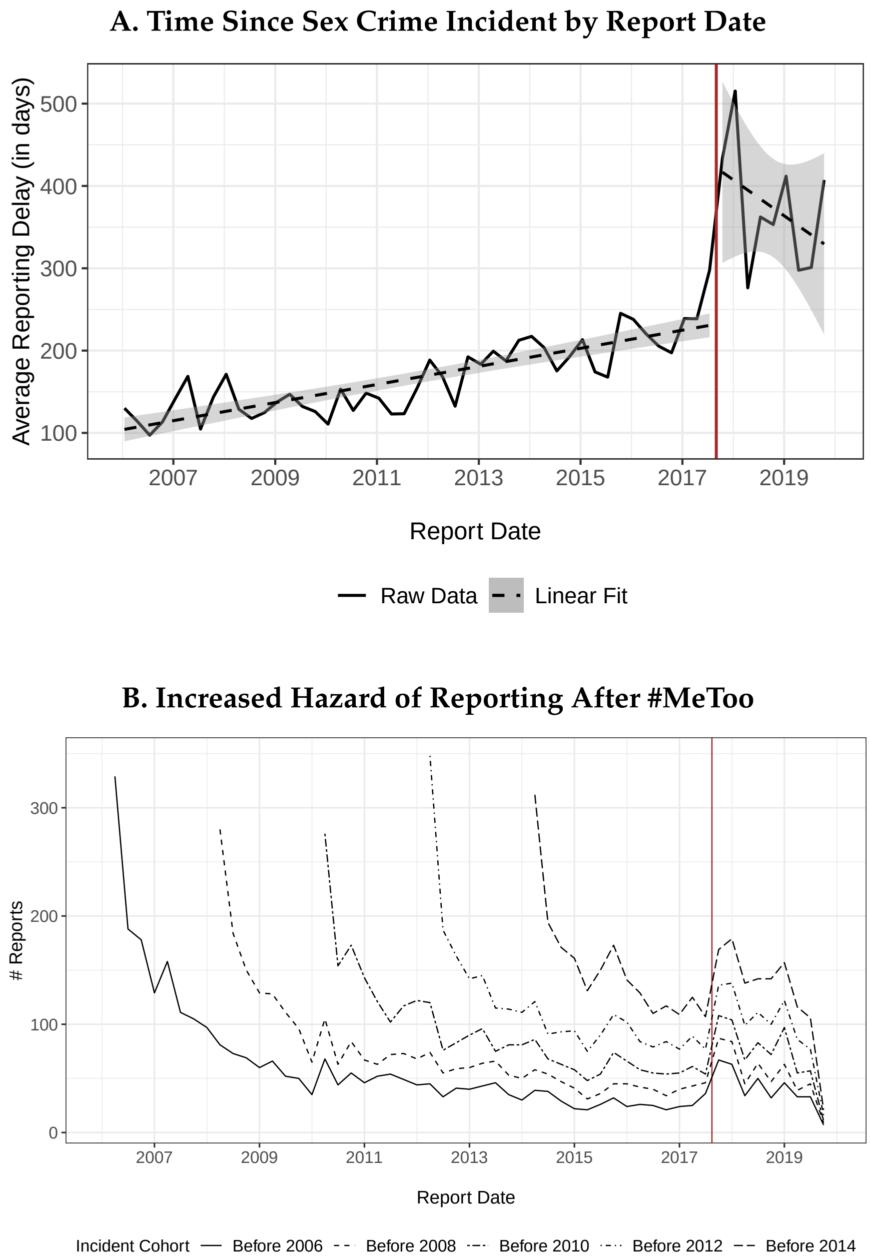

Figure 2: Raw Evidence of an Increased Propensity to Report Sex Crimes

Notes: Panel A displays average reporting delays (in days) for sex crimes by report date in the sample. The dashed lines are linear fits before/after #MeToo with 95% confidence intervals in grey. The increase in reporting delays suggests that victims of past crime incidents have increased their likelihood of filing a complaint over the period. Panel B presents police report counts, by report date, for several incident cohorts – sex crimes committed before 2006, 2008, 2010, 2012, and 2014. For each cohort, the stock of past sex crimes is progressively reported to the police. After #MeToo, every cohort shows a clear, unusual increase in reported crimes, suggesting an increase in the reporting rate of victims. The cohort cutoffs are illustrative, and similar figures can be produced for other thresholds. The vertical solid line corresponds to #MeToo (Oct 2017).

4 Empirical Strategy

In this section, I develop a simple econometric framework to decompose changes in reported crimes to the police into two margins: victim reporting and crime incidence. Building on the stylized facts presented above, I exploit reporting delays for identification.[6]

4.1 The Setup

Let $R_{t,\tau}$ denote the number of crimes committed in period t that are reported to the police in period τ, where $k = \tau - t$ is the reporting delay. Let t^* denote the date of an intervention, such as the onset of the Me Too movement. Let $R_{t,\tau}(1)$ and $R_{t,\tau}(0)$ denote the number of crimes committed in period t and reported in period τ under the intervention and under no intervention, respectively. Similarly, let $C_t(1)$ and $C_t(0)$ denote the potential number of crimes committed in period t, and let $p_{t,\tau}(1)$ and $p_{t,\tau}(0)$ denote the potential reporting probabilities. These quantities satisfy the accounting identity:

$$R_{t,\tau}(z) = C_t(z) \times p_{t,\tau}(z) \times \varepsilon_{t,\tau}, \tag{1}$$

where z indicates whether the intervention is implemented ($z = 1$) or not ($z = 0$), and $\varepsilon_{t,\tau}$ is an error term. We wish to identify how the intervention affects the potential outcomes $C_t(z)$ and $p_{t,\tau}(z)$ using an observed panel of reported crimes $R_{t,\tau}$ indexed by incident and report dates. My goal is to understand under which assumptions this is feasible.

4.2 Assumptions

Two assumptions are necessary for identification.

Assumption 1 (No Anticipation) The intervention does not affect reporting behavior or crime incidence prior to t^*. Formally, for all $t, \tau < t^*$:

$$C_t = C_t(0) \qquad \text{and} \qquad p_{t,\tau} = p_{t,\tau}(0).$$

This condition ensures that the pre-intervention data represent untreated counterfactual outcomes. It is straightforward to weaken this assumption, for example, by allowing anticipation for several periods before treatment: this simply requires redefining event dates to earlier ones (as I do in Section 5).

Assumption 2 (Parallel Trends) The untreated potential reported crimes can be written:

$$\mathbb{E}[R_{t,\tau}(0)] = C_t(0) \times \alpha_k.$$

[6] For completeness, in Appendix B, I formalize why simply running regressions on reported crimes to the police aggregated at the incident or the report date – the most prevalent approach in the crime literature – will generally lead to biased and uninterpretable estimates.

In other words, the untreated potential reporting probability is given by:

$$p_{t,\tau}(0) = \alpha_k \quad \text{for all } t, \tau \text{ with } k = \tau - t,$$

where α_k is the probability of reporting at delay k, and the error term has mean one:

$$\mathbb{E}[\varepsilon_{t,\tau}] = 1$$

That is, untreated reporting probabilities depend only on delay k, not on calendar time t. This is the nonlinear analogue of the parallel trends assumption: the untreated potential number of reported crimes would have continued to evolve according to the same multiplicative structure even after the intervention at t^*. We can also see the analogy with two-way fixed effects models for nonlinear difference-in-differences, if we think of incident dates t (or equivalently, crime rates C_t) as "units" and reporting delays k as "time". This suggests a staggered treatment setting, in the sense that different incident dates t are first treated by the intervention at different delays k.

4.3 Identification

I now show how the assumptions above allow identification of changes in reporting behavior without observing total crime incidence C_t. To better understand what is identified from the data, I focus on the multiplicative effect of the intervention on reporting probabilities, denoted by $\gamma_{s,k}$, for each combination of event time s and delay k. Formally, for an incident date $t_b \equiv t^* + s - k$ and a report date $\tau_b \equiv t^* + s$ (so that $\tau_b = t_b + k$), the estimand is

$$\gamma_{s,k} \equiv \frac{p_{t_b,\tau_b}(1)}{p_{t_b,\tau_b}(0)}.$$

This quantity compares the probability that a crime committed at t_b is reported at τ_b under the intervention to the probability that it would have been reported in the absence of the intervention. It can therefore be interpreted as a granular decomposition of the intervention's effect on reporting behavior along both margins of heterogeneity: event time s and delay k. The main challenge in identifying $\gamma_{s,k}$ is that the total number of crimes C_t is unobserved.

As in standard nonlinear difference-in-differences, the identification strategy proceeds in two steps. First, I exploit within-incident-date variation to eliminate the unobserved number of offenses. Second, I use cross-incident-date variation to cancel out baseline reporting probabilities.

Recall that Assumption 1 (No Anticipation) ensures that the intervention does not affect either crime incidence or reporting behavior prior to t^*, so all quantities observed before t^* can safely be treated as untreated counterfactuals. Now consider an offense committed at $t_b < t^*$ that is reported after a delay k such that $\tau_b = t_b + k \geq t^*$. The expected number of reports is

$$\mathbb{E}[R_{t_b,\tau_b}(1)] = C_{t_b}(0)\, p_{t_b,\tau_b}(1),$$

where $C_{t_b}(0)$ is the number of offenses committed at t_b (which remains at its pre-intervention value by the No-Anticipation assumption). Now consider the same incident date but a shorter delay k_0 such that the corresponding report date $\tau_0 = t_b + k_0$ occurs before the intervention, i.e. $\tau_0 < t^*$. For that case,

$$\mathbb{E}[R_{t_b,\tau_0}(0)] = C_{t_b}(0)\, p_{t_b,\tau_0}(0).$$

Taking the ratio of these two expectations removes the unobserved $C_{t_b}(0)$:

$$\frac{\mathbb{E}[R_{t_b,\tau_b}(1)]}{\mathbb{E}[R_{t_b,\tau_0}(0)]} = \frac{p_{t_b,\tau_b}(1)}{p_{t_b,\tau_0}(0)}.$$

To difference out the remaining baseline reporting probability $p_{t_b,\tau_0}(0)$, consider another incident date $t_a < t^*$ for which both the short- and long-delay reports occur before the intervention, i.e. $t_a + k_0 < t_a + k < t^*$. The corresponding expectations are

$$\mathbb{E}[R_{t_a,t_a+k}(0)] = C_{t_a}(0)\, p_{t_a,t_a+k}(0), \quad \mathbb{E}[R_{t_a,t_a+k_0}(0)] = C_{t_a}(0)\, p_{t_a,t_a+k_0}(0).$$

Taking their ratio removes $C_{t_a}(0)$:

$$\frac{\mathbb{E}[R_{t_a,t_a+k}(0)]}{\mathbb{E}[R_{t_a,t_a+k_0}(0)]} = \frac{p_{t_a,t_a+k}(0)}{p_{t_a,t_a+k_0}(0)}.$$

Assumption 2 (Parallel Trends) implies that, in the absence of the intervention, the relative likelihood of reporting at different delays is invariant across incident dates:

$$\frac{p_{t_b,\tau_b}(0)}{p_{t_b,\tau_0}(0)} = \frac{p_{t_a,t_a+k}(0)}{p_{t_a,t_a+k_0}(0)}.$$

Combining the two sets of ratios leads to the identifying equation for $\gamma_{s,k}$:

$$\gamma_{s,k} = \frac{\dfrac{\mathbb{E}[R_{t_b,\tau_b}(1)]}{\mathbb{E}[R_{t_b,\tau_0}(0)]}}{\dfrac{\mathbb{E}[R_{t_a,t_a+k}(0)]}{\mathbb{E}[R_{t_a,t_a+k_0}(0)]}} \qquad \text{for all } k \geq s+1.$$

The ratio across delays eliminates unobserved crime incidence, while the ratio across incident dates removes baseline reporting probabilities, leaving the proportional effect of the intervention on reporting at each cell (s,k). For simplicity, this argument uses a single quadruple of cells for clarity; in practice, many incident dates provide valid short- and long-delay comparisons for each (s,k), and the estimators introduced below aggregate all of them, which improves efficiency.

Figure 3 illustrates this ratio-of-ratios. Panel (a) plots expected reports against calendar time for two incident dates committed before the intervention, t_a and t_b. Since the intervention shifts reporting only once a report date crosses t^*, offenses committed before t^* but reported afterward are the switchers that identify $\gamma_{s,k}$. Panel (b) re-indexes the same reports by delay k, so that

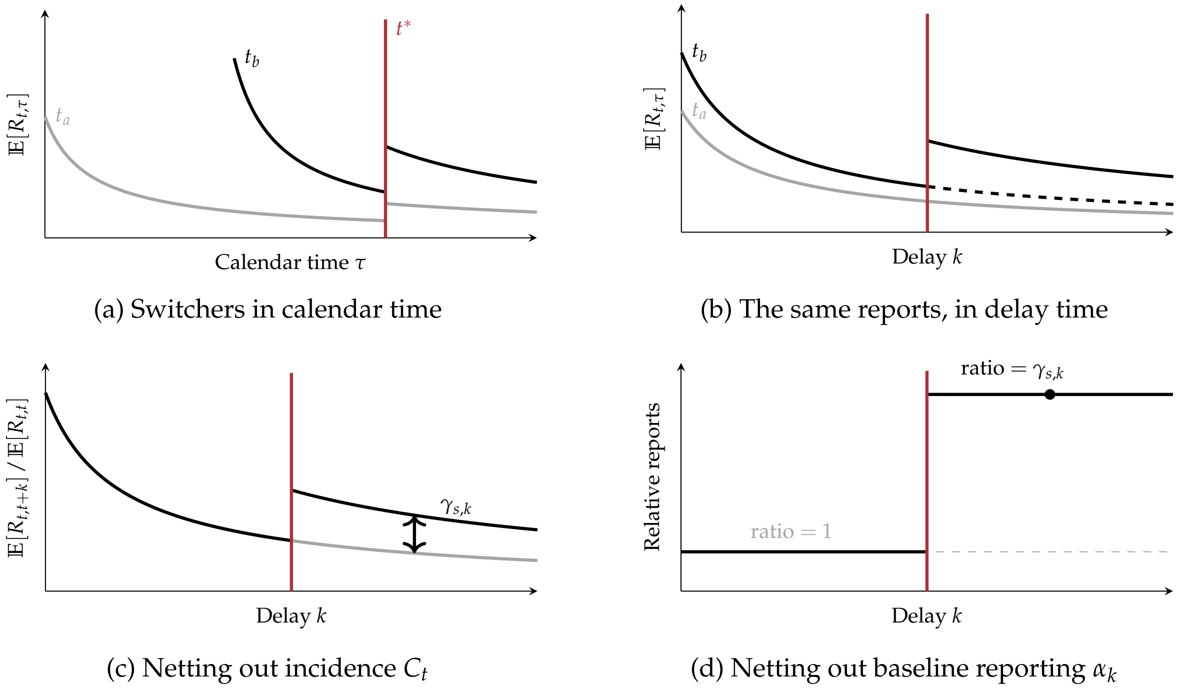

Figure 3: Graphical intuition for the identification of $\gamma_{s,k}$.

Notes: Schematic illustration. Each curve traces expected reports for a given incident date, and the vertical maroon line marks the intervention at t^*. Panels (a)–(b) isolate the switchers: offenses committed before t^* but reported afterward, shown first in calendar time and then re-indexed by reporting delay k (the dashed curve in (b) is the untreated counterfactual). Panels (c)–(d) perform the two cancellations that recover $\gamma_{s,k}$: normalizing each cohort within its own incident date removes incidence C_t, and taking the ratio of the switcher to the not-yet-treated cohort removes the baseline reporting schedule α_k, leaving a relative curve equal to one before t^* and to $\gamma_{s,k}$ afterward.

the switcher cohort t_b separates from its untreated counterfactual exactly when its report date reaches t^*. The levels of these curves still mix reporting with the unobserved number of offenses C_t. Panel (c) divides each cohort by its own count of short-delay reports, which cancels C_t and leaves the baseline reporting schedule α_k; absent the intervention, the normalized curves coincide. Panel (d) then takes the ratio of the switcher to the not-yet-treated cohort, differencing out α_k: the relative curve equals one before t^* and jumps to $\gamma_{s,k}$ afterward.

A necessary condition for this design is that, for a given s, there exists a shorter delay k_0 such that the short-delay report occurs before the intervention, while the long-delay report occurs after. Because $k_0 \geq 0$, this is possible only when $k \geq s+1$. Intuitively, the same incident date must generate both a pre- and post-intervention report to eliminate unobserved incidence. In other words, the effects on crime reporting are identified for switchers in the data. Table 1 illustrates which $\gamma_{s,k}$ cells are identified in an example with three post-intervention periods and four delay categories.

Estimates of $\gamma_{s,k}$ represent proportional changes in reporting probabilities for the set of switcher cells $\Omega_{\text{sw}} = \{(s,k) : k \geq s+1\}$. They can be further aggregated to recover an informative estimand. I focus on the Proportional Treatment effect on the Treated (PTT), which measures the percent

Table 1: Illustration of Identified $\gamma_{s,k}$ Cells

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
$s=0$	$\times$	✓	✓	✓	✓
$s=1$	$\times$	$\times$	✓	✓	✓
$s=2$	$\times$	$\times$	$\times$	✓	✓

change of the average number of reported crimes among treated observations:

$$\text{PTT} = \frac{\sum_{(t,\tau)\in\Omega} \mathbb{E}[R_{t,\tau}(1)]}{\sum_{(t,\tau)\in\Omega} \mathbb{E}[R_{t,\tau}(0)]} - 1,$$

where Ω denotes the set of cells for switchers ($k \geq s+1$). It is the targeted estimand in the canonical 2×2 non-linear difference-in-differences using Poisson pseudo-maximum likelihood (PPML) (Moreau, 2025). It is a proportional treatment effect or, equivalently, a semi-elasticity.

I have just shown that, under standard assumptions, researchers can examine the intervention's impact on the victim reporting rate among switchers without prior knowledge of C_t. However, one may also be interested in studying the effects of the intervention on "always-treated" incident dates (i.e., dates such that $t \geq t^*$). For "always-treated" cells ($k \leq s$), no short delay k_0 exists such that both pre- and post-intervention reports are observed for the same incident date. The within-incident-date anchor is therefore unavailable, and the ratio-of-ratios identification strategy cannot identify $\gamma_{s,k}$.

This has clear implications for identifying changes in crime incidence $C_t(1)$. In the model, observed reports satisfy $R_{t,\tau}(z) = C_t(z) \times p_{t,\tau}(z) \times \varepsilon_{t,\tau}$, so any inference about $C_t(z)$ depends on the reporting probabilities $p_{t,\tau}(z)$. Once reporting behavior changes after t^*, $C_t(z)$ is implicitly a function of $\gamma_{s,k}$, but precisely for those parameters that are not point-identified from the data. For instance, an estimate of C_{t^*} would require an estimate of $\gamma_{0,0}$, $\gamma_{1,1}$, $\gamma_{2,2}$, etc. In practice, this means that the post-intervention incidence series $C_t(1)$ is ultimately an extrapolation exercise that relies on stronger functional form restrictions to map the treatment effects identified for switchers to "always-treated" incident dates. Note that this is a well-known limitation of difference-in-differences approaches and that, for this reason, most recently proposed estimators in the literature exclude "always-treated" observations (Liu et al., 2022).

Several typical functional form restrictions used in applied research easily resolve this issue. For example, one option is to impose a homogeneous effect and assume a single post-intervention parameter γ that does not vary with s or k. A second option is to allow for dynamic treatment effects by letting $\gamma_{s,k}$ vary over event time s while remaining constant across delays, so that $\gamma_{s,k} = \gamma_s$. A third option is to allow for delay-dependent treatment effects by permitting $\gamma_{s,k}$ to vary across delays while remaining constant over event time, so that $\gamma_{s,k} = \gamma_k$. I discuss these variants in the estimation section below.

4.4 Estimation

I propose two approaches to estimation: (i) the traditional Poisson pseudo-maximum likelihood (PPML) estimator and (ii) an imputation estimator that correctly handles heterogeneous treatment effects across event times s and delays k. For clarity, I presented the framework around a single binary (pre/post) intervention, but it extends directly to an event-study specification that traces the dynamics of the treatment effect over event time, as shown below.

A natural and convenient choice to estimate the model is Poisson pseudo-maximum likelihood (PPML) with two-way fixed effects for incident dates ($\delta_t \equiv \log C_t$) and reporting delays ($\lambda_k \equiv \log \alpha_k$), and a functional form restriction on the treatment effects. Standard errors are clustered at both the incident date and the report date. This approach accommodates the large number of zeros in $R_{t,\tau}$ and is consistent under correct specification of the conditional mean, even if the distribution is not Poisson. It also allows researchers to directly recover estimates of changes in C_t in the post-intervention period.

For example, the simplest approach assumes a constant post-intervention effect γ that does not vary with delay or event time. This collapses all coefficients $\gamma_{s,k}$ into a single coefficient β, estimated from:

$$\mathbb{E}[R_{t,\tau}] = \exp\left(\delta_t + \lambda_k + \beta D_\tau\right),$$

where D_τ is an indicator if the crimes were reported in the post-intervention period.

Of course, researchers may be unwilling to assume treatment effect homogeneity. To relax this assumption, the treatment effect can be allowed to vary across delays. Partition delays into bins $\mathcal{B} = \{B_1, \ldots, B_J\}$ and define $G_b = \mathbb{1}\{k \in B_b\}$. The conditional mean becomes:

$$\mathbb{E}[R_{t,\tau}] = \exp\left(\delta_t + \lambda_k + \sum_{b \in \mathcal{B}} \theta_b \left(D_\tau \times G_b\right)\right),$$

where each θ_b measures the change in reporting probability for crimes reported with delays in bin B_b. If the conditional mean is correctly specified, this approach tests whether the intervention affects reporting at short and long delays differently.

Alternatively, one can allow treatment effects to evolve over event time s. Including event-time indicators leads to:

$$\mathbb{E}[R_{t,\tau}] = \exp\left(\delta_t + \lambda_k + \sum_s \beta_s \mathbb{1}\{S = s\}\right).$$

As in traditional event studies, the coefficients β_s trace how reporting probabilities evolve before and after the intervention. If the conditional mean is correctly specified, pre-intervention coefficients provide a diagnostic for anticipatory behavior, while post-intervention coefficients capture the dynamics of the treatment effect over time.

In Appendix C, I run a series of Monte Carlo simulations. Under correct model specification, the PPML estimator shows good finite-sample performance and correctly decomposes variations in reported crimes into variations in crime incidence and victim reporting.

PPML with two-way fixed effects is convenient, but it may produce biased estimates when treatment-effect heterogeneity is present and not sufficiently flexibly modeled (see de Chaisemartin and d'Haultfoeuille (2023); Roth et al. (2023) for linear models and Wooldridge (2022); Moreau (2025) for non-linear models). Saturated regressions that account for heterogeneity in event times s and reporting delays k (see Wooldridge (2021) for the linear case and Wooldridge (2022) for the nonlinear case) or counterfactual imputation estimators (see Borusyak et al. (2024) for the linear case and Moreau (2025) for the nonlinear case) resolve these issues. In what follows, I rely on a "heterogeneity-robust" imputation estimator.

The estimator proceeds in four steps:

1. *Estimate the untreated baseline.*

   Using only pre-intervention cells (i.e., such that $\tau < t^*$), I estimate the untreated conditional mean:

   $$\mathbb{E}[R_{t,\tau}] = \exp(\delta_t + \lambda_k),$$

   where δ_t and λ_k capture, respectively, variation in offense incidence and baseline reporting probabilities. This step yields estimates of $\widehat{\delta_t}$ and $\widehat{\lambda}_k$ for incident dates and delays observed before the intervention.

2. *Impute counterfactual untreated outcomes.*

   For each treated cell (t, τ) with $\tau \geq t^*$, I construct a counterfactual untreated value:

   $$\widehat{R}_{t,\tau}(0) = \exp(\widehat{\delta_t} + \widehat{\lambda}_k).$$

   For switcher cells such that $k \geq s + 1$, I then compute the implied proportional change in reporting probability:

   $$\widehat{\gamma}_{s,k} = \frac{R_{t,\tau}}{\widehat{R}_{t,\tau}(0)}.$$

3. *Aggregate to interpretable estimands.*

   These cell-level estimates are aggregated to form interpretable measures of average treatment effects on the treated – such as the Proportional Treatment effect on the Treated (PTT).

4. *Extrapolate C_t to post-intervention incident dates.*

   While the procedure above identifies reporting effects for switcher cells, it does not identify $\gamma_{s,k}$ for "always-treated" cells such that $k \leq s$. To recover the post-intervention crime incidence series C_t for $t \geq t^*$, we need to make assumptions on the unobserved $\gamma_{s,k}$ for "always-treated" cells. For example, we could model $\gamma_{s,k}$ as:

$$\log(\gamma_{s,k}) = f(s) + g(k) + h(s,k) + u_{s,k},$$

where f, g, and h are flexible functions. Given the previously estimated values of $\alpha_k = \exp(\widehat{\lambda}_k)$ and the additional functional form restriction on treatment effects, we can solve for the post-intervention offense incidence C_t directly from the data on $R_{t,\tau}$. This allows me to extrapolate the full post-intervention path of crime incidence.

Step 4 makes it clear that the imputation estimator does not escape functional-form assumptions. Recovering the post-intervention incidence series C_t still requires extrapolating the unidentified $\gamma_{s,k}$ for always-treated cells through a functional form, exactly as the PPML estimator does. Its role is therefore one of robustness and guidance. As a robustness check, it recovers the treatment effects on the identified switcher cells ($k \geq s+1$) without imposing any restriction across event times or delays. As guidance, the pattern of the cell-level estimates $\widehat{\gamma}_{s,k}$ informs which functional-form restriction to impose when extrapolating to the non-identified cells: for a fixed delay k, one can test whether the effect is constant across event times s (a test for dynamic effects), and for a fixed event time s, whether it is constant across delays k (a test for delay dependence). Plotting $\widehat{\gamma}_{s,k}$ against s and k provides an informative visual diagnostic.

I follow the literature and use a block bootstrap over incident dates to construct confidence intervals (see, for example: Liu et al., 2022; Borusyak et al., 2024; Moreau, 2025).

4.5 Alternative Approach: Duration Modeling

The TWFE approach has several attractive features: it is reduced-form and does not make behavioral assumptions, it does not require knowledge about the share of never-reporters at any point in time, and it requires transparent identification assumptions that are both familiar and in part testable (via the analysis of pre-trends). However, there are two limitations of working on aggregate incident-report cell counts. First, the change in reporting propensity β is a reduced-form estimate that is not micro-founded from a victim's decision-making process, so it cannot be interpreted as the average change in an individual victim's hazard of reporting a crime to the police. This does not affect the extrapolation of changes in sex crime incidence. Second, there is a loss in statistical power due to aggregation: ideally, one would like to work at the individual level. To address these concerns, I develop an alternative empirical framework based on duration modeling, which I briefly explain below (see Online Appendix G for technical details).

This alternative approach derives an individual victim's hazard of reporting from a modified mixed proportional hazards (MPH) model that explicitly accounts for the share of victims who never report to the police. The model can be understood as a Poisson counting process during which victims of sex crimes are exposed to a certain number of potential decisive arguments to report to the police, and the time for each argument to trigger a report to the police is drawn from a probability distribution. It incorporates three essential features to model victim reporting:

(i) a share of victims report with a delay, (ii) a share of victims will never report, (iii) covariates may influence reporting delays, and ultimately, the share of victims who will never report (e.g., indicators for policy interventions or public awareness campaigns such as #MeToo).

The model requires two identification assumptions, analogous to the TWFE model, plus additional structure to handle never-reporters. The identification assumptions are: (i) no anticipation – the intervention does not affect reporting behavior before t^*, identical to Assumption 1 in the TWFE model; and (ii) conditional independence – time-to-report is independent of incident date, conditional on covariates (including treatment indicators). This is the analog of Assumption 2 (Parallel Trends) in the TWFE model: conditional on the intervention dummy, the distribution of reporting delays does not depend on calendar time. These assumptions would suffice if we observed all victims. Because we only observe reporters, two additional components are required: (iii) a mixed proportional hazards functional form that links the observed distribution of reporting delays to the unobserved share of never-reporters; and (iv) external calibration of the share of never-reporters for at least one time period. The TWFE model avoids this last requirement by working with aggregate cell counts rather than individual-level data, and focusing on recovering *variations* in crime reporting and crime incidence. Under these assumptions, the duration model can reconstruct the evolution of the share of never-reporters over time (in *levels*).

In practice, the model is estimated via maximum likelihood, with a likelihood function that accounts for double-truncation in the data, a non-trivial sample-selection issue that arises because only crimes reported within the study window are observed. Monte Carlo simulations mirroring those of the TWFE estimator validate the approach (see Online Appendix G.3). Section 5 discusses the resulting decomposition alongside the TWFE estimates.

5 Main Results

This section studies sex crime incidence and reporting in New York City between 2006 and 2019. I first decompose reported sex crimes into separate series for incidence and victim reporting, and trace how each evolved over the decade. I then isolate the effect of the #MeToo movement on each margin: first on victim reporting, then on sex crime incidence.

All estimates in this section share a common baseline sample, which I state here once. The unit of observation is a quarterly incident-by-report cell for sex crimes in New York City, with reports filed between January 2006 and October 2019. The baseline applies three restrictions to the raw records: I exclude incidents dated the first of a month, a value the police enter as a placeholder when the exact date is unknown; I exclude left-truncated cells whose incident date precedes the start of the sample; and I cap reporting delays at ten years. The pooled effect is stable across all of these choices and in the full sample without restrictions (see Online Appendix Table D.1).

5.1 Trends in Crime Incidence and Reporting (2010–2019)

To separate the reporting and incidence components, I estimate the baseline reporting-delay profile λ_k over the first four years of the sample, 2006 through 2009, and identify later changes in reporting as deviations from it, estimated from 2010 onward.

Concretely, the decomposition reported below comes from the dynamic two-way fixed effects Poisson regression

$$\mathbb{E}[R_{t,\tau}] = \exp\left(\delta_t + \lambda_k + \sum_s \beta_s \mathbb{1}\{S = s\}\right), \tag{2}$$

estimated by PPML with standard errors clustered by incident and report date, where $S = \tau - t^*$ denotes event time. The incident-date effects $\delta_t \equiv \log C_t$ recover sex crime incidence, the delay effects $\lambda_k \equiv \log \alpha_k$ the baseline reporting-delay profile, and the event-time effects β_s the reporting response, normalized to zero over the 2006–2009 baseline window. The reporting index is $\exp(\beta_s)$, plotted in green in Figure 4, and the incidence series is $\exp(\delta_t)$, plotted in red; both are shown relative to 2010, alongside reported sex crimes in black.

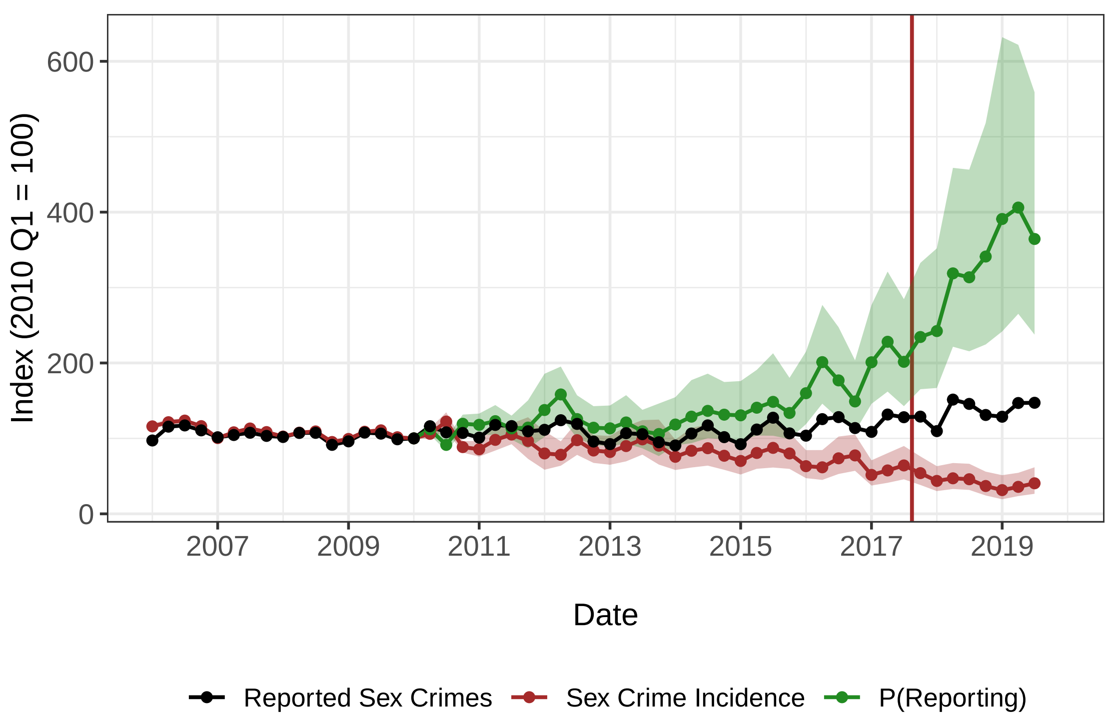

Figure 4: Trends in Sex Crime Reporting and Incidence

Notes: This figure decomposes reported sex crimes to the police by incident date into crime reporting probability and crime incidence estimates based on Equation 2. The TWFE model identifies reporting effects from delayed reports that cross the treatment date, then extrapolates these effects to recover the full incidence series. Sex crime reporting behaviors are plotted in green, sex crime incidence in red, and reported sex crimes to the police in black. All series are normalized to 100 in 2010 Q1. Shaded areas represent 95% confidence intervals constructed with a block bootstrap over incident dates. The vertical solid red line corresponds to the Me Too movement's mediatization.

The two estimated components move in opposite directions over the decade. The reporting probability (green) rose throughout: it increased at approximately 8% per year between 2010 and 2017 and more than tripled between 2010 and 2019, a roughly 3.7-fold rise. Sex crime incidence (red) declined over the same period by roughly two-thirds, approximately 66% between 2010 and 2019. Because these two forces pull in opposite directions, reported sex crimes (black, by report date) rose by only approximately 40% over the decade, far less than the movement in either underlying component.

These trends carry a clear methodological implication. Research that uses reported crime to evaluate an intervention almost always assumes, explicitly or implicitly, that one of the two margins is held fixed, attributing any movement in reported counts entirely to incidence or entirely to reporting. The decade studied here shows how tenuous that assumption can be: both margins moved, and by large amounts, over the same period. As many have suggested since Quêtelet (1831), reported crime statistics are likely the tip of the iceberg, and separating incidence from reporting is essential for empirical research on crime. They also carry a substantive lesson by placing the #MeToo movement in its broader historical context. Although the precise date on which it went viral was unforeseen, the movement arrived amid a deeper, decade-long change in both sex crime reporting and incidence.

For robustness, I also perform a similar decomposition exercise using the duration modeling approach described in Subsection 4.5. I calibrate the model so that 80% of victims would not have reported a sex crime committed in 2006. This value is bracketed by external estimates of the share of victims who do not report: national victimization surveys indicate that 60 to 70% of rape and sexual assault victimizations are not reported to the police (Planty et al., 2013), while the Campus Climate Survey implies a non-reporting rate closer to 90% for reports to university programs (Levy and Mattsson, 2025). I report results for the bracketing values of 60% and 90% never-reporters. The baseline hazard is flexibly estimated via a piecewise function with breaks after 1, 30, 90, 180, and 365 days. I assume no unobserved heterogeneity (as it seems very weakly identified in the data). Despite substantial methodological differences with the TWFE approach, the duration model's estimates corroborate my results on trends in sex crime reporting and incidence over the period (see Appendix Section G for technical details and Subsection G.4 for the decomposition exercise). The clear convergence of results across two very different empirical strategies is perhaps not surprising given the compelling raw empirical evidence provided in Section 3, but it strongly suggests that the main findings are not artifacts of any particular modeling choice.

5.2 Did #MeToo increase victim reporting?

Despite these large pre-trends, the time series suggests that #MeToo substantially increased victim reporting. A single post-#MeToo indicator implies an average increase in the reporting probability of approximately 47% (0.38 on the log scale; see Table 2, Column 1). To fix ideas about the magnitude, I linearly extrapolate the 2010–2017 trend in Figure 4 as a rough no-#MeToo counterfactual: relative to it, reporting in 2019 stands approximately 81% higher, accounting for roughly 61% of the total increase over the decade.

Table 2: #MeToo Effect on Sex Crime Reporting and Incidence

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Proposed Approach						**Usual Approach**	
	Reporting Probability (in logs)		**Estimated Sex Crimes (in logs)**				**Reported Sex Crimes (in logs)**	
							Incident date	**Report date**
#MeToo (indicator)	0.38***	0.41***	-0.48***	-0.44***	-0.49***	-0.38***	0.16***	0.22***
	(0.10)	(0.07)	(0.14)	(0.13)	(0.15)	(0.08)	(0.04)	(0.04)
Estimator	PPML	Imputation	MC	IFE	OLS	OLS	OLS	OLS
Fixed Effects								
Incident date	✓	✓	✓	✓	✓			
Crime type			✓	✓	✓	✓	✓	✓
Reporting delay	✓	✓						
Seasons						✓	✓	✓
Time Controls								
Crime-specific linear trends						✓	✓	✓
Common post-#MeToo indicator						✓	✓	✓
Control Crimes								
Murder / Assault / Robbery / Burglary			✓	✓	✓	✓	✓	✓
Standard Errors	Clustered	Bootstrap	Bootstrap	Bootstrap	Bootstrap	Bootstrap	Robust	Robust
N Observations	1,540	1,540	275	275	275	275	275	275

Notes: The table reports the Me Too movement's effect on sex crime reporting and incidence in New York City, on a quarterly panel; all estimates are on the log scale. Columns 1–6 use my proposed approach; Columns 7–8 the usual approach. Columns 1–2 report the effect on the reporting probability: Column 1 is the pooled two-way fixed effects PPML estimate, with incident-date and reporting-delay fixed effects; Column 2 is the heterogeneity-robust Proportional Treatment effect on the Treated (PTT) recovered from the imputation estimator, aggregating the cell-level effects over the identified switcher cells. Columns 3–6 report the effect on the recovered sex crime incidence series: matrix completion (MC, Column 3), interactive fixed effects (IFE, Column 4), a two-way fixed effects OLS regression (Column 5), and an OLS specification that replaces time fixed effects with crime-specific linear trends and crime-specific seasonal fixed effects (Column 6). Columns 6–8 also include a common post-#MeToo indicator shared by all crime types, so the coefficient reported in these columns is the additional post-#MeToo shift specific to sex crimes. Columns 7–8 report the observed effect on reported sex crimes, aggregated by incident date and by report date, respectively; both are estimated with the same specification as Column 6. Standard errors are in parentheses: Column 1 is two-way clustered by incident and report period; Columns 2–6 are bootstrap standard deviations (described in the text); Columns 7–8 are heteroskedasticity-robust. Columns 1–2 are estimated on the full quarterly incident-by-report grid ($N = 1{,}540$ cells; cells whose reports are excluded by the sample restrictions enter as zeros); Columns 3–8 on the quarterly crime-by-period panel ($N = 275$). All columns apply the baseline sample restrictions described in the text (excluding left-truncated cells, first-of-month placeholder dates, and reporting delays beyond ten years). The control group comprises non-sexual assaults, murders, robberies, and burglaries, excluding female murder victims, whose deaths may be sexual in nature. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The increase in reporting is broad-based across victim and suspect characteristics, as well as across misdemeanors and felonies (see Online Appendix Figure D.2). The point estimates differ somewhat across groups, but the confidence intervals overlap considerably, so the evidence on which groups responded most strongly is largely inconclusive. My reading is that the movement raised reporting across all groups.

This estimate is not sensitive to how the sample is constructed (see Online Appendix Table D.1). Applying the three sample restrictions, excluding left-truncated cells, excluding first-of-month placeholder dates, and capping reporting delays at ten years, each in isolation and in combination, the estimated increase in reporting ranges from approximately 40% to 53% and remains statistically significant throughout.

How does the reporting response vary across cells? The imputation estimator answers this question without functional-form restrictions, recovering a separate effect $\widehat{\gamma}_{s,k}$ for every switcher cell, unrestricted across event time s and delay k. Two clear patterns emerge. First, the effect varies strongly with event time: it builds over the quarters following the movement's mediatization before gradually receding (see Online Appendix Figure D.3). Second, within the ten-year delay window of the baseline sample, it bears no clear relationship to the reporting delay: an OLS fit of the cell-level effects on delay yields a large and significantly positive constant (approximately 0.63 in logs, $p = 0.01$) but a flat slope (approximately 0.007 per quarter, $p = 0.47$), so the response is a large, roughly uniform upward shift across delays, and victims reporting incidents up to a decade old responded roughly as strongly as victims of recent incidents (see Online Appendix Figures D.4–D.5). This pattern serves two purposes. First, it dictates the functional form: the event-study specification in Equation 2 lets the reporting effect vary over event time while holding it constant across delays, exactly the structure the unrestricted estimates support, and it is this restriction that extrapolates the reporting response to the always-treated cells when recovering post-intervention incidence. Second, the cell-level estimates confirm the magnitude: aggregating them yields a Proportional Treatment effect on the Treated of +51% (see Table 2, Column 2), close to the pooled PPML estimate in Column 1, so heterogeneity across cells does not materially bias the pooled effect.

The duration model reproduces these findings (see Online Appendix G). Under the baseline calibration, the estimated reporting rate more than doubles over the decade, reaching approximately 33% just before #MeToo and approximately 50% by 2019, with the movement reinforcing the pre-existing upward trend (see Online Appendix Figure G.7). The same exercise for alternative values of the never-reporter share, from 60% to 90%, shifts the level of the reporting rate but leaves the substantial increase over the decade intact (see Online Appendix Figure G.9).

5.3 Did #MeToo have a deterrent effect?

The preceding decomposition indicates that victim reporting rose sharply while sex crime incidence decreased over the period. In what follows, I isolate the contribution of #MeToo to the decline in incidence. As #MeToo potentially affected all cities in the United States and worldwide, credible control groups for causal inference are limited. However, the crime literature suggests that crime categories are subject to cyclical fluctuations, some of which have been explained by weather conditions, economic downturns, labor market conditions, alcohol consumption, and sports events (e.g., Markowitz, 2005; Jacob et al., 2007). For these reasons, reported non-sexual crimes are a plausible control group. I therefore construct several counterfactual models for quarterly sex crime incidence from quarterly non-sexual crime reports, and report the average effect of #MeToo on sex crime incidence under each. The simplest is a two-way fixed effects difference-in-differences. For crime i in quarter t,

$$\log(\text{Crimes})_{it} = \beta \text{MeToo}_{it} + \delta_i + \delta_t + \varepsilon_{it}. \tag{3}$$

where δ_i and δ_t are crime and time fixed effects and MeToo_{it} equals one for sex crimes after October 2017, so that β is the average effect of #MeToo on sex crime incidence. I also estimate two more flexible counterfactual models: an interactive fixed effects model (Xu, 2017) and a matrix completion model (Athey et al., 2021). Finally, following Levy and Mattsson (2025), I estimate a specification that replaces the time fixed effects δ_t in Equation 3 with a common post-#MeToo indicator, crime-specific linear time trends, and crime-specific seasonal fixed effects (Column 6 of Table 2). The control group comprises the most common non-sexual categories in the data, non-sexual assaults, murders, robberies, and burglaries, excluding female murder victims, since some homicides against women may be sexual in nature.

The incidence series that serves as the outcome is itself estimated, so I propagate its sampling uncertainty into the treatment effect with a bootstrap of 1,000 replications that combines two sources. To reflect uncertainty in the estimated incidence, each replication draws a new vector of incidence coefficients from the estimated sampling distribution of the decomposition (a multivariate normal centered at the point estimates with their estimated covariance) and recomputes the implied incidence series. To reflect uncertainty in the counterfactual, each replication resamples the control crime categories with replacement. For each replication, I re-estimate the treatment effect under each counterfactual model on the resulting panel. The reported standard errors are the standard deviations of the bootstrap distribution, with p-values from the implied t-ratios, and the confidence intervals in the event-study figures are the 2.5th and 97.5th percentiles of the bootstrap distribution in each quarter.

To interpret the results as causal evidence, I rely on two assumptions. The first is that, absent #MeToo, trends in sex crime incidence would have evolved like the counterfactual. The two-way fixed effects difference-in-differences assumes parallel trends in outcomes between sexual and non-sexual crimes; matrix completion and interactive fixed effects weaken this requirement, recovering the counterfactual incidence path from the joint movement of sex and non-sexual crimes without requiring their trends to coincide. The analysis of the pre-trends in the event study figures adjudicates between these different counterfactual models: a joint test of the pre-#MeToo coefficients is consistent with no differential trend for matrix completion and interactive fixed effects, but rejects it for the plain difference-in-differences (despite only one pre-treatment quarter's confidence interval being statistically significantly different from zero). I therefore take matrix completion as the main specification (see Figure 5). The second condition is that #MeToo did not change the reporting of non-sexual crimes, so that the control series reflects incidence rather than reporting. This is plausible because the movement concerned sexual offenses specifically.

Table 2 presents the results. My preferred matrix completion specification implies that #MeToo reduced sex crime incidence by approximately 38% (-0.48 in logs; Column 3), a large and statisti-

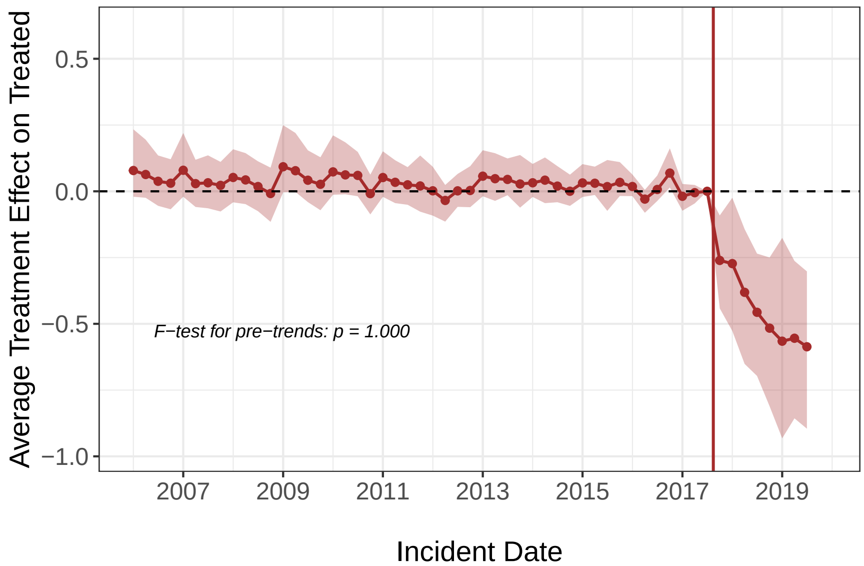

Figure 5: #MeToo Effect on Sex Crime Incidence

Notes: Quarterly estimates of the average treatment effect using matrix completion (MC). The control group includes non-sexual assaults, murders, robberies, and burglaries. The main results are presented in Table 2. 95% confidence intervals are the 2.5th and 97.5th percentiles of the bootstrap distribution described in the text, which accounts for uncertainty in both the estimated incidence and the counterfactual model. The vertical solid red line corresponds to the Me Too movement's mediatization. For alternative counterfactual models, see Online Appendix Figure E.1.

cally significant decline.

These quarterly effects cumulate to a sizable aggregate impact. Let $\widehat{C}_t$ denote estimated sex crime incidence and $\widehat{C}_t^0$ its matrix completion counterfactual. The share of sex crimes prevented by the movement over the decade is the cumulated counterfactual gap relative to the total number of crimes that would have been committed absent the movement:

$$\text{Prevented share} = \frac{\sum_{t \geq t^*} \left(\widehat{C}_t^0 - \widehat{C}_t\right)}{\sum_t \widehat{C}_t + \sum_{t \geq t^*} \left(\widehat{C}_t^0 - \widehat{C}_t\right)},$$

where the sums over $t \geq t^*$ run over the post-#MeToo quarters (2017 Q4 to 2019 Q3) and the first sum in the denominator runs over 2010 Q1 to 2019 Q3. Because the level of the estimated incidence series is identified only up to scale, this share, and not the absolute number of prevented crimes, is the interpretable quantity. The point estimate is 6.4%, with a 95% bootstrap confidence interval of 3 to 12% obtained by applying the bootstrap draws of the pooled effect to the observed post-period incidence. Despite operating only over the final two years of the decade, the movement thus prevented approximately 6% of the sex crimes that would otherwise have been committed between 2010 and 2019.

This decline is hidden in the raw data, which move the other way. Applying the linear-trend specification of Column 6 to reported sex crimes rather than to estimated incidence (Columns 7 and 8) reverses the sign: reported counts rise after #MeToo, by approximately 17% (+0.16 in logs) when aggregated by incident date and by approximately 25% (+0.22 in logs) when aggregated by report date. The gap between these two figures is itself informative. Aggregating by report date assigns the post-#MeToo surge of often-delayed reports to recent quarters, whereas aggregating by incident date attributes them back to earlier incident dates, so the choice of date alone moves the estimate by eight percentage points.

Both increases have been read as evidence that victim reporting rose: using the same New York City records, Levy and Mattsson (2025) obtain a reduced-form increase of approximately 20% (their Appendix Table A.3), within the 17 to 25% range here. They are better understood as a lower bound on the movement's effect: because the rise in reporting and the decline in incidence partly cancel within reported counts, a design that equates reported sex crimes with sex crimes understates the increase in reporting, which I estimate at approximately 51%, and entirely misses the roughly 38% decline in incidence.

These deterrent estimates are robust to the choice of counterfactual model. Matrix completion, interactive fixed effects, two-way fixed effects, and the linear specification yield declines ranging from approximately 32% to 39% (-0.38 to -0.49 in logs), all significant at the 1% level (see Table 2; event-study plots in Online Appendix Figure E.1).

The effect is also not driven by any single control category. I re-estimate the model dropping each non-sexual crime from the control group in turn (see Online Appendix Table E.1); in every case the estimated decline in incidence remains negative and significant at the 1% level, ranging from approximately 32% to 41% (-0.39 to -0.52 in logs).

Finally, Online Appendix G replicates the analysis with the duration model, which rests on different modeling assumptions. Applying the same counterfactual designs to the incidence series implied by that model, #MeToo reduced sex crime incidence by approximately 25% (-0.29 in logs under matrix completion), significant at the 1% level and robust across counterfactual models and leave-one-out control sets (see Online Appendix Tables G.1 and G.2). For alternative values of the never-reporter share, from 60% to 90%, the magnitude of this estimate depends on the calibration (see Online Appendix Figure G.10).

6 Discussion and Mechanisms

6.1 Comparison with Traditional Approaches Disentangling Both Margins

Various approaches are commonly used in the literature to disentangle sex crime reporting and incidence. Below, I list them and discuss how they compare with my results. Note that all these exercises are conducted at the national level, whereas my main results are based on a single large city (New York City).

I first turn to victimization surveys for direct estimates of sex crime incidence and reporting, as those are often used in research on crime. The National Crime Victimization Survey (NCVS) estimates that increased sex crime incidence mainly drives the rise in sex crime reports (up roughly

145% at its 2018 peak on the 2006-indexed series; see Online Appendix Figure F.2). The increase is particularly large after #MeToo, as noted in previous work (Worthen and Schleifer, 2024).

Another common approach is to rely on emergency records to proxy sexual violence (Aizer, 2010). Emergency department visits indicate that consultations for sexual assaults increased by approximately 28% over the period (peaking at about 41% in 2017), whereas consultations for non-sexual assaults decreased by approximately 12% (see Online Appendix Figure F.3). The increase is particularly large in 2017 and 2018.

Thus, both approaches suggest an increase in sex crime incidence that may be understood as a backlash effect following the Me Too movement's mediatization, and are inconsistent with my main results. A plausible explanation is that seeing a doctor and declaring a sexual assault in a survey are the result of a victim's decision. Even if victims are more likely to see a doctor or disclose in a survey than to report to the police, this higher level does not insulate these measures from variation in the propensity to disclose over time. Identifying changes in incidence requires a stable disclosure rate, not a high one. Thus, as for reported crime statistics, emergency records may jointly reflect variations in victim reporting behaviors and crime incidence.

I now show trends for sex crime incidence and reporting at the national level using homicides (see Online Appendix F.1). Homicides are extremely likely to be recorded by law enforcement agencies and are thus the only crime category with a reporting rate of virtually 100%. Though it is rarely explicitly stated in applied work, the critical assumption of this alternative empirical strategy is that the ratio of sexual (non-sexual) homicides to sexual (non-sexual) violent crimes is constant over time (Levitt, 1998). Under this assumption, changes in sexual (non-sexual) homicides reflect changes in sexual (non-sexual) crimes. Of course, a clear limitation of this empirical strategy is that homicides are much rarer than other crimes. For instance, in 2017, fewer than 20 homicides were associated with a sex crime in the United States. Despite this caveat, I find qualitatively similar trends at the national level. My estimates suggest a substantial (roughly five-fold) increase in sexual crime reporting and a 70 to 80% decrease in sexual crime incidence between 2010 and 2018, while the reporting rate and incidence of non-sexual crimes remain stable over the same period.

6.2 Recording Guidelines and Practices

An alternative interpretation of my results is that police officers' recording guidelines changed over the period. The legal definitions of sex crimes did not change between 2006 and 2019 in my sample. However, the FBI changed its definition of rape in 2013.[7] City-level police records rely on a different categorization of offenses, which should not affect my results. To address this concern, I re-estimate the pooled reporting effect restricting the sample to crimes reported after 2014: the

[7] On the FBI's website, one may read: *"The old definition was 'The carnal knowledge of a female forcibly and against her will.' Many agencies interpreted this definition as excluding a long list of sex offenses that are criminal in most jurisdictions, such as offenses involving oral or anal penetration, penetration with objects, and rape of males. The new Summary definition of Rape is: 'Penetration, no matter how slight, of the vagina or anus with any body part or object, or oral penetration by a sex organ of another person, without the consent of the victim."'*

estimate falls from 0.38 to 0.16 in logs, as expected since the shorter window starts from an already-elevated level of reporting and thus absorbs less of the pre-#MeToo trend, but it remains positive and significant at the 5% level (see Column 6 of Online Appendix Table D.1).

Similarly, police officers could have become more likely to accept and record complaints for sex crimes. This is plausible and difficult to assess directly. Arrest rates offer an indirect test: if officers had lowered the bar for recording complaints, the marginal complaints should be weaker cases and arrest rates should fall. Instead, arrest rates remained broadly stable over the period (see Online Appendix Figure F.6).

6.3 Complementary Policy Responses

The Me Too movement was also followed by legislative responses that may themselves have affected reporting. In New York, the Child Victims Act, signed in February 2019, extended the criminal statute of limitations for child sex crimes and opened a heavily publicized one-year window for previously time-barred civil claims beginning in August 2019; separate legislation of September 2019 extended the limitation periods for second- and third-degree rape. These reforms fall within the last three quarters of the sample. I do not view them as confounds but as part of the treatment: they were plausibly accelerated by the movement itself, and they operate through the same channel of lowering the cost, and raising the expected benefit, of coming forward.[8] The estimated post-#MeToo effects should therefore be read as the combined effect of the movement and the policy responses it helped set in motion. The timing nonetheless indicates that the movement's direct effect predates these reforms: the break in reporting is dated 2017 Q4, and the estimated effects are already large throughout 2018, before the first reform was enacted.

6.4 Unfounded Allegations

A remaining concern is that my main results assume truthful and founded crime reports. In practice, false allegations of a crime are a prevalent concern for the criminal justice system – in particular when it comes to sex offenses. It is notoriously difficult to assess the incidence of false accusations. Recent estimates suggest that baseless rape allegations represented approximately 5% of total rape charges in the United States between 2006 and 2010 (De Zutter et al., 2017). Though a precise estimate of such allegations is out of reach of researchers, one can still ponder their implications for interpreting my estimates. If the rate of false allegations is positively correlated to #MeToo, then the model will overestimate the movement's effects on the victim reporting rate. As a result, it will also inflate the size of its extrapolated deterrent effect. This is a plausible scenario, particularly if the expected benefits of filing a charge increase for complainants after #MeToo (through larger financial compensations or higher probabilities of sentencing, for instance). At the same time, arrest rates remain stable over the period suggesting that the allegations following #MeToo were not considered unreliable by police officers (see Online Appendix Figure F.6).

[8] Reporting a crime to the police is not itself subject to a statute of limitations; the reforms bind prosecution and civil litigation, so any effect on police reports operates through incentives and publicity rather than eligibility.

6.5 Attitudes Toward Sexual Violence

The recurring public debate over the incidence of false allegations is a reminder that social norms and beliefs also influence the decision to report a crime to the police. Victims may incur social costs for reporting a sex crime to the police in at least two ways. First, narratives portraying accusers as liars can undermine the credibility of the charges and question the motives behind them. In turn, beliefs about the prevalence of sex crimes and of false accusers in one's society may weigh on a victim's decision.[9] Second, social conformity concerns may also influence a victim's decision. If victims care about what other victims do or about what society expects of them, then a repeated coordination game may easily result in persistent, suboptimal equilibria. Unfortunately, there is no systematic empirical evidence on the discourse surrounding sex crimes or on the attitudes of the broader public on the matter.[10] To proxy attitudes towards sexual violence, I collect tweets mentioning sex crimes between 2011 and 2019, and plot the time series of tweet counts, in logs (see Online Appendix Figure F.4). The hashtag #MeToo was virtually never used before October 2017, which might suggest that public discussion of sex crimes was similarly limited. In fact, the topic was already discussed on Twitter, and its share of total tweets has been increasing since 2011. Similar observations are made with Google Trends data (see Figure F.5). This suggestive evidence is consistent with the pre-trends in crime reporting uncovered in Section 5.

6.6 Odds of Punishment for Sex Offenders

Higher odds of reporting crimes to the police mechanically increase the probability of arrest of sex offenders (unless arrest rates are zero). Thus, a plausible mechanism for the estimated decrease in sex crime incidence between 2006 and 2019 is deterrence. Between 2006 and 2019, arrest rates in New York City remained relatively stable, fluctuating between 37% and 56% for sexual offenses (see Online Appendix Figure F.6). Combining arrest rates with my TWFE estimates of reporting probability changes, I compute an index of the probability of arrest conditional on committing a sex crime, normalized to 2006. The index increases from 100 in 2006 to 168 by 2016 and roughly quadruples to 408 by 2019 (see Online Appendix Figure F.7). The decrease in sex crime incidence is thus consistent with a Beckerian deterrence argument: as reporting becomes more likely, so does arrest, which in turn discourages potential offenders. To quantify this channel, I compute the implied elasticity of sex crime incidence with respect to the probability of arrest, using #MeToo as the source of variation and measuring the #MeToo-induced change in each margin as a deviation from its pre-2017 trend. The matrix completion estimates imply a decline in log incidence of about 0.48, while the log probability of arrest rose by about 0.47 (roughly 61%) above its extrapolated pre-2017 path. Their ratio is an arc elasticity of approximately -1.0 (-0.80 to -1.0 across specifications), in line with the certainty-of-punishment elasticities in the deterrence literature (Chalfin

[9] Given the considerable uncertainty on the matter, these beliefs need not be accurate or unbiased.

[10] To the best of my knowledge, the Views of the Electorate Research (VOTER) survey is the only national survey with explicit questions on attitudes towards sexual violence. Levy and Mattsson (2025) find inconclusive empirical evidence of a change in attitudes towards sexual violence among respondents. Note that the questions that were asked before and after #MeToo investigated sexual harassment in the workplace, which is not the focus of this paper.

and McCrary, 2017; Anker et al., 2021). Of course, this back-of-the-envelope estimate attributes the entire incidence response to the probability of arrest – abstracting from other channels such as the shift in social norms discussed above – and so should be understood as an upper bound on this elasticity.

6.7 External Validity

My analysis covers a single city, and the magnitudes need not generalize to the rest of the United States or to other countries. If anything, the point estimates suggest that New York is indeed a special case. My reduced-form effect of #MeToo on reported sex crimes, between 17% (by incident date) and 25% (by report date) in Columns 7 and 8 of Table 2, is larger than the average treatment effects Levy and Mattsson (2025) estimate elsewhere: roughly 11% across the United States, whether measured on their seven-city panel or on NIBRS, and roughly 8% across 31 OECD countries. New York thus appears to sit at the upper end of the distribution of reported-crime responses, so its estimates likely overstate the movement's average effect.

The reduced-form comparison, however, speaks only to reported crimes, which conflate reporting and incidence. The one source I am aware of that credibly separates the two is the Campus Climate Survey, which records both whether an assault occurred and whether it was reported. Using it, Levy and Mattsson (2025) find that #MeToo raised the rate at which victims reported to their university by approximately 40% and lowered the incidence of sexual assault by approximately 20%. This is the same opposing pattern my decomposition recovers from police records, but it is smaller on both margins than my New York estimates of approximately $+51\%$ on reporting and approximately -38% on incidence. The reduced-form and survey evidence thus point in the same direction: New York is likely an upper bound on the #MeToo movement's effects on victim reporting and crime incidence alike.

7 Conclusion

Underreporting has been a major empirical challenge in interpreting crime statistics over the past two centuries. In this paper, I proposed a methodology to separate crime incidence and reporting. My empirical strategy exploits delayed reports, a pervasive but largely understudied feature of police records. These delays allow researchers to work with familiar TWFE models to study variations in crime reporting and incidence over time under transparent assumptions.

This methodology has a straightforward implication for data collection practices. Law enforcement agencies should systematically record both the date of an incident and the date it was reported to police. While this information is available in some jurisdictions – such as New York City, studied here – it is far from universal. Recording these two dates is low-cost and would allow researchers and policymakers to disentangle reporting behavior from actual crime trends, enabling more informed policy responses.

I then studied sex crime incidence and reporting surrounding the Me Too movement's mediatization. Three key results emerged. First, #MeToo significantly and persistently increased victims' propensity to report. The response was broad-based across victim and suspect groups. Second, according to my estimates of sex crime incidence over the period, the movement also had a deterrent effect on sexual offenders. Third, both effects arose amid a decade-long rise in sex crime reporting and decline in sex crime incidence. These substantial trends partly cancel each other out and are less apparent in the time series of reported crimes.

This last finding highlights the importance of disentangling crime incidence and reporting in police data. Though many competing explanations may rationalize my estimates, they are unlikely to be driven by false allegations or changes in law enforcement agencies' recording guidelines. Instead, I presented suggestive evidence that rapidly changing social norms increased the reporting rate of victims and, ultimately, the likelihood of arrest for sex offenders.

My results suggest that public awareness campaigns and social movements may significantly deter criminal behavior. They further highlight that social norms can successfully enforce socially desirable behaviors when the legal system fails to do so on its own. The Me Too movement's broader legal, political, and economic consequences remain largely unknown and represent an opportunity for future research.

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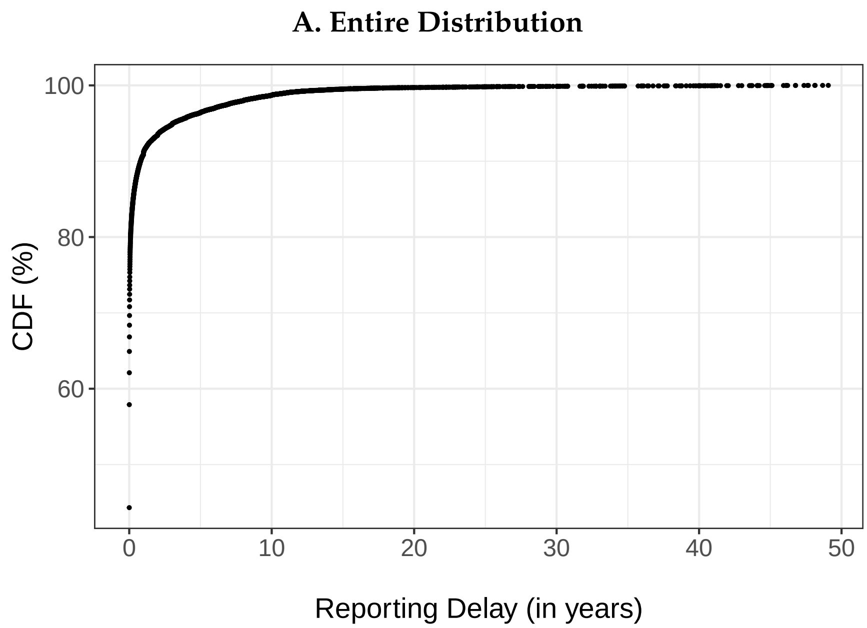
Online Appendix

A Data Sources

A.1 New York City Police Records



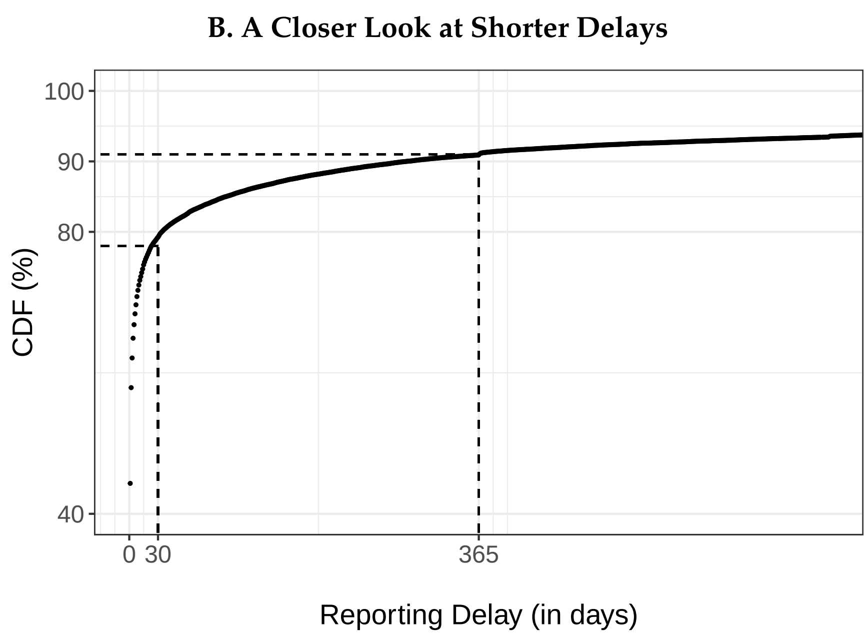


Figure A.1: Distribution of Reporting Delays for Sex Crimes

Notes: Distribution of observed reporting delays for sex crimes. Approximately 44% of complainants report on the day of the incident, 79% within the first month, and 91% within the first year.

Table A.1: Classification of Offenses

Panel A: Non-Sexual Offenses

Classification	High-Level Category	Granular Category
Assault	assault 3 & related offenses, felony assault	assault 3, menacing peace officer, menacing unclassified, obstruction breathing circulation, assault 2/1 unclassified, assault other public service employee, assault police peace officer, assault school safety agent, assault traffic agent, strangulation 1st
Burglary	burglary	burglary truck unknown time, burglary commercial day, burglary commercial night, burglary commercial unknown time, burglary residence day, burglary residence night, burglary residence unknown time, burglary truck day, burglary truck night, burglary unclassified day, burglary unclassified night, burglary unclassified unknown, burglary unknown time
Murder	homicide negligent unclassified, homicide negligent vehicle, murder & non-negligent manslaughter	–
Robbery	robbery	robbery chain store, robbery payroll, robbery ATM location, robbery bank, robbery bar restaurant, robbery begin as shoplifting, robbery bicycle, robbery bodega convenience store, robbery car jacking, robbery check cashing business, robbery clothing, robbery commercial unclassified, robbery delivery person, robbery doctor dentist office, robbery dwelling, robbery gas station, robbery hijacking, robbery home invasion, robbery licensed for hire vehicle, robbery licensed medallion cab, robbery liquor store, robbery neckchain jewelry, robbery of truck driver, robbery on bus or bus driver, robbery open area unclassified, robbery personal electronic device, robbery pharmacy, robbery pocketbook carried bag, robbery public place inside, robbery residential common area, robbery unlicensed for hire vehicle

Panel B: Sexual Offenses

Status	High-Level Category	Granular Category
Excluded	loitering, loitering for drug purposes, loitering deviate sex, loitering gambling, miscellaneous penal law, offenses against public order sensibility, prostitution & related offenses, sex crimes	loitering school, loitering unclassified, loitering 1st degree for drug, loitering gambling other, promoting a sexual performance, use of a child in a sexual performance, disseminating a false sex offender, loitering to promote prostitution, prostitution, prostitution 1 under 11, prostitution 2 compulsory, prostitution 2 under 16, prostitution 3 promoting under, prostitution 3 promoting business, prostitution 4 promoting & security, prostitution patronizing 2/1, prostitution patronizing 4/3, prostitution permitting, sex trafficking, abortional articles issuing, obscene material under 17 years, obscenity 1, obscenity material 3, obscenity performance 3, sodomy consensual
Included	felony sex crimes, rape, sex crimes	stalking commit sex offense, rape 1, rape 1 attempt, rape 2, rape 3, aggravated sexual abuse, child endangering welfare, course of sexual conduct against child, facilitating sexual offense with controlled substance, incest, incest 3, sexual abuse, sexual abuse 3/2, sexual misconduct deviate, sexual misconduct intercourse, sodomy 1, sodomy 2, sodomy 3

Notes: This table presents the classification of offenses in the New York City police records used for the empirical analysis. Panel A shows non-sexual offenses used as control groups. Panel B shows sexual offenses; I exclude offenses related to *loitering*, *prostitution*, *obscenity*, and *consensual acts*.

Table A.2: Descriptive Statistics on Police Records

	Crime Type:				
	Sex Crime	**Murder**	**Assault**	**Robbery**	**Burglary**
Number of Observations	99,398	6,022	935,302	250,197	236,356
Report Type					
Delayed	56%	6.0%	19%	16%	45%
Direct	44%	94%	81%	84%	55%
Time to Report (days)					
Mean	204.30	189.66	4.02	2.90	5.17
Median	2.00	1.00	1.00	1.00	1.00
Standard Deviation	886.75	1,265.82	57.25	58.50	64.62
Victim Sex					
Female	85%	18%	51%	30%	49%
Male	15%	82%	49%	70%	51%
Victim Age					
Adult	50%	92%	89%	84%	100%
Juvenile	50%	7.7%	11%	16%	0.3%
Victim Race					
White	20%	8.8%	15%	20%	41%
Black	40%	63%	46%	37%	31%
Hispanic	41%	28%	39%	44%	28%
Suspect Sex					
Female	8.7%	8.2%	24%	5.8%	6.7%
Male	91%	92%	76%	94%	93%
Suspect Age					
Adult	96%	96%	97%	97%	100%
Juvenile	3.9%	4.0%	2.7%	3.1%	0.3%
Suspect Race					
White	13%	7.1%	11%	5.1%	15%
Black	47%	63%	55%	70%	52%
Hispanic	40%	30%	34%	25%	33%

Notes: Descriptive statistics for incident-level police records of New York City between 2006 and 2019. Time to report is measured in days, with same-day reports coded as one day. Percentage shares are computed over non-missing categories (complaints with unknown victim or suspect sex, age, or race are excluded from the corresponding denominators).

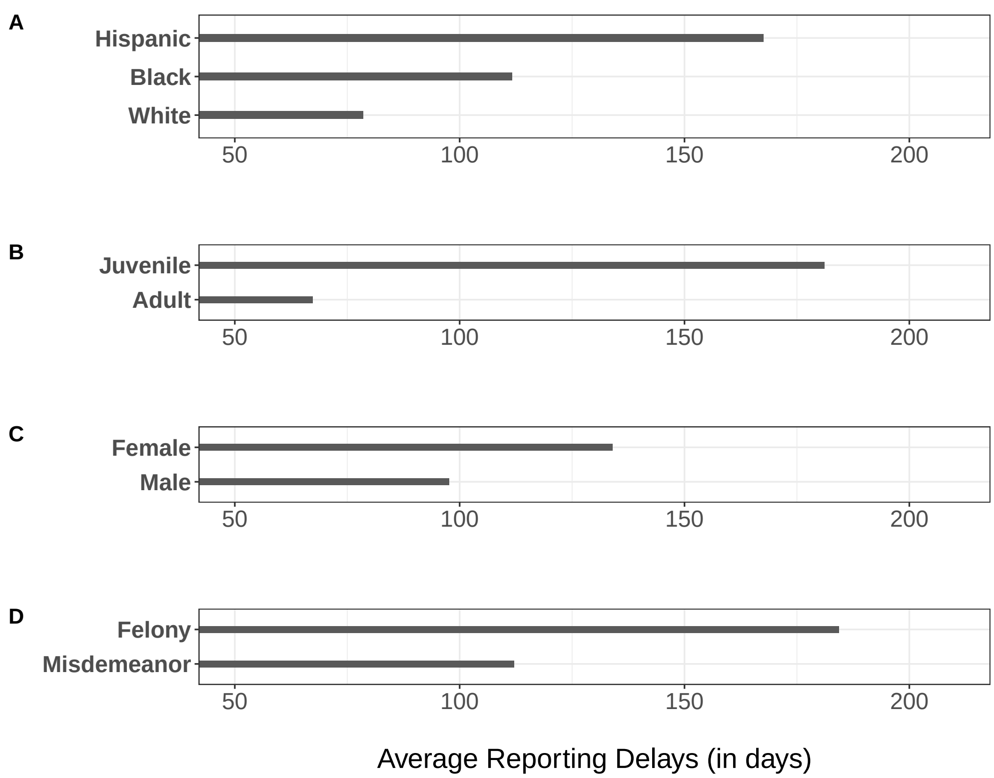


Figure A.2: Reporting Delays per Incident and Victim Characteristics

Notes: average reporting delays for sex crime incidents per incident characteristics based on the New York City Police Department database (2006 – 2019). I restrict the sample to incidents reported in less than ten years (to avoid inflating the means with outliers). Socio-demographic groups and incidents with lower reporting rates are usually associated with longer average reporting delays.

B Common Pitfalls of Reported Crime Statistics

This section formalizes the empirical issues researchers encounter when studying reported crime data. I focus specifically on the large share of crimes that are not reported or reported with a delay and their consequences for causal inference. My conclusion is that, in general, simply running regressions aggregated at the incident or the report date – the most common empirical approach in the crime literature – will lead to biased and uninterpretable estimates.

B.1 The Canonical Problem

Consider an analyst with access to reports recorded by the police between τ_1 and τ_2, respectively the first and last calendar data collection dates. She aims to study the impact of an intervention (e.g., the Me Too movement, an increase in the number of police officers, a harsher institutional penalty) on the number of crimes C_t. The intervention takes place in period $t^* \in [\tau_1, \tau_2]$. For each period t, she observes R_t, the number of crimes recorded by the police. As a share of victims does not report the incident to the police, reported crimes R_t generally do not equate to the total number of crimes C_t. Let r_t denote the victim reporting rate. Assuming no delayed reporting, we have

$$R_t = r_t \times C_t. \tag{4}$$

It becomes apparent that reported crimes R_t are a function of two unobserved variables and that a simple linear regression framework will be subject to an omitted variable bias. Let D_t be an indicator variable that takes the value one in periods after the intervention (i.e., $t \geq t^*$) and zero otherwise. We have

$$\log(C_t) = a + bD_t + \varepsilon_t - \log(r_t), \tag{5}$$

where a is an intercept term, b is the coefficient associated with D_t, and ε_t is an error term.

If the share of unreported crimes r_t is correlated with the treatment D_t, then estimates of b will be biased. In many applications, researchers explicitly or implicitly assume that the reporting rate r_t is orthogonal to D_t to conduct inference. Though this is a convenient assumption, it is also unlikely to hold in practice. On the contrary, interventions aimed at fighting crime often increase the probability of arrest or the severity of sentencing. One would expect crime rates to drop, but also victims to increase their reporting rate as their odds of "seeing justice served" increase.[11] Despite these obvious pitfalls, and for lack of a better alternative, regressions taking log counts of reported crimes as an outcome variable remain the default approach in the crime literature.

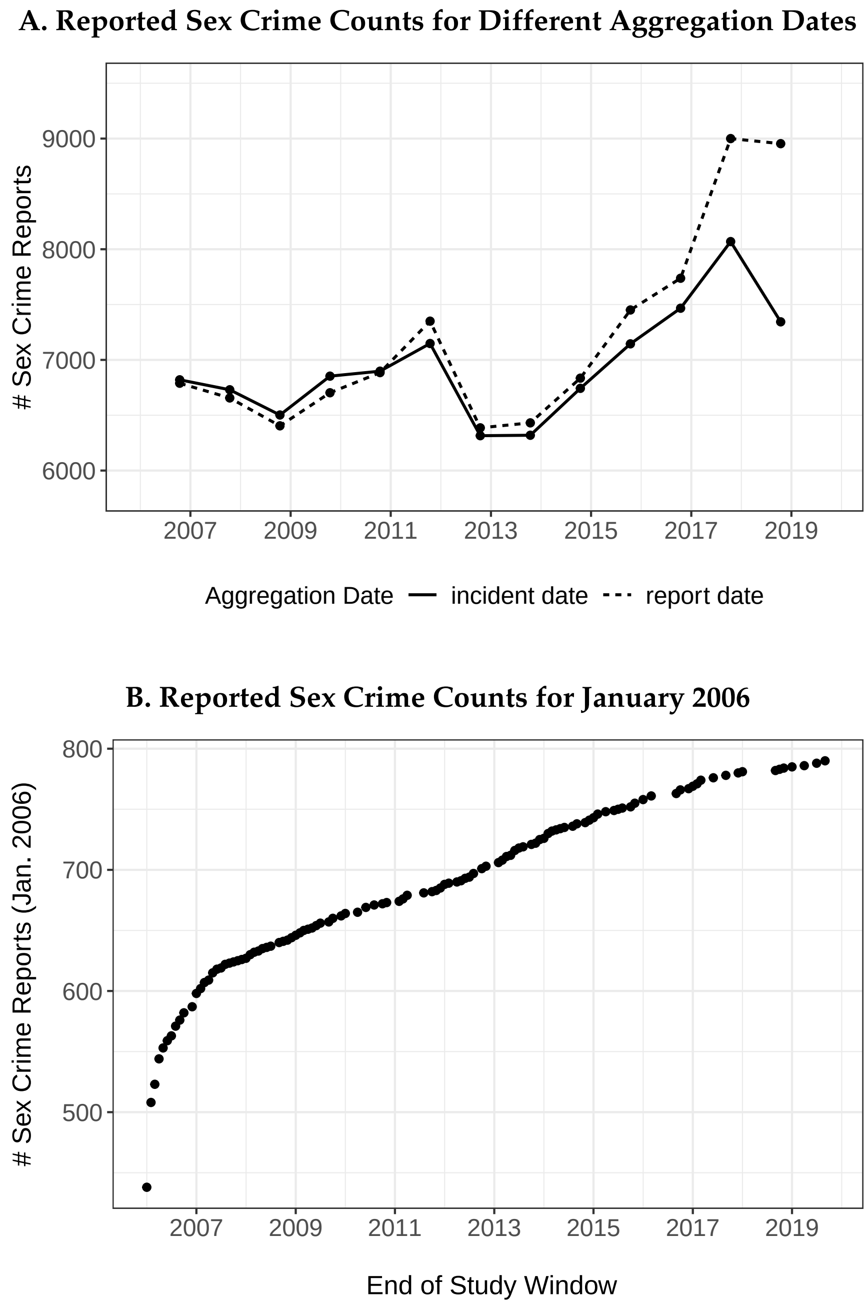


Figure B.1: Delayed reports make reported crime statistics unreliable.

Notes: This figure demonstrates issues with police records with delayed reports. Panel A presents reported sex crime counts for different aggregation dates. The solid line aggregates reports by incident date. The dashed line aggregates reports by the report date. Overall, depending on the study window, counts may vary substantially. Panel B presents reported sex crime counts for the same incident date but different study periods. Each data point corresponds to reported sex crime counts for January 2006 for various end-of-data collection dates. The closer the incident date to the end of the study period, the more biased downwards the reported sex crime counts. This is because victims often report with a delay, and may report outside of the study window.

B.2 The Econometric Implications of Delayed Reports

Equation 4 assumes the absence of delayed reports. In practice, however, delayed reports are common in police data (see Section 2). These reports complicate the analysis of reported crimes, because there are now two dates to consider: the incident date and the date of its report to the police. Aggregating reported crimes on the date of their report or of their occurrence will generally produce very different time series. Furthermore, researchers now face a sample selection issue, as crimes reported outside of the study window are unobserved (because they have not been reported yet at the time of data collection).

I briefly illustrate these empirical issues based on the police records later used to study the Me Too movement (see Section 2). Panel A of Figure B.1 plots reported sex crimes aggregated at the incident or the report date. Counts markedly vary across measures – by over a thousand in some years – particularly at the end of the study window. Panel B of Figure B.1 plots reported sex crime counts for the same incident date (January 2006) but different cutoff dates for the end of data collection (ranging from January 2006 to October 2019). Counts are clearly dependent on the study window considered, with a downward bias on the counts of incidents occurring closer to the end of the study window. Once again, these differences are not benign: the number of incidents recorded for January 2006 varies between approximately 440 and 790 depending on the data collection dates.

I now turn to the econometric implications of these observations. First, consider the date of the report as the main date for the analysis of reported crime statistics. Let f denote the victim population density function of times to report Y and χ_{t_1,t_2} the history of interventions between dates t_1 and t_2. Then R_t is a sum over all previous dates $j \leq t$, where each term is the product of the number of crimes committed in period j and the probability of a victim reporting an incident $t-j$ periods later:

$$R_t = \sum_{j=0}^{t} f(t-j \mid \chi_{0,t-1}) \times C_j. \tag{6}$$

Turning to the linear regression analysis, we have

$$\log\Big(\sum_{j=0}^{t} f(t-j \mid \chi_{0,t-1}) \cdot C_j\Big) = \alpha + \beta_1 D_t + \epsilon_t.$$

There is a sum in the log transform, which makes it very difficult to measure the effect of the intervention D_t on crime incidence C_t with this outcome variable.

Next, consider the date of the incident for the analysis of reported crime statistics. In this case, one can extend Equation 4 to account for delayed reports. Let R_{t,τ_1,τ_2} denote the number of crimes

[11] In some cases, the analyst is interested in understanding the impact of the policy on the victim reporting rate r_t. In this case, C_t acts as the omitted variable, and similar issues may arise. If a public awareness campaign encourages victims to file complaints to the police, this mechanically increases the probability of arrest of offenders, and should ultimately lower crime rates.

committed in period t that were reported between τ_1 and τ_2, and F the cumulative distribution function of times to report Y. R_{t,τ_1,τ_2} satisfies

$$R_{t,\tau_1,\tau_2} = p_{t,\tau_1,\tau_2} \times C_t, \tag{7}$$

where p_{t,τ_1,τ_2} is simply the probability of reporting a crime that occurred at date t within the study period:

$$p_{t,\tau_1,\tau_2} = \begin{cases} F\left(\tau_2 - t \mid \chi_{t,\tau_2}\right) - F\left(\tau_1 - t \mid \chi_{t,\tau_1}\right) & \text{if } \tau_1 > t, \\ F\left(\tau_2 - t \mid \chi_{t,\tau_2}\right) & \text{if } t \geq \tau_1. \end{cases}$$

The omitted variable bias in the classical regression analysis remains:

$$\log(C_t) = a + bD_t + \varepsilon_t - \log(p_{t,\tau_1,\tau_2}).$$

Contrary to R_t in Equation 6, R_{t,τ_1,τ_2} depends on the study period considered, because p_{t,τ_1,τ_2} is a function of τ_1 and τ_2. In fact, p_{t,τ_1,τ_2} is mechanically correlated to D_t. For instance, the closer t is to the end of the study period τ_2, the smaller the probability of reporting a crime that occurred in period t before the end of data collection. This implies a spurious decreasing time trend in observed reported crimes R_{t,τ_1,τ_2}. As a result, in the classical regression framework, estimates of the marginal effect of D_t will *necessarily* be biased. They will notably be biased even if the intervention D_t has no effect on crime incidence C_t or crime reporting p_{t,τ_1,τ_2}. This is clearly illustrated in Panel B of Figure B.1. The closer the incident date to the end of the study period, the smaller the number of reported sex crimes for this date. This is an entirely spurious correlation related to the structure of the data, as many incidents simply have not been reported yet by the end of data collection and are thus unobserved.

C Monte Carlo Simulations

This section validates the TWFE Poisson estimator developed in the Empirical Strategy section through Monte Carlo simulations. The simulations demonstrate that the estimator correctly identifies treatment effects on reporting probabilities while remaining robust to simultaneous changes in crime incidence. This decomposition is central to the paper's identification strategy, as it allows us to disentangle whether observed changes in reported crimes stem from shifts in victim reporting behavior or actual changes in crime incidence.

Data Generating Process The observed count $R_{t,\tau}$ of crimes committed in period t and reported in period τ is generated as:

$$R_{t,\tau} \sim \text{Poisson}\left(C_t \times \alpha_k \times \gamma_s \times \varepsilon_{t,\tau}\right)$$

where C_t denotes crime incidence in incident period t, α_k is the baseline reporting probability at delay $k = \tau - t$, γ_s is the multiplicative treatment effect at event time $s = \tau - t^*$ (where t^* is the treatment period), and $\varepsilon_{t,\tau} = \exp(\eta_{t,\tau})$ with $\eta_{t,\tau} \sim \mathcal{N}(0, 0.04)$ captures cell-level variation.

The baseline delay distribution follows an exponential decay with "never-reporters":

$$\alpha_k = (1 - \alpha_{\text{never}}) \times \frac{\exp(-\rho k)(1 - \exp(-\rho))}{\sum_j \exp(-\rho j)(1 - \exp(-\rho))}$$

where $\rho = 0.15$ governs the decay rate and $\alpha_{\text{never}} = 0.30$ represents the fraction of crimes never reported.

I simulate 160 quarterly periods (40 years), of which the first 40 form an unobserved burn-in: reports dated before period 41 are dropped, so that the observation window opens part-way through and still contains long-delay reports of older incidents. Treatment occurs at period 120, using 500 baseline crimes per incident period. I run 1,000 Monte Carlo replications for each scenario.

Simulation Scenarios I consider four scenarios to evaluate the estimator's performance:

1. **No treatment effect:** $\gamma = 1$ and C_t constant. This null scenario verifies correct size and the absence of false positives.

2. **Reporting only:** $\gamma = 1.30$ for $s \geq 0$ (30% increase in reporting post-treatment), C_t constant. Tests identification of a pure reporting effect.

3. **Incidence only:** $\gamma = 1$, C_t drops by 20% for $t \geq t^*$. This is the critical test: the estimator should find $\hat{\gamma} \approx 0$ despite the change in total reported crimes.

4. **Both effects:** $\gamma = 1.30$ for $s \geq 0$ and C_t drops by 20% for $t \geq t^*$. Verifies simultaneous identification of both channels.

Estimation Each simulated dataset is estimated by PPML with incident-period and reporting-delay fixed effects, with standard errors two-way clustered by incident and report period. The pooled specification includes a single indicator for reports filed from period t^* onward, whose coefficient is $\hat{\gamma}$. The event-study specification replaces this indicator with event-time dummies for $s \in \{-39, \ldots, 40\}$: observed report periods with $s < -39$ carry no event dummies and form the omitted baseline block, so the $\hat{\gamma}_s$ are measured relative to the average of that block. Omitting a block of event times, rather than a single reference period, is required for identification: because $s = t + k - t^*$, a saturated set of event-time dummies is collinear with the incident-period and delay fixed effects. This mirrors the specification used in the empirical application, where the event-time indicators start in 2010 and the 2006–2009 window forms the baseline.

Results Figure C.1 displays the sampling distributions of the pooled reporting effect estimate $\hat{\gamma}$ across 1,000 simulations. In all four scenarios, the distributions are centered on the true parameter values, demonstrating the estimator's unbiasedness. Crucially, in the "incidence only" scenario (bottom-left panel), the distribution remains centered at zero despite the 20% drop in crime incidence – confirming that the incident-period fixed effects successfully absorb incidence variation without contaminating the reporting effect estimate.

Figure C.2 presents the event study coefficients $\hat{\gamma}_s$ by event time. The individual simulation paths (faint lines) cluster around the mean path (solid line with 95% confidence band). Pre-treatment coefficients ($s < 0$) are indistinguishable from zero across all scenarios, validating the parallel trends assumption embedded in the estimator. Post-treatment coefficients correctly track the true effect (dashed line): flat at zero for the null and incidence-only scenarios, and stepping up to $\log(1.30) \approx 0.26$ for scenarios with a reporting effect.

Finally, Figure C.3 verifies that the incident-period fixed effects correctly recover crime incidence. The estimated fixed effects $\hat{\delta}_t$ (normalized so the pre-treatment mean equals zero) accurately track the true incidence path. In scenarios with an incidence drop, the mean path steps down to approximately $\log(0.80) \approx -0.22$ after treatment, matching the true effect. This confirms that the two-way fixed effects structure successfully decomposes observed crime reports into their reporting and incidence components.

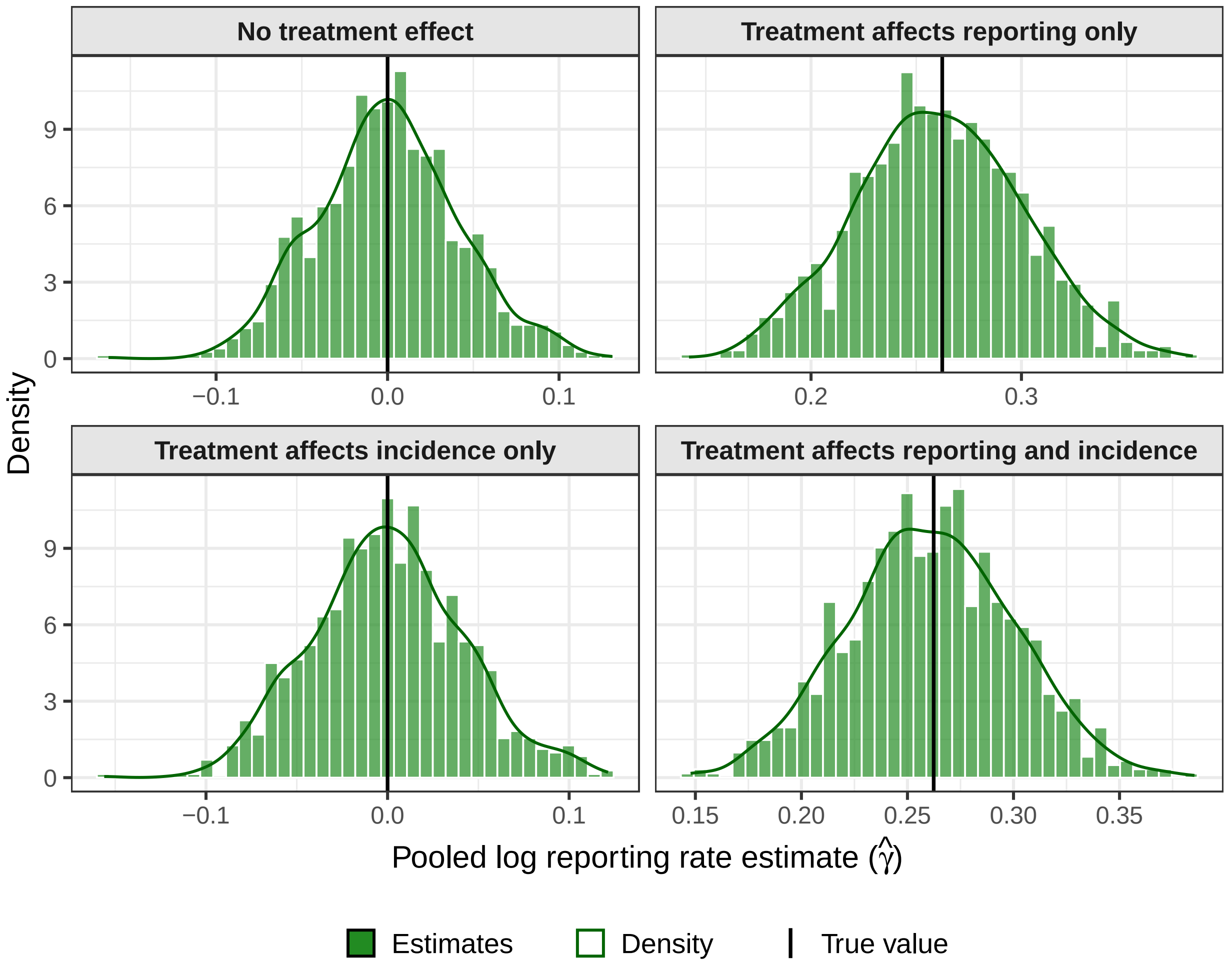


Figure C.1: Monte Carlo Simulations: Sampling Distributions of Reporting Effect Estimates

Notes: Sampling distributions of the pooled log reporting rate estimate $\hat{\gamma}$ across 1,000 Monte Carlo simulations. The vertical black line indicates the true parameter value. Each panel corresponds to a different scenario: no treatment effect (top-left), treatment affecting reporting only (top-right), treatment affecting incidence only (bottom-left), and treatment affecting both reporting and incidence (bottom-right). The distributions are centered on the true values in all scenarios, demonstrating unbiasedness. In the "incidence only" scenario, the estimator correctly finds no reporting effect despite the 20% drop in crime incidence.

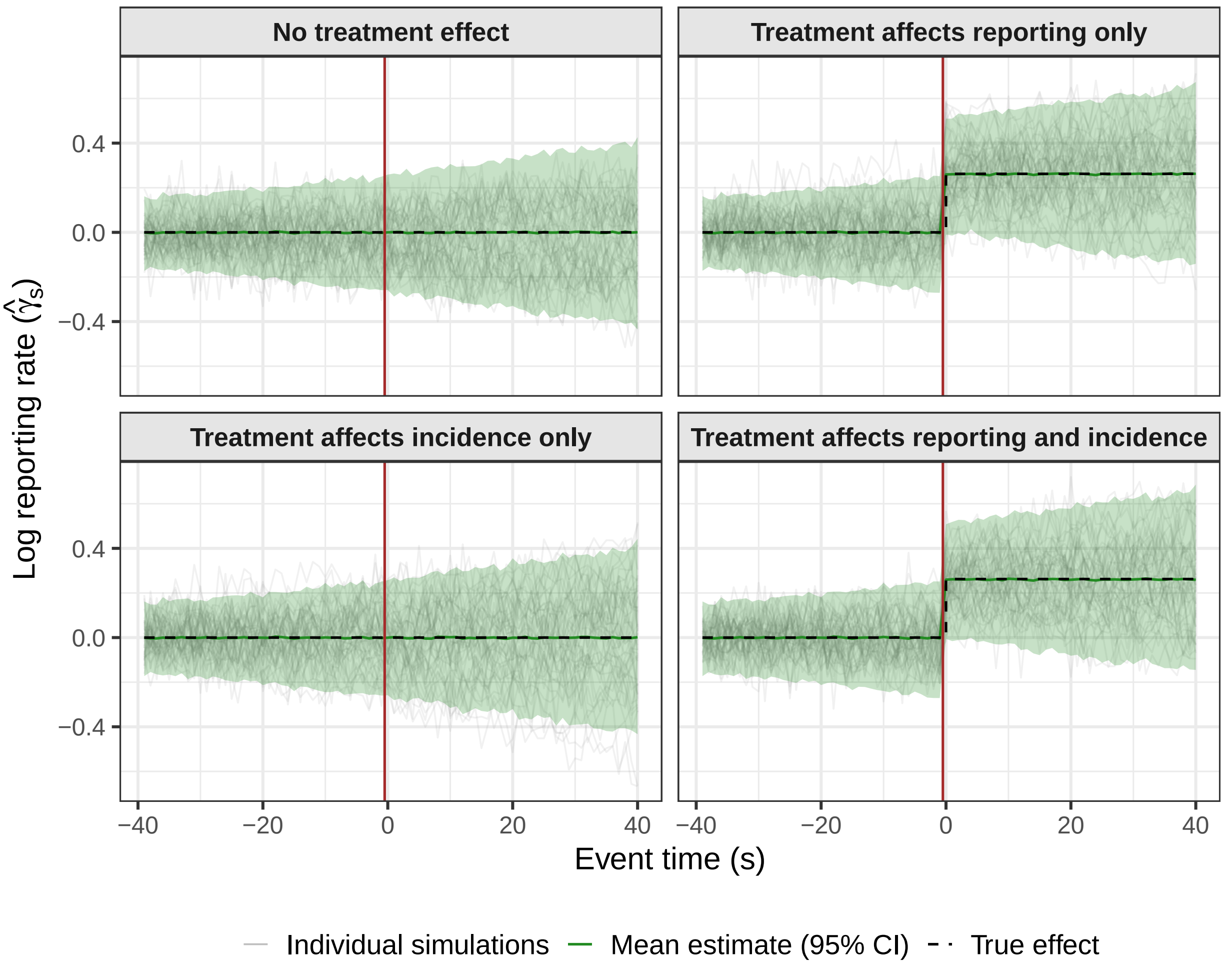


Figure C.2: Monte Carlo Simulations: Recovery of Changes in Reporting

Notes: Event study coefficients $\hat{\gamma}_s$ by event time across 1,000 Monte Carlo simulations. Gray lines show individual simulation paths; the solid green line shows the mean estimate with 95% confidence bands; the dashed black line shows the true effect. The vertical red line marks the treatment period. Pre-treatment coefficients are centered at zero across all scenarios, validating the parallel trends assumption. Post-treatment coefficients correctly identify the true reporting effect.

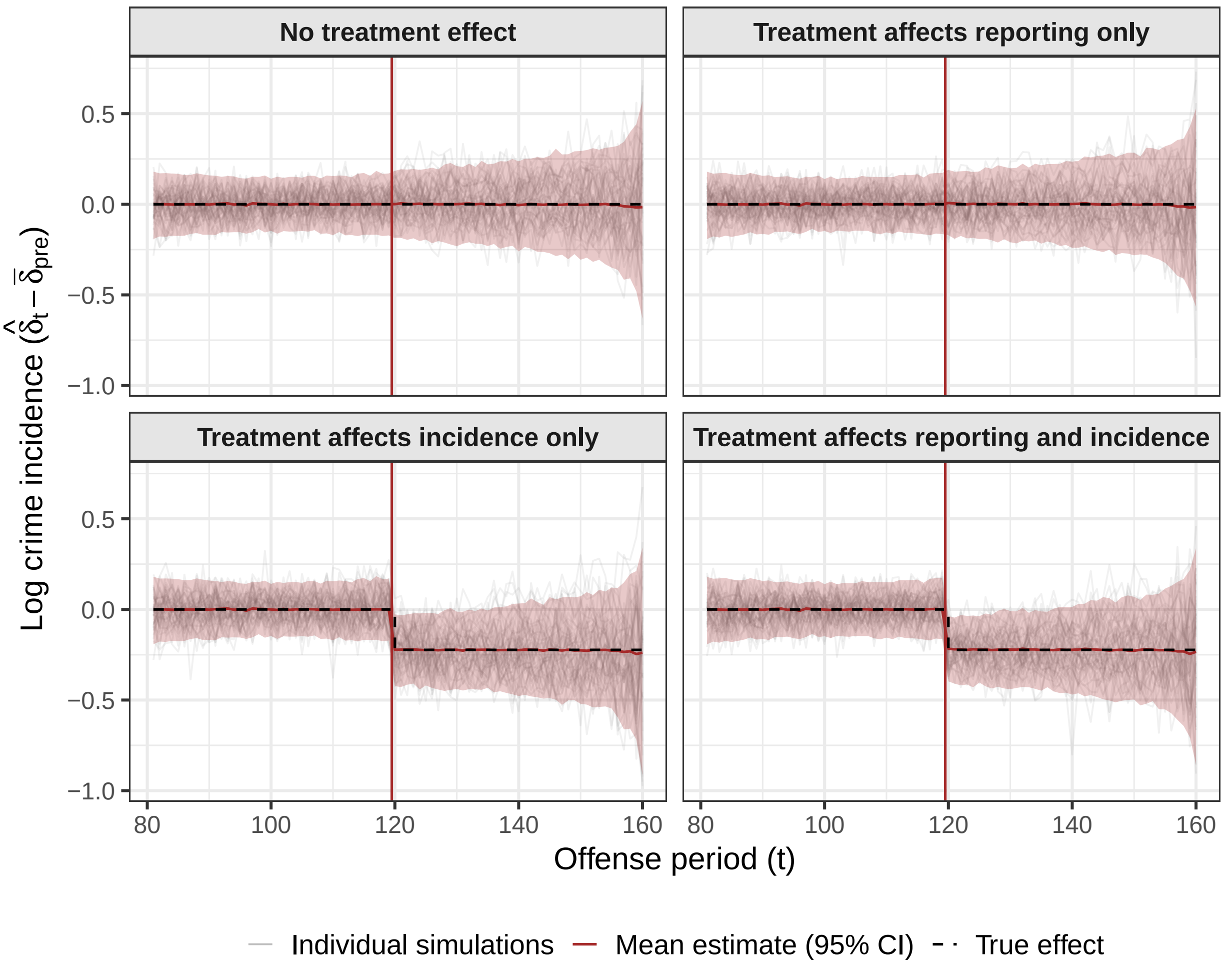


Figure C.3: Monte Carlo Simulations: Recovery of Crime Incidence

Notes: Incident-period fixed effects $\hat{\delta}_t$ (normalized so pre-treatment mean equals zero) across 1,000 Monte Carlo simulations. Gray lines show individual paths; the solid brown line shows the mean with 95% confidence bands; the dashed black line shows the true incidence change. The vertical red line marks the treatment period. In scenarios with an incidence change (bottom-left and bottom-right), the fixed effects correctly recover the 20% drop in crime incidence ($\log(0.80) \approx -0.22$).

D The #MeToo Effect on Victim Reporting: Additional Results

D.1 Decomposition Normalized at 2017 Q3

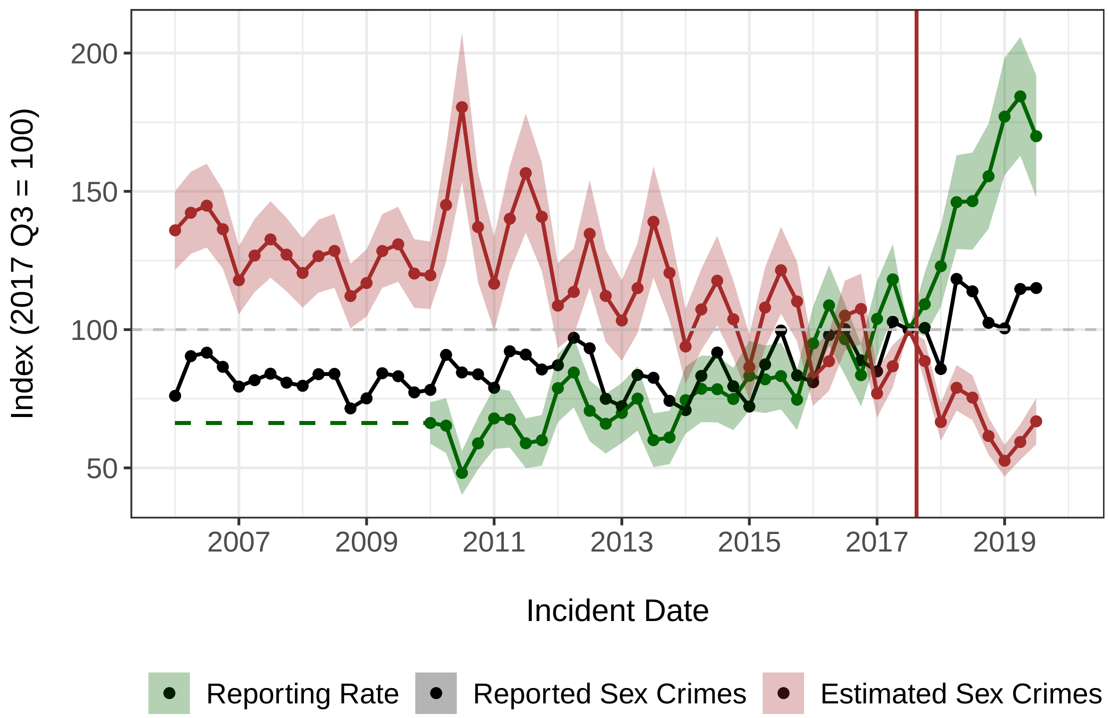


Figure D.1: Trends in Sex Crime Reporting and Incidence

Notes: This figure decomposes reported sex crimes to the police by incident date into crime reporting probability and crime incidence estimates based on the duration model. All series are normalized to 100 at 2017 Q3 (immediately before the Me Too movement's mediatization). The duration model assumes 20% of victims would have eventually reported sex crimes committed in 2006 (i.e., 80% of never-reporters at baseline). Breaks in the baseline hazard are set after 1, 30, 90, 180, and 365 days. The likelihood appropriately corrects for double-truncation. 95% confidence intervals are constructed with a non-parametric bootstrap procedure (1,000 replications). The vertical solid red line corresponds to the Me Too movement's mediatization.

D.2 Heterogeneity Analysis

I investigate heterogeneity in the #MeToo effect across victim and suspect characteristics using two complementary approaches.

The first approach augments the baseline TWFE specification with interaction terms. For each demographic dimension d (e.g., victim race), I estimate:

$$\mathbb{E}[R_{t,\tau,d}] = \exp\left(\beta \cdot \mathbf{1}\{s \geq 0\} + \gamma_d \cdot \left(\mathbf{1}\{s \geq 0\} \times \mathbf{1}\{D = d\}\right) + \delta_t + \lambda_k + \delta_d\right), \tag{8}$$

where D denotes the demographic group of the cell and γ_d captures the differential effect of #MeToo for group d relative to the baseline category. This approach estimates heterogeneity within a single model, allowing for direct statistical comparison across groups.

The second approach estimates separate TWFE regressions for each subgroup (e.g., separate regressions for White, Black, and Hispanic victims). This stratified analysis allows the entire model – including incident-period and delay fixed effects – to vary flexibly across groups, at the cost of reduced precision due to smaller sample sizes.

Figure D.2 presents estimates from both approaches. Hollow circles show coefficients from the interaction model; solid circles show estimates from stratified regressions. The two methods yield broadly consistent patterns, providing reassurance that the results are not driven by functional form assumptions.

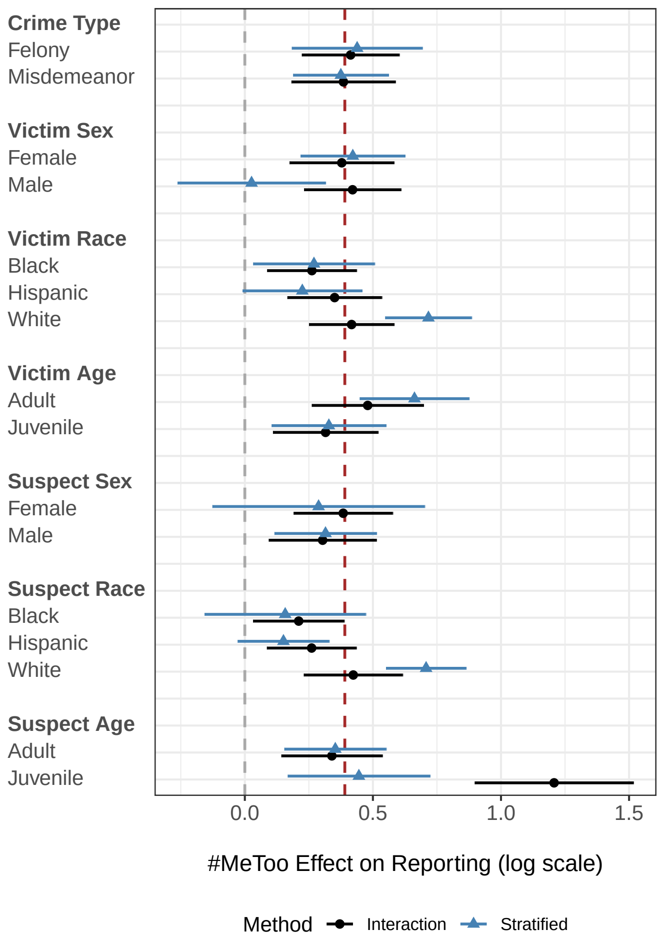


Figure D.2: #MeToo Effects on Crime Reporting – Heterogeneity Analysis

Notes: Heterogeneity in the #MeToo effect on reporting probability by victim and suspect characteristics. Hollow circles show interaction coefficients from the pooled TWFE model; solid circles show estimates from separate stratified regressions. The grey vertical dashed line corresponds to a null effect. The brown vertical dashed line corresponds to the pooled #MeToo effect (the post-#MeToo coefficient) in the full sample. 95% confidence intervals computed with standard errors clustered at the incident-period and report-period level.

D.3 Robustness – Different Samples

Table D.1 presents robustness checks for the baseline pooled treatment effect across six sample specifications. Columns 1–5 progressively apply three restrictions to address potential sources of bias; Column 6 additionally restricts the sample to reports filed after 2014.

The first restriction removes *left truncation*: I exclude cells where the incident period precedes the earliest report period in the data. This addresses potential bias from crimes that occurred before the data recording began but were reported afterward.

The second restriction removes incidents recorded on the *first day of each month*. Police officers sometimes record the first of the month as a placeholder when the exact incident date is unknown. This measurement error could bias estimates if it correlates with reporting behavior.

The third restriction removes *old crimes*: I exclude reports filed more than ten years after the incident (delay > 40 quarters). Very long delays may be subject to different reporting dynamics and could introduce time-dependent confounding with the #MeToo effect.

Across the first five specifications, the estimated #MeToo effect on the reporting probability remains strong, positive, and statistically significant: point estimates range from 0.337 to 0.423 in logs (an increase of approximately 40 to 53%), with Column 5 applying all three restrictions simultaneously. Column 6 further restricts the baseline sample to reports filed after 2014, addressing the FBI's 2013 revision of its definition of rape; the estimate falls to 0.158 but remains positive and significant at the 5% level. The smaller magnitude is expected: the pooled coefficient partly reflects the secular pre-#MeToo rise in reporting, and a comparison window that starts in 2014, when reporting was already elevated, mechanically yields a smaller contrast. The stability of the estimates across specifications indicates that the main findings are not driven by any particular sample selection concern.

D.4 Imputation Estimator to Investigate Heterogeneous Treatment Effects

The imputation estimator yields cell-level treatment effect estimates $\hat{\gamma}_{s,k}$ that reveal how the #MeToo effect varies across the (event-time, delay) grid. Figures D.3–D.5 present diagnostic plots of these estimates.

Aggregating these cell-level effects into a single summary yields a Proportional Treatment effect on the Treated (PTT) of $+51\%$ – that is, #MeToo raised the number of reported sex crimes among treated incidents by roughly half. This heterogeneity-robust estimate is close to the pooled two-way fixed effects estimates (see Table D.1), indicating that treatment-effect heterogeneity across cells does not materially bias the pooled effect.

Figure D.3 shows the treatment effect by event time s (quarters since #MeToo), aggregating across delays as a ratio of sums (a Proportional Treatment effect on the Treated restricted to each event time). The estimates reveal substantial dynamic heterogeneity: the effect on reporting builds over the quarters following the movement before receding.

Table D.1: Robustness – Sample Selection

	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent Variable: Reported Sex Crimes (cell counts)					
#MeToo (indicator)	0.388***	0.337***	0.423***	0.354***	0.384***	0.158**
	(0.063)	(0.065)	(0.105)	(0.066)	(0.099)	(0.070)
Sample Restrictions						
Remove left truncation		✓			✓	✓
Remove day 1			✓		✓	✓
Remove old crimes				✓	✓	✓
Reported after 2014						✓
Standard Errors	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered
N Complaints	99,075	96,046	86,414	97,893	85,219	38,355
N Observations	7,920	1,540	6,325	3,740	1,540	1,012

Notes: Robustness of the baseline pooled treatment effect on the reporting probability to alternative sample restrictions. Column 1 imposes no restriction; Columns 2–4 apply each restriction separately; Column 5 applies all three simultaneously; Column 6 further restricts the baseline sample (Column 5) to crimes reported after 2014, addressing the FBI's 2013 revision of its definition of rape. *Left truncation* excludes cells whose offense period precedes the earliest report period. *Day 1* excludes incidents recorded on the first day of the month. *Old crimes* excludes reports filed more than ten years after the incident. *N Complaints* is the number of underlying complaints; *N Observations* is the number of quarterly offense-by-report-period cells in the estimation grid, which spans the offense and report periods observed after each restriction (cells whose complaints are excluded by a restriction enter as zeros). Standard errors (in parentheses) are two-way clustered by offense and report period. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Figure D.4 shows the treatment effect by delay k, aggregating across event times. The relationship is essentially flat: the slope with respect to delay is small and not significantly different from zero (0.007 per quarter, $p = 0.47$), against a significantly positive intercept (0.631, $p = 0.01$). The treatment effect therefore does not vary systematically with delay, which supports the modeling assumption that allows event-time heterogeneity while treating delay effects as homogeneous.

Figure D.5 presents a heatmap of the full $\hat{\gamma}_{s,k}$ grid, confirming that variation in treatment effects is primarily driven by event time rather than delay. This empirical pattern justifies the specification choices in the main analysis.

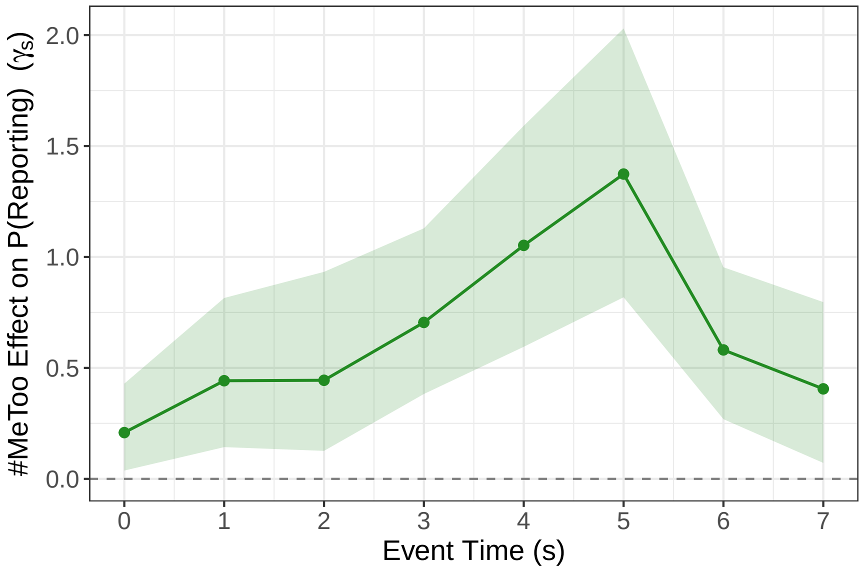


Figure D.3: Treatment Effect by Event Time

Notes: Treatment effect by event time s (quarters since #MeToo), computed as $\sum_k N_{s,k} / \sum_k \widehat{R}_{0,s,k} - 1$ over the identified switcher cells at each s (a Proportional Treatment effect on the Treated restricted to event time s). The shaded band is a 95% percentile confidence interval from a block bootstrap over incident periods (1,000 replications).

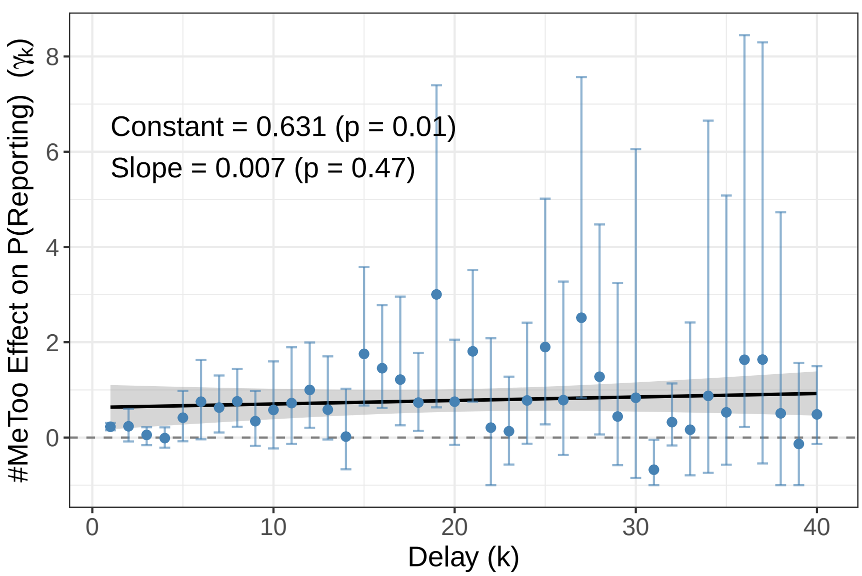


Figure D.4: Treatment Effect by Delay

Notes: Treatment effect by delay k, computed as $\sum_s N_{s,k} / \sum_s \widehat{R}_{0,s,k} - 1$ over the identified switcher cells at each delay. Blue error bars are 95% percentile intervals from a block bootstrap over incident periods (1,000 replications). The solid line is an OLS fit (constant 0.631, $p = 0.01$; slope 0.007, $p = 0.47$) with its 95% confidence band; the slope is not significantly different from zero, so the effect does not vary systematically with delay.

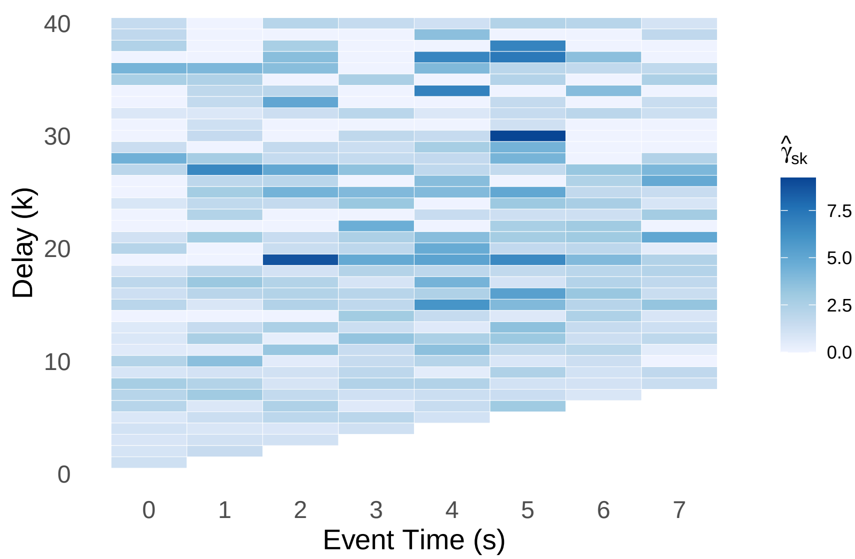


Figure D.5: Heatmap of Cell-Level Treatment Effects

Notes: Cell-level treatment effect estimates $\hat{\gamma}_{s,k}$ across the (event-time, delay) grid. Darker shades indicate larger treatment effects. The pattern confirms that variation is primarily driven by event time (horizontal axis) rather than delay (vertical axis).

E The #MeToo Effect on Sex Crime Incidence: Additional Results

E.1 Robustness – Alternative Counterfactual Models

A. Two-Way Fixed Effects (TWFE)

B. Interactive Fixed Effects (IFE)

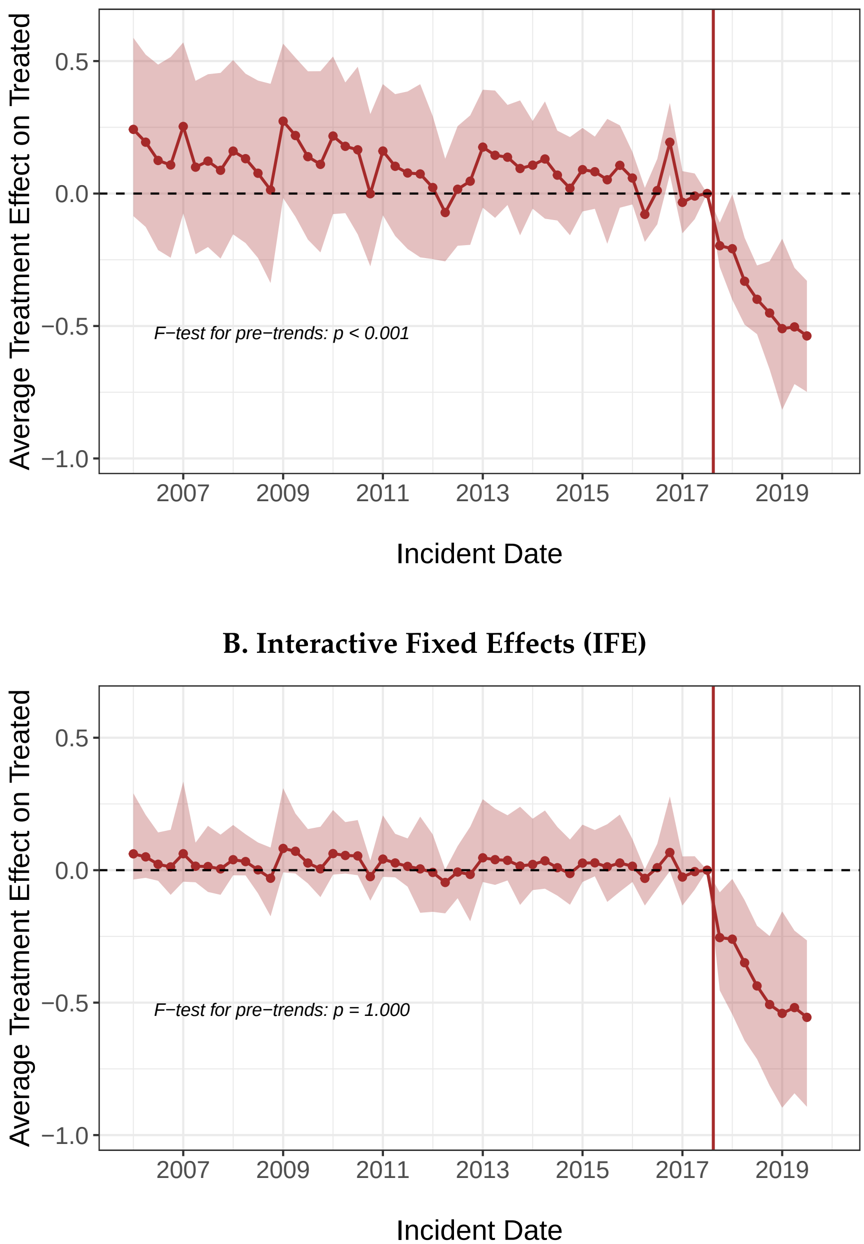


Figure E.1: Alternative Counterfactual Models of Sex Crime Incidence

Notes: Quarterly estimates of the Average Treatment Effect for alternative counterfactual models of sex crime incidence. Panel A uses a standard two-way fixed effects (TWFE) model. Panel B uses an interactive fixed effects (IFE) model with additional latent factors (Xu, 2017). The control group includes non-sexual assaults, murders, robberies, and burglaries. 95% confidence intervals are bootstrap percentile intervals (see the main text). The vertical solid red line corresponds to the Me Too movement's mediatization. The main text presents the matrix completion (MC) results.

E.2 Robustness – Dropping Control Crimes

Table E.1 presents robustness checks for the main deterrent effect estimate. I sequentially exclude each non-sexual crime category from the control group to assess whether the results depend on any particular crime type.

The baseline specification using all control crimes (murders, assaults, robberies, and burglaries) yields an estimated deterrent effect of -0.48 in logs. When dropping each control crime individually, the estimates range from -0.39 to -0.52 in logs, all strong, negative, and statistically significant. The consistency of results across specifications provides reassurance that the findings are not driven by any single crime category in the counterfactual.

Table E.1: Robustness – Dropping One Control Crime

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: Estimated Sex Crimes (in logs)				
#MeToo (indicator)	-0.48***	-0.49***	-0.39***	-0.49***	-0.52***
	(0.14)	(0.16)	(0.14)	(0.17)	(0.15)
Counterfactual Model	MC	MC	MC	MC	MC
Control Groups					
Murders	✓		✓	✓	✓
Assaults	✓	✓		✓	✓
Robberies	✓	✓	✓		✓
Burglaries	✓	✓	✓	✓	
Standard Errors	Bootstrap	Bootstrap	Bootstrap	Bootstrap	Bootstrap
N Observations	275	220	220	220	220

Notes: Robustness of the estimated #MeToo effect on sex crime incidence to excluding individual control crimes. All columns use the Matrix Completion counterfactual and the bootstrap standard errors described in the text (incidence draws plus control resampling, 1,000 replications). Columns 2–5 sequentially exclude one control crime. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

F Mechanisms and Discussion

F.1 Alternative Measures of Sex Crime Incidence and Reporting

The FBI's Uniform Crime Reporting (UCR) program provides the general public with a broad range of statistics from local law enforcement agencies. To compare my results to national reported crime statistics, I download official consolidated crime databases between 2006 and 2019. Reported crimes are harmonized into the Uniform Crime Reporting Summary Reporting System (SRS) (Kaplan, 2021). The SRS is a crime report database aggregated by month, agency, and crime category. It is the most comprehensive database on offenses known and clearances by arrest in the United States.[12] The UCR also provides supplementary reports on homicides (Kaplan, 2019). This allows me to compute sexual and non-sexual homicide rates.

Homicides are extremely likely to be recorded by law enforcement agencies and are thus the only crime category with a reporting rate of virtually 100%. Under the assumption that homicides are a constant fraction of violent crimes, one can recover variations in the reporting rate over time. More formally, let H_t denote the number of homicides, C_t the number of violent crimes, and R_t the number of violent crimes reported to the police, and r_t the reporting rate of victims. If we assume that $H_t = aC_t$ for some unknown constant value a, and that $R_t = r_t C_t$, then we have

$$r_t = a \times \frac{R_t}{H_t}. \tag{9}$$

That is, the ratio of reported crimes to homicides provides an alternative estimate of variations in victim reporting over time. The critical assumption of this empirical strategy is that the ratio of homicides to violent crimes is assumed to be constant over time. Another clear limitation of this empirical strategy is that homicides are much rarer than other crimes. For instance, in the United States, the average number of sexual homicides per year over the period was approximately 20. This makes such an approach unsuitable for studying smaller geographical areas such as cities and counties (contrary to the duration modeling approach that relies on many observations per geographical unit). Furthermore, we expect larger uncertainty in estimates.

Relying on Equation 9, Figure F.1 presents the results of this alternative decomposition exercise. Note that the FBI's definition of rape changed in 2013 and led to a mechanical increase in the number of reported sex crimes. From 2013 to 2015, the FBI provided estimates using the old and the new definition. On average, the old definition accounted for approximately 72% of sex crimes reported in these three years. To make trends comparable over time, I rescale reported sex crime reports from 2013 to 2019 for each US state by the average share of sex crime reports that were accounted for by the old definition in these three years. Panel A suggests a substantial increase in sexual crime reporting and a substantial decrease in sexual crime incidence between 2006 and

[12] The National Incident-Based Reporting System (NIBRS) is a more recent data collection effort implemented to improve the overall quality of crime data collected by law enforcement. I do not rely on this database because it has limited geographical coverage relative to the SRS. According to the FBI, in 2017, it covered 33% of the US population.

2019, while Panel B indicates that this is not the case for non-sexual crimes. Overall, the uncovered trends are consistent with the main analysis at the city level.

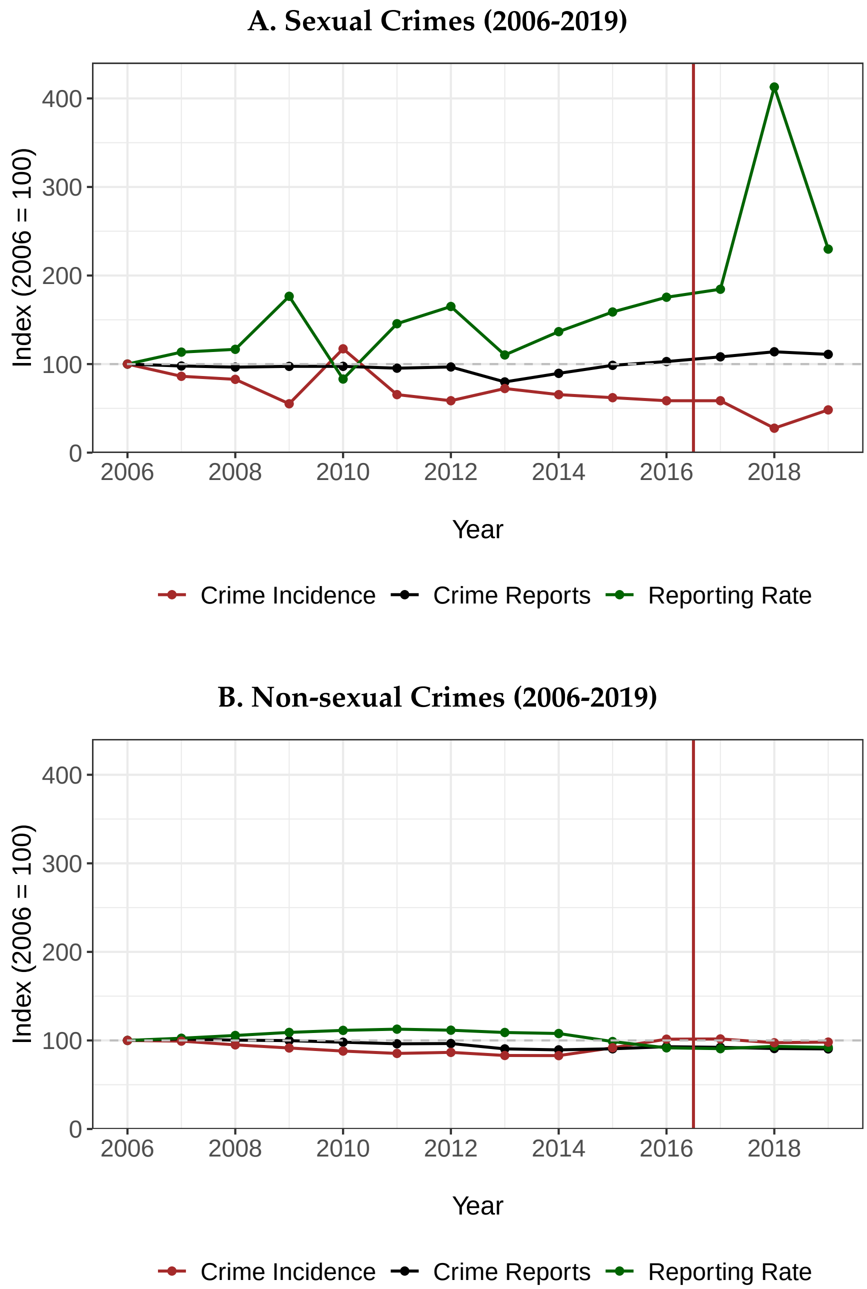


Figure F.1: National Trends in Crime Reporting and Incidence Based on Homicides

Notes: This figure presents national trends in crime reporting and incidence using homicides as a proxy for crime incidence based on Equation 9. The analysis relies on the FBI's Uniform Crime Reporting Summary Reporting System and its supplementary reports on homicides between 2006 and 2019 (Kaplan, 2019, 2021). Panel A plots trends for sexual crimes, and Panel B trends for non-sexual crimes. The solid vertical line separates years pre- and post-#MeToo.

For comparison, I turn to estimates of crime incidence and reporting based on the National Crime

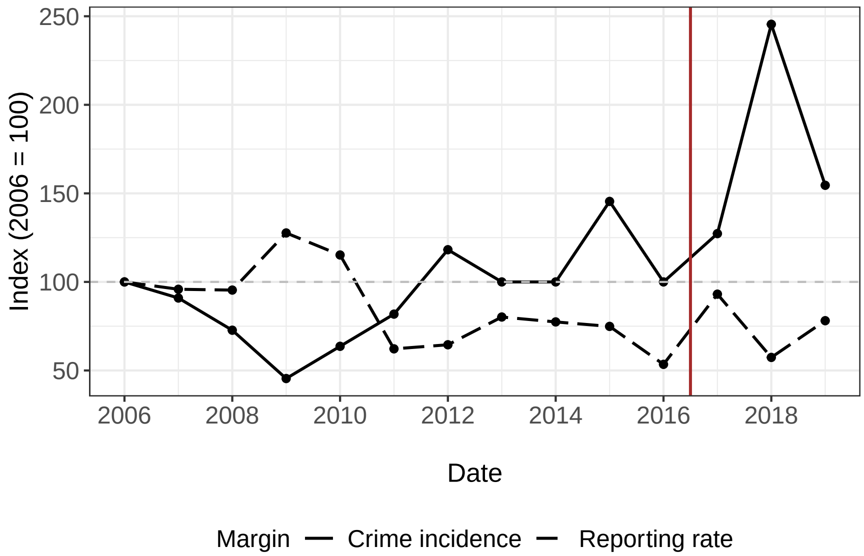


Figure F.2: National Crime Victimization Survey – Official Sexual Violence Estimates

Notes: Official national estimates of sex crime reporting and incidence produced by the Bureau of Justice Statistics based on the National Crime Victimization Survey (NCVS). The solid black line is for crime incidence. The dashed black line is for the reporting rate. Values are normalized at 100 for 2006. The solid vertical line separates years pre- and post-#MeToo.

Victimization Survey (NCVS). The NCVS is a comprehensive, nationally representative survey conducted by the U.S. Bureau of Justice Statistics to gather detailed information about nonfatal criminal victimization in the United States. It collects data on both personal crimes, such as assault, robbery, and sexual violence, and property crimes, including burglary, theft, and motor vehicle theft. Using a rotating panel design, the survey interviews individuals aged 12 and older in selected households over several waves. It provides insights into the frequency, characteristics, and consequences of victimization, as well as victims' interactions with law enforcement and their reasons for reporting or not reporting crimes. Figure F.2 presents official NCVS estimates normalized at 100 for 2006. It indicates a large increase in sexual violence following #MeToo. If anything, the reporting rate of victims decreased over the period. This is very much at odds with my main results as well as my estimates based on homicide data.

I also consider emergency department visits as a proxy for sexual violence (Aizer, 2010). I rely on the National Electronic Injury Surveillance System All Injury Program (NEISS-AIP), which collects data on all types and causes of injuries treated in a representative sample of U.S. hospitals with emergency departments (EDs). Hospital personnel declare confirmed and suspected assaults, and further distinguish between sexual and non-sexual assaults. Thus, I can compute the number of emergency department visits in the United States for sexual assaults, non-sexual assaults, and other causes. Figure F.3 presents national estimates of emergency department visits between 2006 and 2019 for sexual assaults (solid line), non-sexual assaults (dashed line), and other conditions (dotted line). Observations are weighted using the official weights provided by the NEISS-AIP

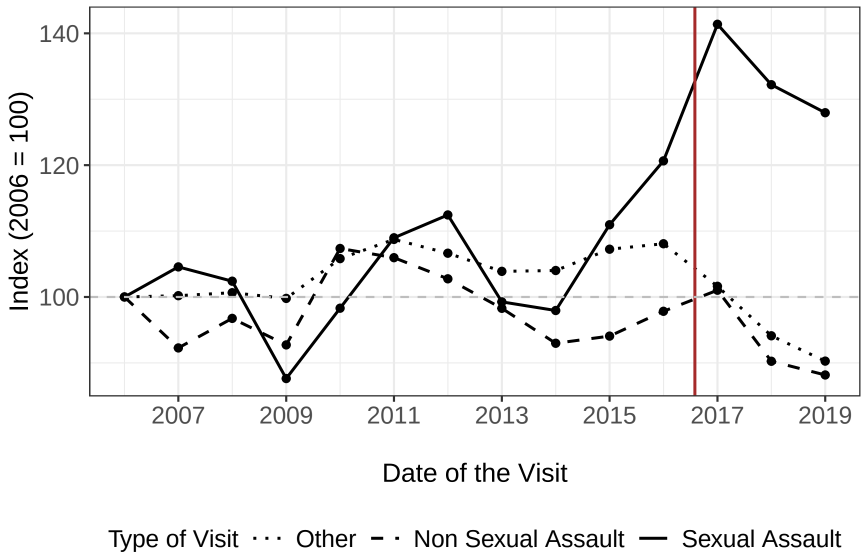


Figure F.3: Emergency Department Visits by Type of Visit

Notes: National estimates based on the National Electronic Injury Surveillance System All Injury Program (NEISS-AIP). Observations are weighted using the official weights provided by the NEISS-AIP dataset. I distinguish between visits for sexual assaults (solid line), non-sexual assaults (dashed line), and other reasons (dotted line). Values are normalized at 100 for 2006. The solid vertical line separates years pre- and post-#MeToo.

dataset. It indicates that ED visits flagged as sexual assaults increase substantially over the period. Once again, if we were to interpret this increase as an increase in sexual violence, these results would be at odds with my main results as well as my estimates based on homicide data.

In practice, neither ED visits nor victimization surveys are fundamentally different from police records. They require the victim to be willing to report the crime and/or a practitioner to be willing to record it as a crime. One might expect these probabilities of reporting and recording to increase following #MeToo. Ultimately, unless one is willing to make very strong assumptions (i.e., fully transparent reporting in victimization surveys or perfect recording of sexual assaults during ED visits), it remains unclear to what extent such alternative data sources allow researchers to disentangle variations in crime reporting and incidence.

F.2 Sexual Violence Awareness on Google and Twitter

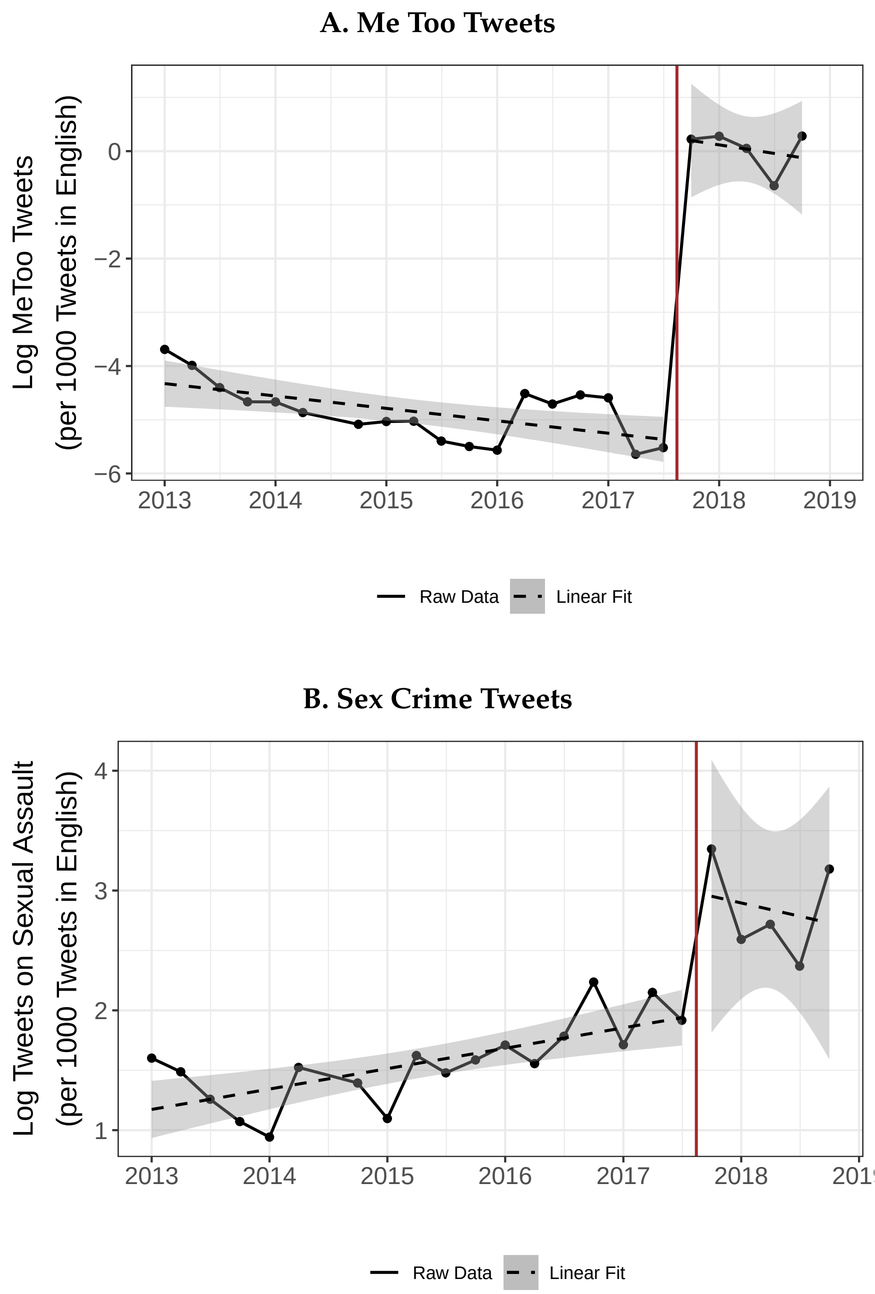


Figure F.4: Sexual Violence Awareness on Twitter

Notes: This figure presents trends in tweets with #MeToo (Panel A) and referring to sex crimes more generally (Panel B). Values are in log scale and normalized per 1,000 tweets in English. The dashed line in Panel B is a linear fit. The structural break in Panel A indicates that the Me Too movement's sudden mediatization brought sex crimes to the forefront of the public debate. However, Panel B nuances this interpretation, as there were clear pre-trends in the number of tweets related to sex crimes before #MeToo was used as a coordination device to combat sexual violence. Before October 2017, the hashtag was marginal on Twitter and rarely referred to sex crimes. The vertical solid line is set one period before #MeToo (Oct 2017).

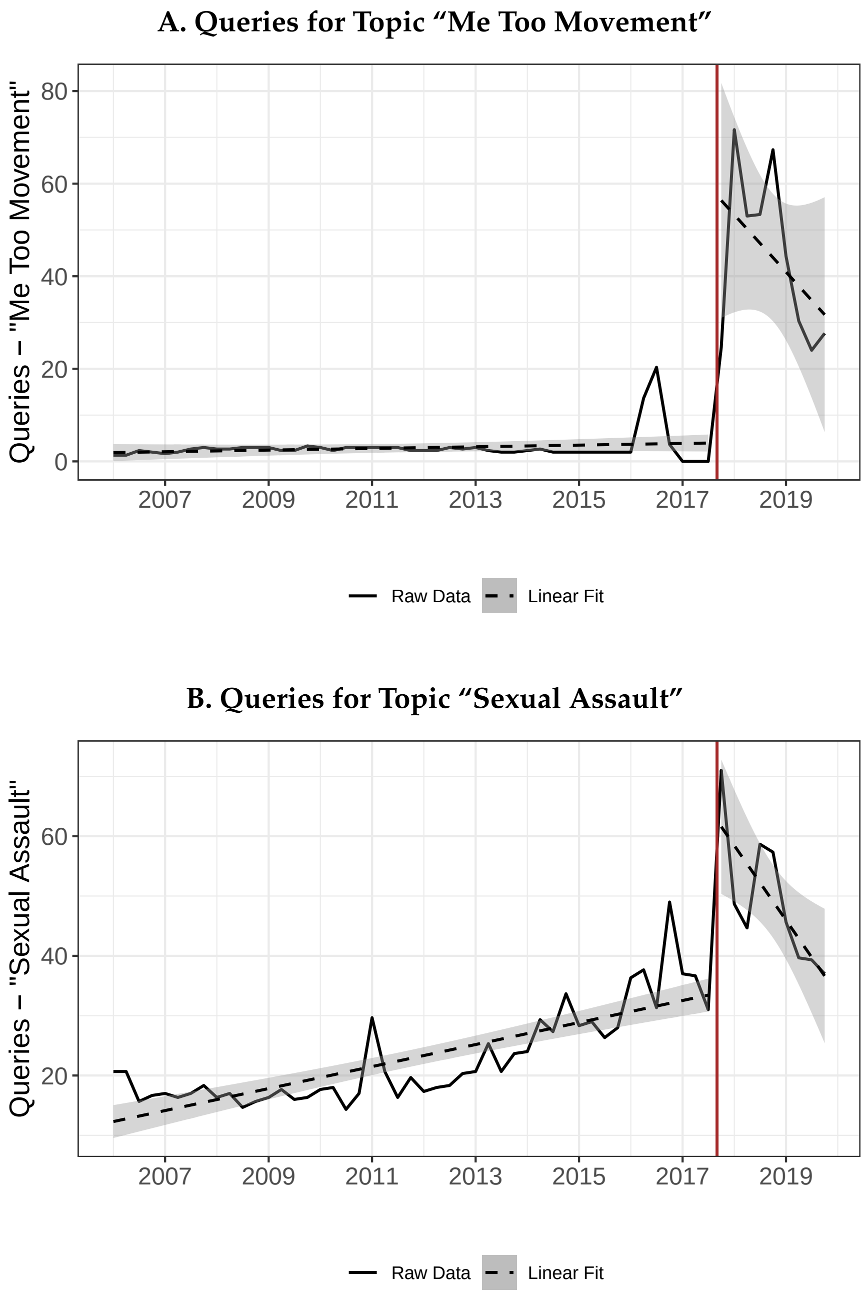


Figure F.5: Sexual Violence Awareness on Google

Notes: This figure presents trends in the number of queries for the topic "Me Too Movement" (Panel A) and for the topic "Sexual Assault" (Panel B). The dashed line in Panel B is a linear fit. The solid vertical line separates months pre- and post-#MeToo. The structural break in Panel A indicates that the Me Too movement's sudden mediatization brought sex crimes to the forefront of the public debate. However, Panel B nuances this interpretation, as there were clear pre-trends in the number of queries related to sex crimes before #MeToo was used as a coordination device to combat sexual violence.

F.3 Arrest Rates and Probabilities of Arrest

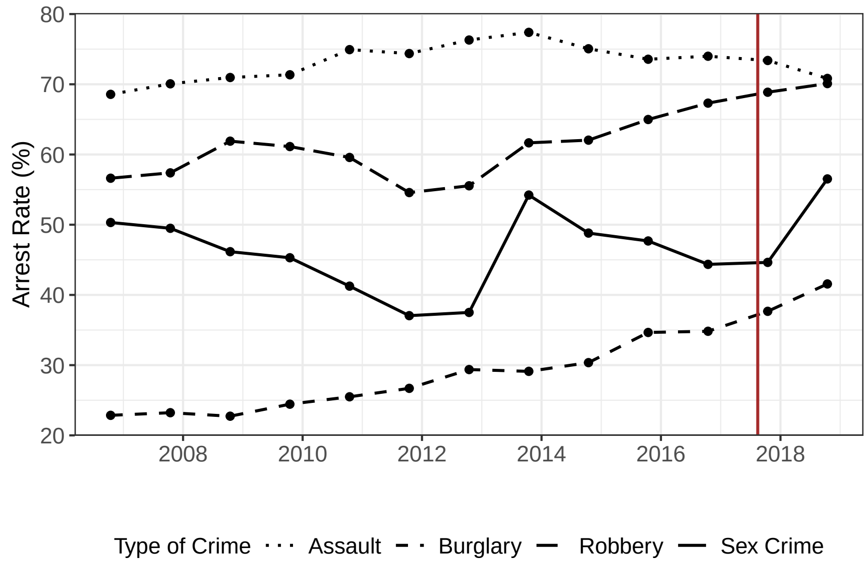


Figure F.6: Arrest Rates Over Time

Notes: Probabilities of arrest conditional on a complaint being filed (i.e., arrest rates) across crime categories for New York City. The vertical solid red line corresponds to the Me Too movement's mediatization.

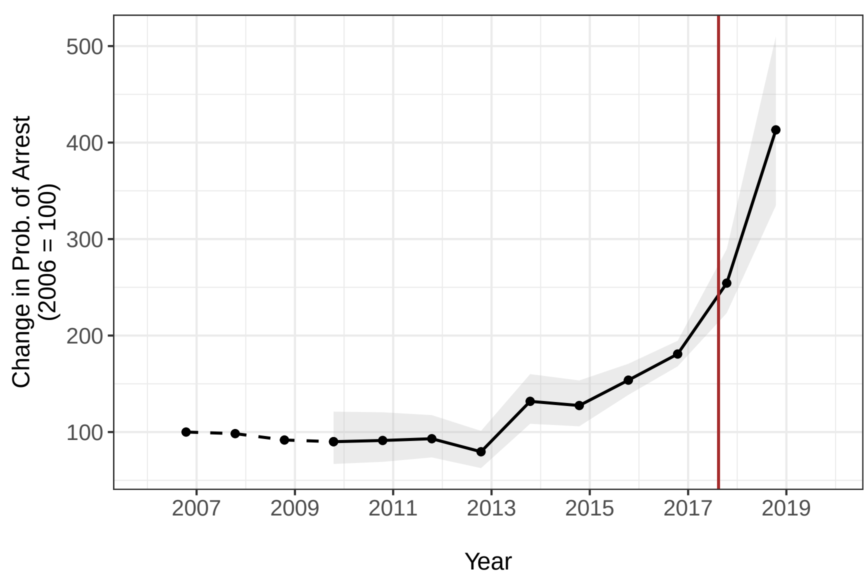


Figure F.7: Yearly Estimates of the Probability of Arrest for Sex Offenders

Notes: Index of the probability of arrest conditional on committing a sex crime for New York City, normalized to 2006. The index combines arrest rates (arrests/complaints) with TWFE estimates of reporting probability changes. The dashed line shows the pre-estimation period where reporting probability is assumed constant; the solid line shows the estimated period (2010 onwards). 95% confidence intervals (in grey) are computed using the delta method. The vertical solid red line corresponds to the Me Too movement's mediatization.

G Alternative Empirical Strategy: Duration Modeling Approach

This appendix presents an alternative empirical strategy to separate crime reporting from incidence using duration modeling. While the main text uses a two-way fixed effects (TWFE) approach, this section develops a mixed proportional hazards (MPH) duration model that explicitly accounts for delayed reporting and never-reporters.

I proceed in four steps. I first present the duration model and its assumptions. I then derive the model's likelihood for estimation. Next, I assess the performance of my estimator in finite samples via Monte Carlo simulations. Finally, I replicate the paper's main results.

G.1 A Duration Model of Time-to-report to the Police

This section develops a duration model to estimate the effect of #MeToo on victim reporting behavior. I first present the identification assumptions – what would be required if we observed all victims. I then explain why we need additional structure to handle the fact that never-reporters are unobserved, and present the model specification that accommodates this challenge.

G.1.1 Identification Assumptions

Two assumptions are necessary for identification, analogous to the TWFE assumptions in the main text. These assumptions would suffice to non-parametrically identify the effect of #MeToo on reporting hazards if we observed all victims, including those who never report.

Let $h_{it}(y \mid \gamma_i, \boldsymbol{x_{ity}})$ denote the hazard of reporting a crime y days after the incident, for individual i who experienced an incident at time t, with unobserved heterogeneity γ_i and covariates $\boldsymbol{x_{ity}}$.

Assumption 1 (No Anticipation) The intervention does not affect reporting behavior prior to t^*. Formally, for all observations where $t + y < t^*$:

$$h_{it}(y \mid \gamma_i, \boldsymbol{x_{ity}}) = h_{it}(y \mid \gamma_i, \boldsymbol{x_{ity}}, D = 0).$$

This is identical to Assumption 1 in the TWFE model. It ensures that pre-intervention observations can be treated as untreated counterfactuals.

Assumption 2 (Conditional Independence) Time-to-report Y is independent of incident date T, conditional on covariates X and unobserved heterogeneity γ:

$$Y \perp T \mid X, \gamma.$$

Equivalently, conditional on covariates (including treatment status) and unobserved heterogeneity, the distribution of reporting delays does not depend on calendar time. This is the analog of Parallel Trends in the TWFE model: both assumptions restrict how untreated outcomes vary with calendar time. Investigating pre-trends allows researchers to assess whether this assumption is likely to hold in practice.

G.1.2 The Challenge of Never-Reporters

Unfortunately, we do not observe all victims – we only observe those who report to the police. Never-reporters are unobserved. This creates a fundamental challenge: even with Assumptions 1 and 2, we cannot identify treatment effects from police data alone without additional structure.

We only observe the time-to-report distribution conditional on reporting:

$$f^{obs}(y) = \frac{f(y)}{\Pr(\text{ever report})}.$$

To recover the unconditional hazard $h(y)$ in the victim population, we must know what fraction never reports. This differs from a standard Cox model, in which hazard ratios can be estimated without knowing the baseline hazard. Here, selection into "reporter" depends on the overall hazard level, so even relative comparisons require knowing the share of never-reporters.

To address this challenge, I adopt a modified Mixed Proportional Hazards (MPH) specification and calibrate the share of never-reporters using external data.

G.1.3 Model Specification

Let (Y, X) denote a random vector where $Y \in \mathbb{R}^+$ is the time-to-report of a complainant and $X \in \mathbb{R}^d$ contains observed covariates. For individual i, victim of an incident at time t, I model the hazard of reporting a crime to the police y days after the incident as a mixed proportional hazards (MPH) duration model:

$$h_{it}(y \,|\gamma_i, \boldsymbol{x_{ity}}) = h_0(y) \exp\left(\boldsymbol{\beta'} \boldsymbol{x_{ity}}\right) \gamma_i. \tag{10}$$

$h_0 : \mathbb{R}^+ \to \mathbb{R}^+$ is the baseline hazard and models the influence of time since the incident on the probability of report. $\boldsymbol{\beta'} \in \mathbb{R}^d$ is the vector of regression coefficients capturing covariate effects on reporting. Covariates $\boldsymbol{x_{ity}}$ may be time-varying: their values can depend on the calendar incident date t and the duration y (e.g., if $t + y \geq$ Oct.2017, then $\text{MeToo}_{ity} = 1$, otherwise $\text{MeToo}_{ity} = 0$). $\gamma_i \in \mathbb{R}$ is a time-invariant random effect to account for unobserved heterogeneity across individuals. The core parametric assumption is the multiplicative decomposition between the baseline hazard, observed covariate effects, and individual-specific unobserved heterogeneity (Van den Berg, 2001).[13] When $\gamma_i = 1$ for all observations, the model reduces to the canonical Cox model (Cox, 1972).

To account for never-reporters, I modify the MPH model by enforcing the baseline hazard as a

[13]MPH models can be derived from economic theory if one assumes myopic individuals (Van den Berg, 2001). The hazard of reporting does not spike around the statute of limitations deadline, suggesting victims display myopic behavior.

density function and adding an intercept:

$$h_{it}(y \mid \gamma_i, \boldsymbol{x_{ity}}) = f_0(y) \exp\left(\alpha + \boldsymbol{\beta}'\boldsymbol{x_{ity}}\right)\gamma_i, \tag{11}$$

where $f_0 : \mathbb{R}^+ \to \mathbb{R}^+$ is a proper density function.[14] The biostatistics literature refers to this specification as a promotion time model. It ensures the survival function has positive mass at infinity, representing never-reporters. In the simplest case of no time-varying covariates, the proportion of victims who will never report is:[15]

$$\lim_{y\to\infty} S_i(y \mid \gamma_i, \boldsymbol{x_i}) = \exp\Big(- \gamma_i \exp(\alpha + \boldsymbol{\beta}'\boldsymbol{x_i})\Big).$$

The model incorporates three essential features: (i) victims report with heterogeneous delays, (ii) a share of victims never report, and (iii) covariates – including treatment indicators – influence both reporting delays and the probability of ever reporting. Crucially, this functional form creates a mathematical link between observables and unobservables: the shape of the observed delay distribution (among reporters) determines how changes in the hazard translate into changes in the share of never-reporters.

When all victims are observed, Equations 10 and 11 yield numerically similar estimates for β.[16] However, when never-reporters are unobserved, Equation 11 provides a way to incorporate the share of never-reporters at baseline via α. This parameter governs the baseline share of never-reporters and must be calibrated, as it is not identified from the observed reporting delays alone. I calibrate it externally: in the application, I set the share of never-reporters at baseline to 80%, a value bracketed by survey-based estimates of non-reporting (see Section G.4 for the baseline calibration and robustness to alternative values). Figure G.1 provides a graphical intuition of the role of α.

G.1.4 Theoretical Motivation

Equation 11 has a theoretical interpretation as a Poisson counting process that naturally accommodates never-reporters. If a crime is committed, victim i chooses whether to report the incident in subsequent periods. The psychology literature has highlighted several influential factors in the decision to report (see Tavarez (2021) for a review). Barriers include internal psychological factors (e.g., trauma, guilt, fear), social interactions (e.g., stigma, relationship with perpetrator), and criminal justice system factors (e.g., negative police interactions, low perceived odds of success). In addition, victims may be unaware of resources or take time to recognize that the incident constitutes a crime. Juvenile victims, in particular, may lack the knowledge to recognize and articulate

[14] I systematically distinguish the baseline distribution's functions (F_0, h_0, f_0, S_0) from those of the time-to-report distribution Y of victims (F, h, f, S).

[15] See Section G.2.3 for the extension to time-varying covariates.

[16] *Proof:* Equation 11 implies the survival function $S_i(y|\gamma_i, \boldsymbol{x_i}) = \exp\big(- \gamma_i \exp(\boldsymbol{\beta}'\boldsymbol{x_i}) \exp(\alpha) F_0(y)\big)$. Equation 10 implies $S_i(y|\gamma_i, \boldsymbol{x_i}) = \exp\big(- \gamma_i \exp(\boldsymbol{\beta}'\boldsymbol{x_i}) H_0(y)\big)$. The baseline cumulative hazard H_0 is unspecified, so it can take the form $\exp(\alpha)F_0(y)$.

that a sex crime occurred.

To capture these deliberations tractably, I assume victim i is exposed to K_t potential decisive arguments to report:[17]

$$K_t \sim Pois(\theta_t).$$

These arguments are independent and identically distributed. The time for each argument to trigger a report is drawn from distribution F_0.[18] The hazard and survival functions are then:

$$h_t^{(v)}(y) = \theta_t f_0(y) \quad \text{and} \quad S_t^{(v)}(y) = \exp\Big(-\theta_t F_0(y)\Big).$$

The survival function has positive mass as $y \to \infty$, representing never-reporters:

$$S_t^{(v)}(+\infty) = \exp\big(-\theta_t\big).$$

Formal Proof Let $W_1, W_2, \ldots$ denote the i.i.d. times for each argument to trigger a report, each drawn from F_0. Then

$$\begin{aligned}
S_t^{(v)}(y) &= P(Y > y) \\
&= P(K_t = 0) + \sum_{n=1}^{\infty} P\big(W_1 > y, \ldots, W_n > y\big)\, P(K_t = n) \\
&= \sum_{n=0}^{\infty} \big(1 - F_0(y)\big)^n \exp\big(-\theta_t\big) \frac{\theta_t^n}{n!} \\
&= \exp\big(-\theta_t F_0(y)\big)
\end{aligned}$$

G.1.5 Intuition: From Observed Delays to Victim Reporting Rates

Figure G.1 illustrates how the model rescales observed delays to account for never-reporters. Consider a researcher interested in a policy implemented at t^*. Let D_{ity} indicate the policy ($D_{ity} = 1$ if $t + y \geq t^*$). She estimates:

$$h_{it}\big(y \mid \gamma_i,\, D_{ity}\big) = f_0(y) \exp\big(\alpha + \beta D_{ity}\big)\gamma_i. \tag{12}$$

The model compares the hazard of reporting y days later before the policy to the hazard after the policy. We do not observe never-reporters in either period. However, if the researcher calibrates α to match the pre-policy never-reporter share, the model can infer reporting probabilities for all

[17] An alternative is to formulate the decision as an optimal stopping problem, where victim i knows circumstances may change over time and chooses whether to file a complaint or wait for more favorable conditions. This is reminiscent of job search models (Mortensen, 1986).

[18] Similar models have been used in fertility and cancer studies (Lambert and Bremhorst, 2019).

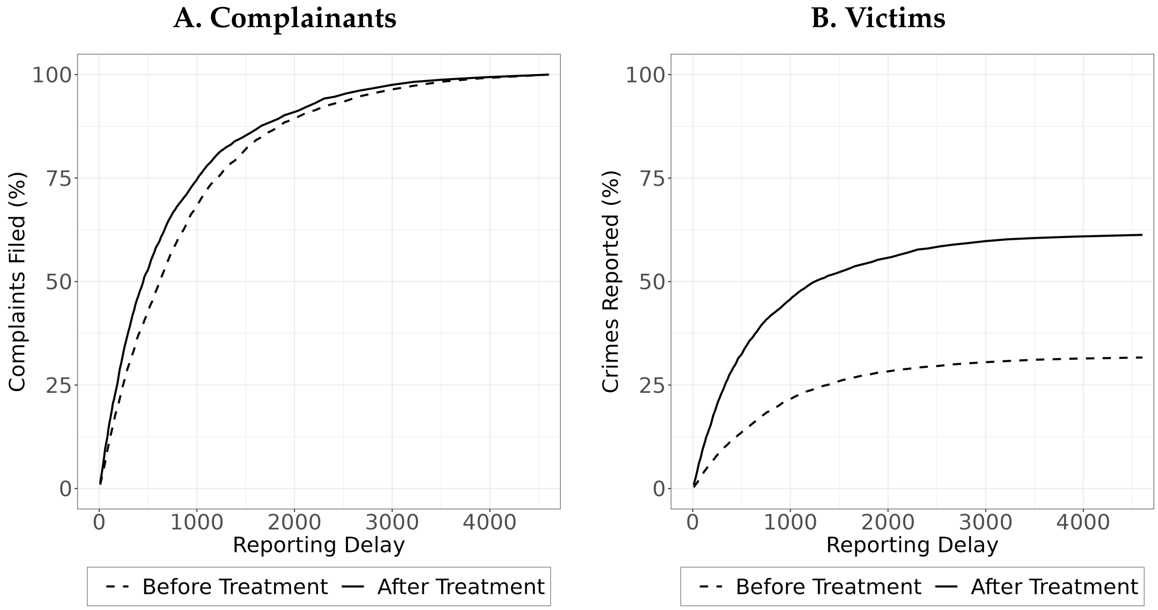


Figure G.1: From Observed Delays to Victim Reporting Rates

Notes: These simulated distributions provide graphical intuition on the duration models developed in the paper. A treatment/intervention increases the propensity to report of victims. The plots compare cumulative distribution functions (CDFs) before (dashed line) and after (solid line) the treatment. Panel A plots the CDF of observed reporting delays in police records. Panel B plots the CDF of the entire victim population and corresponds to Equation 11. The CDFs do not sum to 100% because a share of victims will never report to the police. Those victims are unobserved. However, suppose one knows the victim reporting rate before the treatment (here 30%). In that case, the observed distribution of reporting delays is sufficient to infer the victim reporting rate after the treatment (here around 60%). This intuition can be generalized to multiple treatments/interventions. For instance, fitting a linear spline over time recovers variations of the reporting rate over time (see the application to #MeToo in Section G.4).

delays before the policy. For crimes committed n days before t^*, these n days reveal the CDF of reporting delays before treatment, $F(n \mid \gamma_i, D_{ity} = 0)$. This in turn identifies $h_{it}(y \mid \gamma_i, D_{ity} = 1)$ for $y \geq n$. Conditional on α, $F(n \mid \gamma_i, D_{ity} = 1)$ is identified for all $n \geq 2$, so parametric assumptions only extrapolate $F(1 \mid \gamma_i, D_{ity} = 1)$.

G.2 Double-Truncation and Estimation

I now turn to the estimation of the model. In contrast to common applications in the economics literature, police records present an additional empirical challenge as they raise the issue of double-truncation: crime incidents are observed if they are reported within the study period. Some reports may occur after the end of the study (i.e., right-truncation), and others may occur before its start (i.e., left-truncation). Right truncation implies an oversampling of shorter durations. Conversely, left truncation means an oversampling of longer durations. Without an appropriate correction, a naive estimation will lead to biased estimates (Dörre and Emura, 2019).

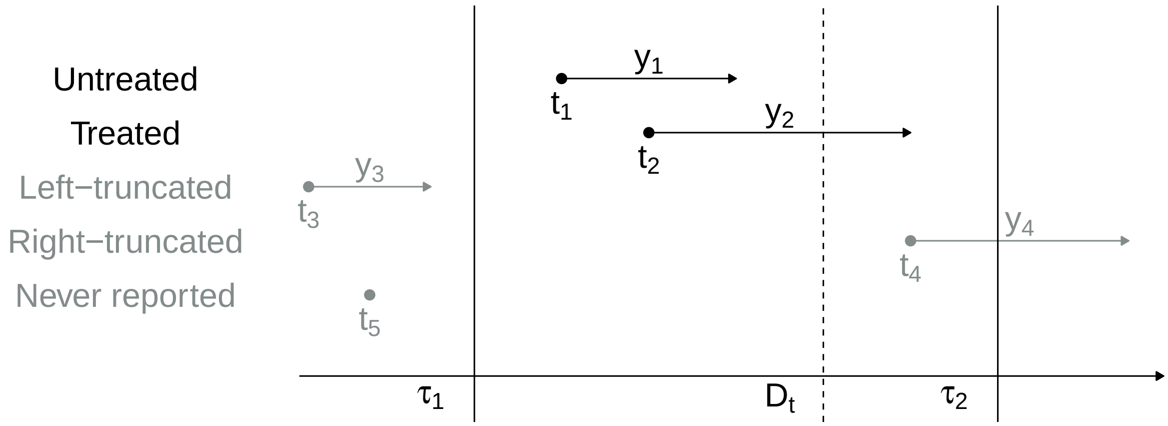


Figure G.2: The Structure of (Delayed) Crime Reports

Notes: A graphical depiction of time to report data based on police records. The study window goes from τ_1 to τ_2 (solid vertical lines), with an intervention D_t in between represented by a dashed vertical line (e.g. #MeToo). Elements in gray are unobserved. Elements in black are observed. Some complainants will have reported before the intervention's implementation and form the control group (non-treated observations). Others will be potentially affected by the intervention and form the treated group (treated observations). Some complainants have reported before the start of the study and are unobserved (left-truncated data points). Some complainants have not yet reported a crime to the police by the end of the study period but will in the future and are unobserved (right-truncated data points). Finally, a fraction of victims decide to never report and are never observed (i.e., never-reporters).

G.2.1 The Structure of Crime Reports

Figure G.2 presents a graphical summary of police data as duration data. The study window goes from τ_1 to τ_2 (solid vertical lines), with an intervention D_t in between represented by a dashed vertical line. Some complainants report before the intervention's implementation and form the control group (non-treated observations). Others are affected by the intervention and form the treated group (treated observations). Some complainants report before the start of the study and are unobserved (left-truncated data points). Some complainants have not yet reported a crime to the police by the end of the study but will in the future and are unobserved (right-truncated data points). Finally, some victims may decide never to report and are unobserved. I call them never-reporters.

G.2.2 An Example of Right-truncation Bias

Figure G.3 highlights how right-truncation bias may lead researchers to seriously overestimate the impact of an intervention against crime. The estimates of a naive Cox model that does not account for right-truncation are heavily dependent on the study window. The closer the intervention to the end of the study window (in this case, #MeToo), the larger the estimate. This is a spurious result: right-truncation leads to an oversampling of shorter durations as we move closer to the end of the study window.

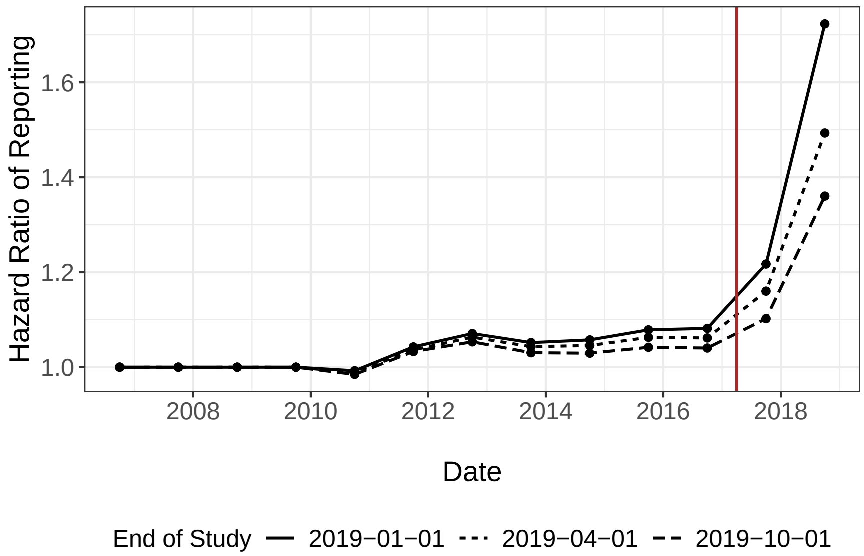


Figure G.3: An Example of Right-truncation Bias

Notes: Estimates of the complainant reporting hazard using a naive Cox regression model (i.e., that does not account for right-truncation) for New York City. Breaks in the baseline hazard are set after 1, 30, 90, 180, and 365 days. There is no unobserved heterogeneity. 95% confidence intervals. Without an appropriate correction for right-truncation, estimated hazard ratios are heavily dependent on the end of the study period τ_2.

G.2.3 Derivation of the Likelihood

Formally, let T denote the incident date, Y the time-to-report to the police, $U = \tau_1 - T$ the left-truncation time, and $V = \tau_2 - T$ the right truncation time. Recall that τ_1 and τ_2 are, respectively, the start and end of the study period. Note also that $V = U + d$ where $d = \tau_2 - \tau_1$.[19] Finally, let f and g denote the density functions of Y and U. Under double-truncation, I observe n incidents indexed by i from the probability distribution (T, Y) given $U \leq Y \leq V$. The density of each data point (u_i, y_i, v_i) is

$$P(U = u_i, Y = y_i \,|\, U \leq Y \leq U + d).$$

In general, when subjects have unequal probabilities of selection, the observed sample will not be representative of the underlying target population. The associated likelihood is

$$L_n(f, g) = \prod_{i=1}^{n} \frac{f(y_i)g(u_i)}{\int_u \left(\int_u^{u+d} f(y)dy \right) g(u)du}.$$

This likelihood is complex, but under the assumption of independence between Y and U, one can

[19] In the case of police records, V is thus entirely determined by U and d. This is referred to as *fixed-length* double-truncation, but the results presented below also hold for more general double-truncation schemes.

decompose it into two, somewhat more tractable conditional likelihoods:

$$L_n(f,g) = \prod_{i=1}^{n} \frac{f(y_i)}{\int_{u_i}^{u_i+d} f(y)dy} \times \prod_{i=1}^{n} \frac{\left(\int_{u_i}^{u_i+d} f(y)dy\right)g(u_i)}{\int \left(\int_{u}^{u+d} f(y)dy\right)g(u)du}.$$

I use the first conditional likelihood to make an inference on f. This first term is relatively intuitive. It is an inverse-probability weighting approach in which observations are weighted by the inverse of their sampling probability. Furthermore, a major advantage of focusing on the first conditional likelihood is that I do not specify the distribution of the truncation time U. The likelihood to maximize eventually simplifies to[20]

$$L_n(f) = \prod_{i=1}^{n} \frac{f(y_i)}{F(u_i+d) - F(u_i)}.$$

Conditioning on observed covariates is straightforward. The covariates can be time-varying (e.g., policy interventions) and thus relax the independence assumption between U and Y to a more realistic, conditional independence assumption.[21] If specified, random effects are integrated out. I estimate the models by full maximum likelihood.

In the following, I provide further details on the derivation of the likelihood.

Recall that Y is the distribution of times to report, U is the distribution of left-truncation times, d is the length of the study period (i.e., $\tau_2 - \tau_1$), X is a random vector of observed covariates, and h_0, H_0 are respectively the baseline hazard and the baseline cumulative hazard of the duration model. In the case of the MPH model, the baseline hazard is a piece-wise constant function. In the case of the promotion time model, the baseline hazard is the density of a distribution of which the hazard is modeled as a piece-wise constant function. Sample observations are indexed by i and Θ is the vector of parameters to estimate. Finally, I refer to random variables with upper-case letters and to their realizations with lower-case letters.

[20]Under the nonparametric setting, Shen (2010) show that the MLE based on $L_n(f)$ and the MLE based on $L_n(f,g)$ give an equivalent estimator for f. This suggests that $L_n(f)$ contains sufficient information about f for maximum likelihood estimation.

[21]Recently, inverse-probability weighting approaches have been proposed for fitting the Cox model to doubly-truncated data (Mandel et al., 2018; Rennert and Xie, 2018), of which right-truncated data is a special case (Vakulenko-Lagun et al., 2019). They rely on the non-parametric maximum likelihood estimators of the selection probabilities proposed by Efron and Petrosian (1999) and Shen (2010). The main assumption is that Y, U, and V are *unconditionally* quasi-independent. This would imply that the incident date does not affect the complainants' time-to-report in the context of crime reports. This runs precisely counter to my prior: the time-to-report likely varies with the incident date, as complainants are more or less likely to report a crime to the police over time (e.g., before/after the MeToo movement). I formally test this quasi-independence assumption in my dataset (Martin and Betensky, 2005). I reject the null hypothesis that survival and truncation times are quasi-independent at all standard significance levels. Thus, these methods are not a good fit in this empirical context. My approach relies on a less demanding and more realistic *conditional* independence assumption. However, it comes at the expense of specifying a parametric baseline hazard. To limit the impact of this parametrization, I specify flexible, piece-wise, constant baseline hazards.

Time-invariant covariates I start with the case of time-invariant covariates and no unobserved heterogeneity. I observe a sample of n realizations $\{(x_i, y_i)\}_{i\in\{1,...,n\}}$ for inference. Under the assumption that Y and U are independent conditional on observed covariates X, we have

$$L(\Theta \mid x) = \prod_{i=1}^{n} \frac{P(y_i \mid x_i)}{P(u_i \leq y_i \leq u_i + d \mid x_i)},$$

which gives

$$L(\Theta \mid x) = \prod_{i=1}^{n} \frac{\exp\left(\beta' x_i\right) h_0(y_i) \exp\Big(-\exp\left(\beta' x_i\right) H_0(y_i)\Big)}{\exp\Big(-\exp\left(\beta' x_i\right) H_0(u_i)\Big) - \exp\Big(-\exp\left(\beta' x_i\right) H_0(u_i + d)\Big)},$$

after some factorization, we obtain

$$L(\Theta \mid x) = \prod_{i=1}^{n} \frac{\exp\left(\beta' x_i\right) h_0(y_i) \exp\Big(-\exp\left(\beta' x_i\right)\left(H_0(y_i) - H_0(u_i)\right)\Big)}{1 - \exp\Big(-\exp\left(\beta' x_i\right)\left(H_0(u_i + d) - H_0(u_i)\right)\Big)}.$$

Time-varying covariates Next, I extend the model to time-varying covariates. In this case, as soon as one of the covariate changes, the covariate vector needs to be updated. Assume that those variations occur at $J_i - 1$ occasions $y_{i1}, ..., y_{iJ_{i-1}} \in \mathbb{R}^{J_{i-1}}$. Among those, the first $J'_i + 1$ are observed variations, whereas the remaining are *counterfactual* values for the right-truncation time[22], such that: $y_{i0} = u_i \leq y_{i1} \leq ... \leq y_{iJ'_i} = y_i \leq ... \leq y_{iJ_i} = v_i \leq \infty$, yielding the sequence of covariate vectors: $\chi_i = \{x_{i1}(y_{i0}), ..., x_{iJ_i}(y_{iJ_{i-1}})\} \in \mathbb{R}^{J_i}$. For $j = 1, ..., J_i$, $y_{ij} - y_{ij-1}$ is the time spent by the i^{th} subject in his or her j^{th} covariate configuration $x_{ij}(y_{ij-1})$. Then we obtain

$$L(\Theta \mid \chi) = \prod_{i=1}^{n} \frac{\exp\left(\beta' x_{iJ'_i}\right) h_0(y_{iJ'_i}) \exp\Big(-\sum_{j=1}^{J'_i} \exp\left(\beta' x_{ij}(y_{ij})\right)\left(H_0(y_{ij}) - H_0(y_{ij-1})\right)\Big)}{1 - \exp\Big(-\sum_{j=1}^{J_i} \exp\left(\beta' x_{ij}(y_{ij})\right)\left(H_0(y_{ij}) - H_0(y_{ij-1})\right)\Big)}.$$

Note that these last two equations do not require knowing the history of covariates between the incident date and the beginning of follow-up (i.e., between 0 and the left-truncation time). They require, however, knowledge of the covariates until the end of the study period (i.e., for the follow-up time, but also up to the right-truncation time).

Unobserved Heterogeneity Next, I extend the model to unobserved heterogeneity. I assume the frailty is a random effect γ. Because the frailty term is unobserved at the individual level, it is necessary to consider the population level and to integrate it out of the likelihood. The likelihood

[22] For example, if the victim had not reported to the police in September 2016, she would have been eventually affected by the Me Too movement in October 2017, setting the dummy variable's value to one.

of time-invariant covariates is

$$L(\Theta \mid \gamma, x) = \prod_{i=1}^{n} \mathbb{E}_{\gamma}\Big[P(y_i \mid u_i \leq y_i \leq u_i + d, x_i)\Big].$$

Applying Bayes rule, we have

$$L(\Theta \mid x) = \prod_{i=1}^{n} \frac{\mathbb{E}_{\gamma}\Big[P(y_i \mid x_i)\Big]}{\mathbb{E}_{\gamma}\Big[P(u_i \leq y_i \leq u_i + d \mid x_i)\Big]}.$$

Replacing expressions with the model parameters gives $L(\Theta \mid \gamma, x)$ as

$$\prod_{i=1}^{n} \frac{\mathbb{E}_{\gamma}\Big[\gamma \exp\left(\beta' x_i\right) h_0(y_i) \exp\Big(-\gamma \exp\left(\beta' x_i\right) H_0(y_i)\Big)\Big]}{\mathbb{E}_{\gamma}\Big[\Big\{\exp\Big(-\gamma \exp\left(\beta' x_i\right) H_0(u_i)\Big) - \exp\Big(-\gamma \exp\left(\beta' x_i\right) H_0(u_i + d)\Big)\Big\}\Big]}.$$

Similar (cumbersome) expressions may be obtained for time-varying covariates. Note that unobserved heterogeneity requires the researcher to know the history of covariates between the incident date and the beginning of follow-up (i.e., between 0 and the left-truncated time). For time-varying covariates, this can be challenging and likely involves some speculation. For simplicity, I assume there were no interventions that affected victim reporting before the beginning of the study period. As a robustness check, I also estimate the models without left-truncated observations and find qualitatively similar results.

Parametric Unobserved Heterogeneity In the case of well-known parametric frailty distributions, the terms can be expressed in terms of the Laplace transform $\mathcal{L}_{\gamma}$ and its first derivative $\mathcal{L}_{\gamma}^{(1)}$:

$$L(\Theta \mid \gamma, x) = \prod_{i=1}^{n} \frac{-\exp\left(\beta' x_i\right) h_0(y_i) \mathcal{L}_{\gamma}^{(1)}\Big(\exp\left(\beta' x_i\right) H_0(y_i)\Big)}{\Big\{\mathcal{L}_{\gamma}\Big(\exp\left(\beta' x_i\right) H_0(u_i)\Big) - \mathcal{L}_{\gamma}\Big(\exp\left(\beta' x_i\right) H_0(u_i + d)\Big)\Big\}}.$$

I assume unobserved heterogeneity is gamma distributed with variance Σ (Vaupel et al., 1979; Abbring and Van Den Berg, 2007). To ensure that the model is identifiable, I use a parameter restriction for the gamma distribution, such that its mean equals one. For gamma distributions, we know that

$$\mathcal{L}_{\gamma}(s) = (1 + \Sigma s)^{-\frac{1}{\Sigma}} \text{ and } \mathcal{L}_{\gamma}^{(1)}(s) = -(1 + \Sigma s)^{-(\frac{1}{\Sigma}+1)}.$$

There are theoretical reasons to assume gamma-distributed unobserved heterogeneity. In a large class of frailty models, the frailty distribution among survivors converges to a gamma distribution under mild regularity assumptions (Abbring and Van Den Berg, 2007).

Non-parametric Unobserved Heterogeneity Nonetheless, in practice, parametric frailties are mainly driven by computational efficiency concerns rather than theoretical justifications. An alternative to parametric distributions is a non-parametric estimation of unobserved heterogeneity introduced by Heckman and Singer (1984). Assume that the population under study consists of K sub-populations with different frailties $\{\gamma_k\}_{k\in\{1,\ldots,K\}}$ and respective shares within the population $\{s_k\}_{k\in\{1,\ldots,K\}}$. Further, I impose that all parameters are strictly positive and that the sum of their shares is one. Just like the piece-wise constant function to model baseline hazards, this formulation is a general specification of unobserved heterogeneity, which can account for many distributions. $L(\Theta \mid \gamma, x)$ is then

$$\prod_{i=1}^{n} \frac{\sum_{k=1}^{K} s_k \left[\gamma_k \exp\left(\beta' x_i\right) h_0(y_i) \exp\Big(- \gamma_k \exp\left(\beta' x_i\right) H_0(y_i) \Big) \right]}{\sum_{k=1}^{K} s_k \left[\Big\{ \exp\Big(- \gamma_k \exp\left(\beta' x_i\right) H_0(u_i) \Big) - \exp\Big(- \gamma_k \exp\left(\beta' x_i\right) H_0(u_i + d) \Big) \Big\} \right]}.$$

This last approach is the most flexible. However, it is computationally demanding, prone to converging to local minima, and requires large numbers of observations to estimate the random effect precisely.

G.3 Monte Carlo Simulations

This section validates the proposed duration modeling approach with its double-truncation correction through Monte Carlo simulations that mirror the four scenarios considered for the TWFE estimator in Appendix C. I benchmark the proposed estimator against a naive piecewise-constant hazard (PCH) model that ignores double truncation – i.e., a standard duration regression applied to the observed reports without inverse-probability weighting.

Data Generating Process I simulate crime reports over 365 periods. At each period t, 30 offenses are committed. The baseline hazard h_0 associated with F_0 is modeled as a piecewise-constant exponential with rate $\lambda_1 = 0.2$ for the first period after the incident and $\lambda_2 = 0.01$ for subsequent periods, capturing the empirical regularity that a large share of crimes is reported on the day of the incident. The cure fraction is set to 0.5, meaning that 50% of victims never report the crime to the police. In period 300, a treatment D_1 permanently shifts the reporting hazard. To reproduce the double-truncation scheme, I keep only incidents whose report date falls in the observation window (periods 199–365); the long pre-window head start supplies the latent offense history that the event-study fit requires. This leaves roughly 2,300–2,650 observed reports per replication, depending on the scenario, out of 10,554–10,950 offenses; the remainder are never-reporters (50%) or fall outside the window. I run 1,000 replications for each scenario.

Simulation Scenarios I consider four scenarios to evaluate the estimator's performance:

1. **No treatment effect:** $\beta_1 = 0$ and C_t constant at 30. This null scenario verifies correct size and the absence of false positives.

2. **Reporting only:** $\beta_1 = 0.3$ (increased reporting hazard post-treatment), C_t constant. Tests identification of a pure reporting effect.

3. **Incidence only:** $\beta_1 = 0$, C_t drops by 20% (from 30 to 24) for $t \geq 300$. The critical test: the duration estimator, which models reporting delays, should be unaffected by changes in incidence alone.

4. **Both effects:** $\beta_1 = 0.3$ and C_t drops by 20% for $t \geq 300$. Verifies correct identification when both channels change simultaneously.

Estimation The naive PCH benchmark and the proposed PT estimator each include a single reporting indicator that activates when calendar time reaches the treatment date, and the PT estimator fixes α at its true value, mirroring the empirical application in which α is calibrated externally. The reporting-rate and incidence figures below rely on a third, more flexible event-study version of the PT estimator: the single indicator is replaced with cumulative calendar-time reporting dummies at five-period boundaries spanning event times -40 to $+40$, with a boundary falling exactly on the treatment date. Observed report periods before the first boundary form a dummy-free baseline block, so the estimated reporting path is measured relative to that block. The long-run reporting rates in Figure G.5 and the observation probabilities used to recover incidence in Figure G.6 are computed from this event-study fit; the incidence figure additionally aggregates incident periods into five-period bins with a bin edge at the treatment date.

Results: Treatment Effect Estimates Figure G.4 displays the sampling distributions of $\hat{\beta}_1$ across 1,000 simulations. The proposed PT estimator with double-truncation correction (green) is approximately unbiased in all four scenarios, with bias never exceeding 0.003 in absolute value. By contrast, the naive PCH estimator (red) is severely biased *upward*: even in the null scenario where the true effect is zero, it produces a mean estimate of about $+0.53$, and its bias stays between $+0.28$ and $+0.53$ across scenarios. This substantial positive bias arises because the naive estimator ignores the right-truncation of reporting delays: incidents closer to the end of the observation window can only be observed if they are reported quickly, so their observed delays are mechanically compressed. Because the treatment switches on late in the window (period 300 of 365), post-treatment incidents sit nearer this truncation boundary than pre-treatment ones, and the naive estimator misreads their mechanically shorter delays as a genuine rise in the reporting hazard. This is the same right-truncation artifact illustrated in Figure G.3, where the naive estimate grows the closer the intervention lies to the end of the study window.

Crucially, in the "incidence only" scenario (bottom-left panel), the proposed estimator correctly identifies no reporting effect (bias ≈ 0.002) despite the 20% drop in crime incidence. This confirms that changes in the number of offenses committed do not contaminate the duration model's estimates of reporting behavior, since the model conditions on the incident having occurred.

Results: Long-Run Reporting Rates Beyond the treatment effect coefficient, the duration model also recovers the long-run share of reporters – the probability that a crime is ever reported, given by $\text{LTR}_t = 1 - \exp(-\exp(\hat{\alpha} + x_t'\hat{\beta}))$. Figure G.5 shows the estimated LTR paths across simulations. In the null and incidence-only scenarios, the estimated LTR remains flat at approximately 0.50, matching the true cure fraction. In the reporting-only and both-effects scenarios, the estimated LTR correctly steps up from 0.50 to approximately 0.60 at the treatment date, tracking the true path closely. The 95% confidence bands contain the true value in all scenarios.

Results: Crime Incidence Recovery Figure G.6 shows that the proposed estimator also accurately recovers crime incidence by dividing the count of reported crimes by the estimated probability of observation. In the incidence-only and both-effects scenarios, the proposed estimates (brown) correctly track the 20% drop in true incidence (dashed black line), and in the no-treatment and reporting-only scenarios they stay flat at the true level. (A naive estimator that ignores right-truncation instead overstates the probability of observation – roughly 0.9 against a true value near 0.33 in this design – and would therefore understate incidence; it is omitted from the figure for clarity.)

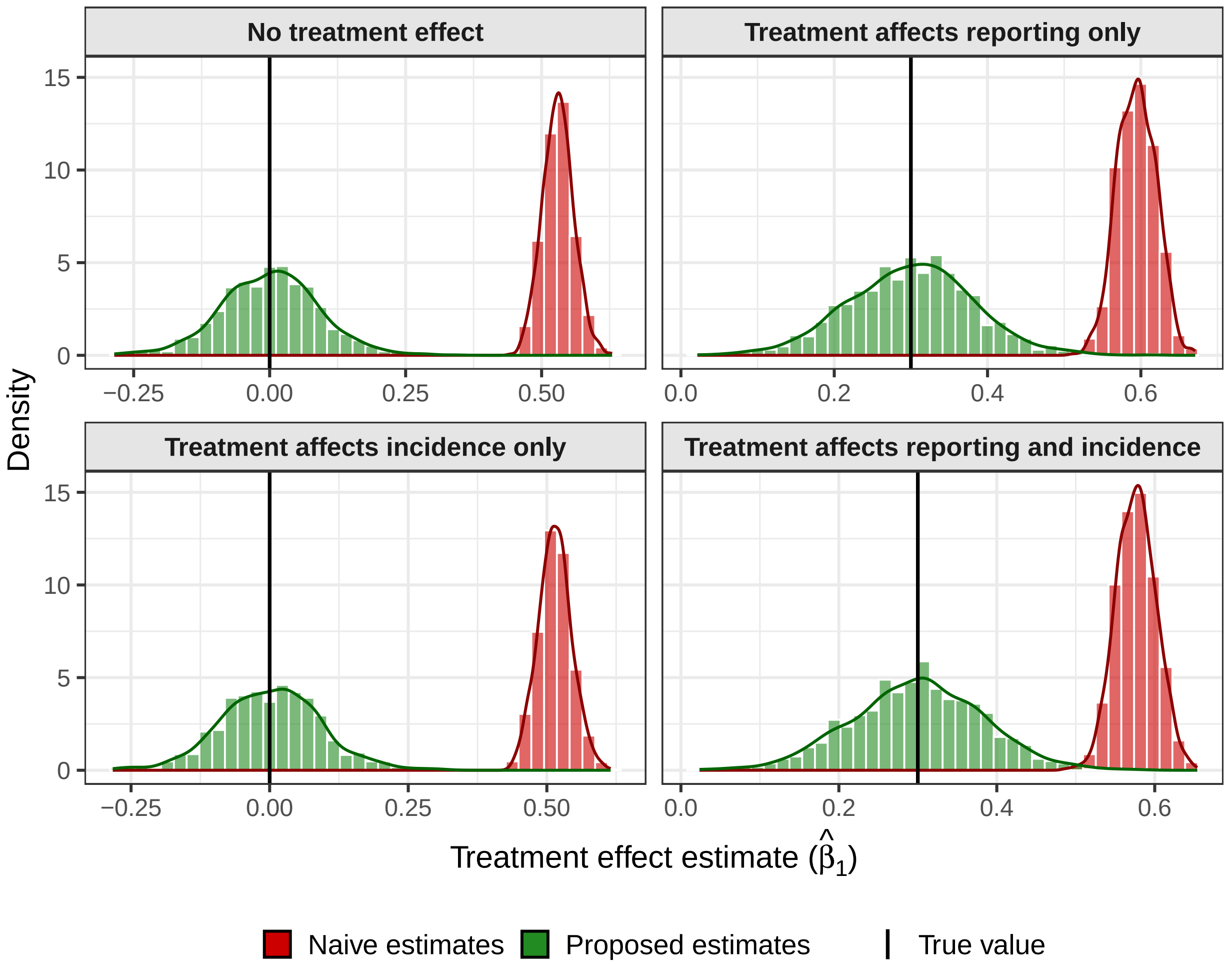


Figure G.4: Monte Carlo Simulations: Duration Model Treatment Effect Estimates

Notes: Sampling distributions of the treatment effect estimate $\hat{\beta}_1$ from the promotion-time model across 1,000 Monte Carlo simulations. The green density corresponds to the proposed estimator with double-truncation (DT) correction; the red density corresponds to the naive piecewise-constant hazard (PCH) estimator without DT correction. The vertical black line marks the true parameter value. The proposed estimator is approximately unbiased in all scenarios, while the naive estimator exhibits severe upward (positive) bias due to right-truncation.

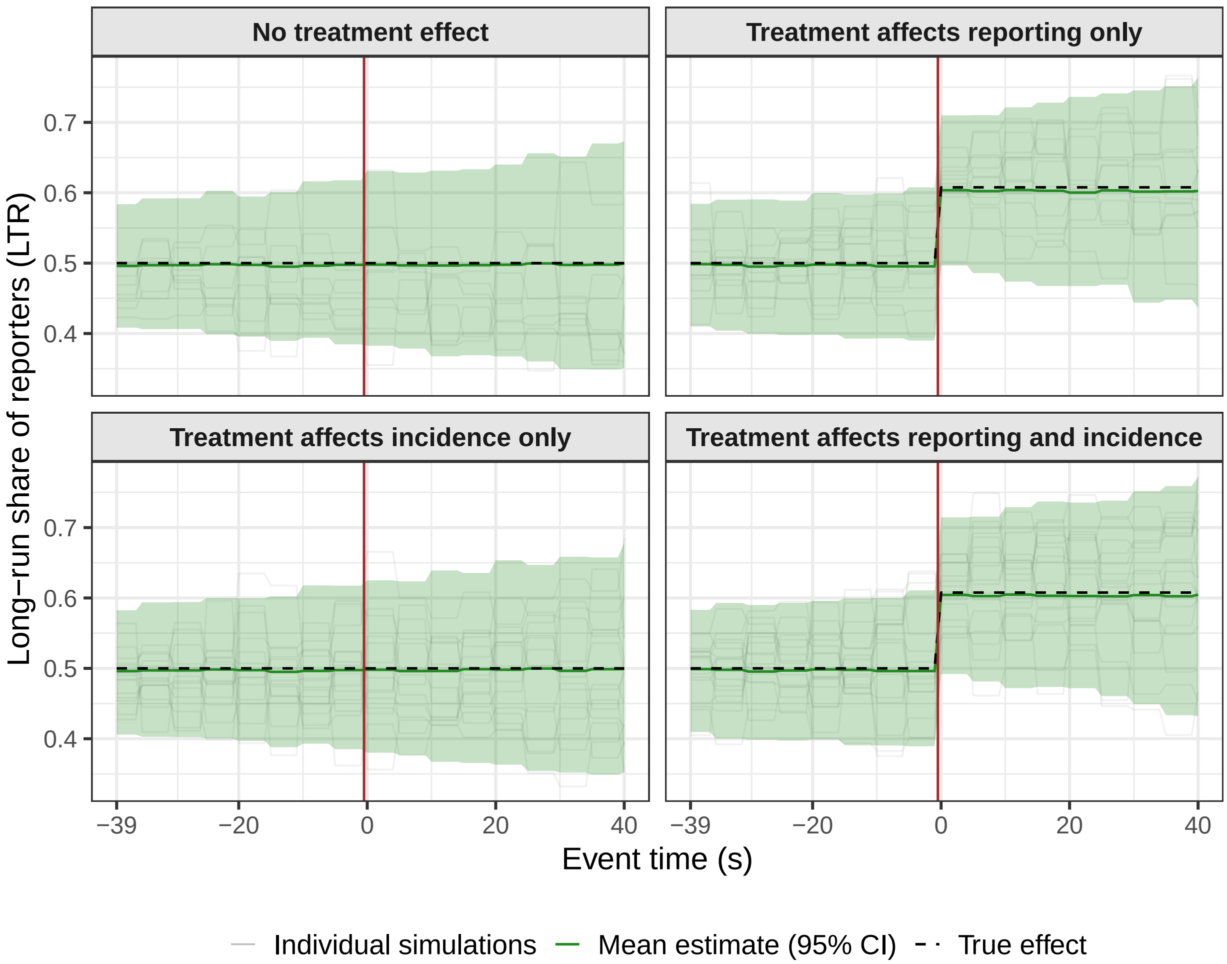


Figure G.5: Monte Carlo Simulations: Recovery of Long-Run Reporting Rates

Notes: Estimated long-run share of reporters (LTR) by incident period across 1,000 Monte Carlo simulations. The green solid line shows the mean estimate with 95% confidence bands; the dashed black line shows the true value; gray lines show individual simulation paths. The horizontal axis is event time s (periods relative to treatment); the vertical red line marks the treatment date ($s = 0$). In scenarios with a reporting effect, the LTR correctly steps up from 0.50 to approximately 0.60 at the treatment date.

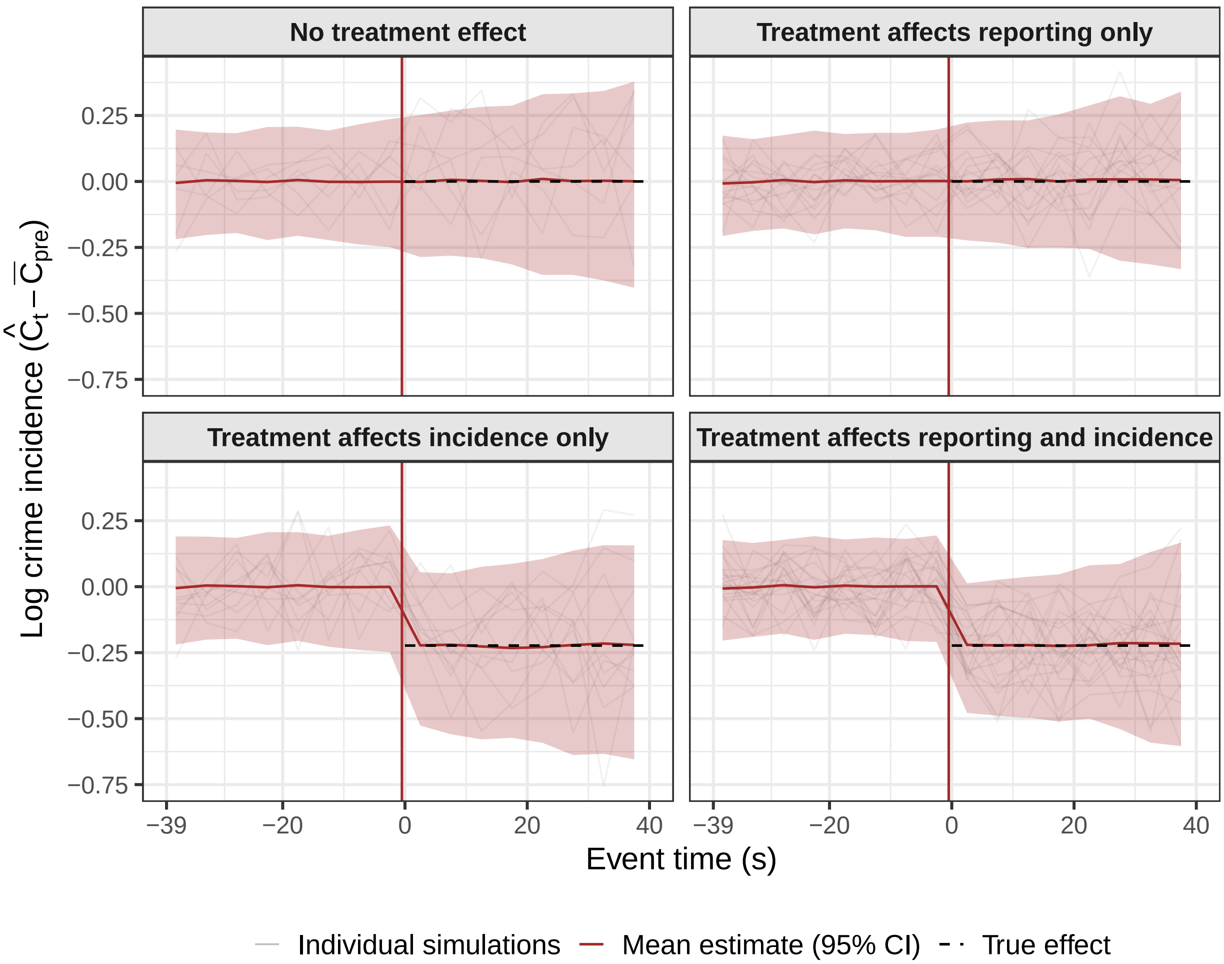


Figure G.6: Monte Carlo Simulations: Recovery of Crime Incidence

Notes: Estimated change in log crime incidence ($\hat{C}_t - C_{pre}$) by event time s across 1,000 Monte Carlo simulations. Gray lines show individual simulation paths; the solid brown line shows the mean estimate with 95% confidence bands; the dashed black line shows the true incidence change. The vertical red line marks the treatment date ($s = 0$). The proposed estimator – which divides reported crimes by the estimated probability of observation – recovers the true incidence path in all scenarios.

G.4 Replication of the Main Results with the Duration Model

This section examines sex crime incidence and reporting between 2006 and 2019 using a duration-modeling approach. I first investigate the reporting rate of victims (see Equation 11). This allows me to compute estimates of sex crime incidence over the period (see Equation 7). Finally, I isolate the effect of #MeToo on sex crime incidence.

G.4.1 Victim Reporting

I estimate the victim reporting hazard between 2006 and 2019. The dependent variable is the number of days elapsed between a sex crime being committed and its report to the police. I specify a piece-wise constant baseline hazard with breaks set after 1, 30, 90, 180, and 365 days.[23] I set α so that the share of never-reporters for crimes committed in 2006 equals 80%, a value bracketed by survey-based estimates of non-reporting (see the main text). With α pinned down, the two empirical strategies – which rely on different identification assumptions and parametric structures – can be compared directly (see Section G.4.3 for robustness to alternative calibrations ranging from 60% to 90%). I specify the hazard as follows:

$$h_{it}(y) = f_0(y) \exp\left(\alpha + \sum_{k=2010Q2}^{2019Q3} \beta_k \mathbb{1}(t + y \geq k)\right) \tag{13}$$

The main coefficients of interest are the quarterly coefficients β_k. I interpret β_k as the additional (higher or lower) propensity to report a sex crime to the police in quarter k among victims. Note that Equation 13 also provides us with the total share of victims who will eventually report a sex crime to the police for a given quarter k (and in the absence of future events that may shift victim reporting behaviors).

Figure G.7 decomposes reported sex crimes to the police by incident date into estimates of sex crime incidence and reporting (see Figure D.1 for the same decomposition normalized to the quarter preceding #MeToo). For each quarter, I report the hypothetical share of victims who would have eventually reported for that quarter absent future changes in reporting behaviors (in green). For quarter J, this corresponds to:

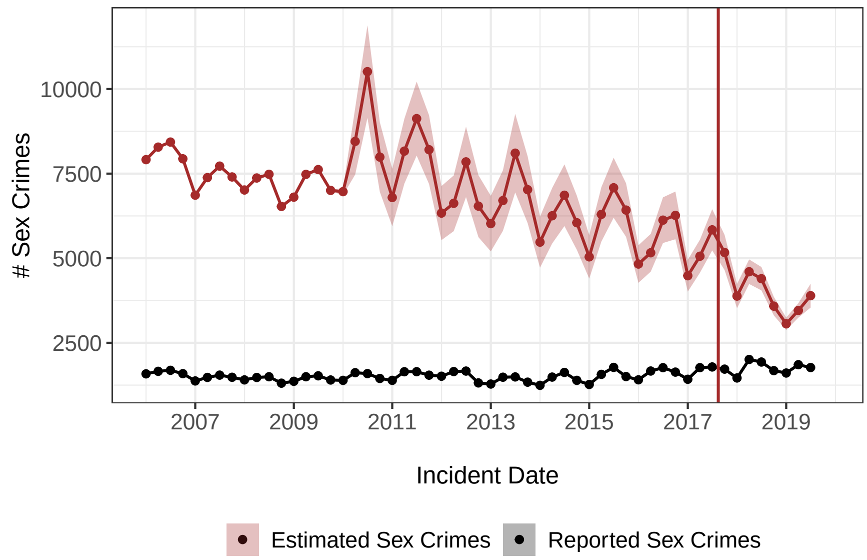
$$\lim_{y\to\infty} F_{it}(y) = 1 - \exp\left(-\exp\left(\alpha + \sum_{k=2010Q2}^{J} \beta_k \mathbb{1}(t + y \geq k)\right)\right). \tag{14}$$

For simplicity, I refer to this as the reporting rate of victims. My estimates indicate that the reporting rate more than doubled during the decade. It increases from approximately 20% in 2011 to 33% just before October 2017. The Me Too movement coincides with a strong reinforcement of these broader trends, as the reporting rate rises further to reach approximately 50% in 2019.

[23]The breaks are chosen by inspecting estimates of a piecewise constant hazard with over 50 breaks. The reporting hazard sharply decreases in the first year and remains flat for longer delays.

Given the duration model, I can also estimate p_{t,τ_1,τ_2} for each incident date t. Using Equation 7, this provides me with direct estimates of sex crime incidence (in red). I find a large decrease in sex crime incidence over the same period – approximately 57% between 2011 and 2019 (measured as annual averages of the quarterly series). Increasing reporting rates and decreasing sex crime incidence largely cancel each other out in the time series of reported crimes and translate into a moderate increase in reported sex crimes.

A. Nominal Values



B. Normalized at 100 for 2010 Q1

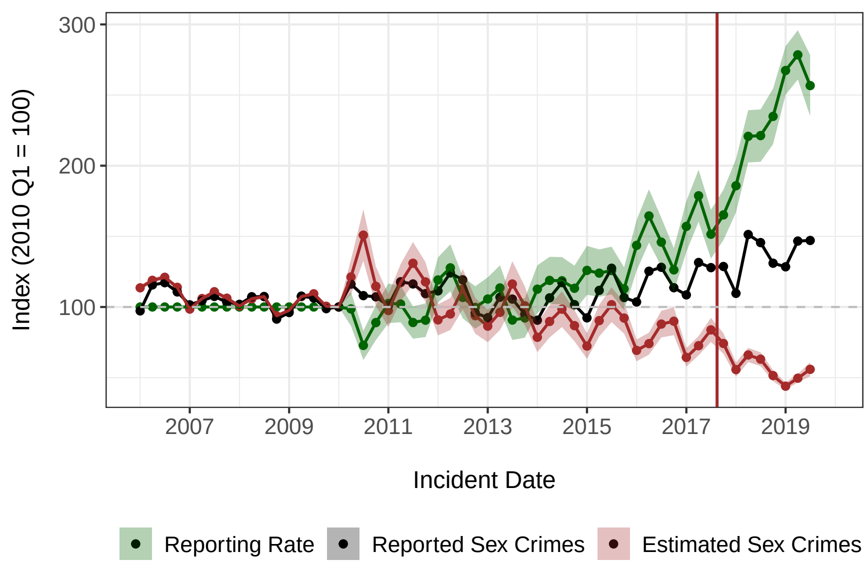


Figure G.7: Trends in Sex Crime Reporting and Incidence – Duration Model

Notes: Decomposition of reported sex crimes by incident date into estimates of victim reporting (green) and sex crime incidence (red), based on the duration model. Panel A reports nominal values; Panel B normalizes all series to 100 in 2010 Q1. The point estimates assume 80% of never-reporters at baseline (i.e., 20% of victims would have eventually reported a sex crime committed in 2006). Breaks in the baseline hazard are set after 1, 30, 90, 180, and 365 days. The likelihood appropriately corrects for double-truncation. There is no unobserved heterogeneity. 95% confidence intervals (shaded) are constructed with a non-parametric bootstrap procedure (1,000 replications). The vertical solid red line corresponds to the Me Too movement's mediatization.

G.4.2 Deterrent Effect on Incidence

Table G.1 presents the main results. Across specifications, I find a negative point estimate of similar magnitude, statistically significant at the 1% level in all four cases. In logs, the estimates are approximately -0.29 (matrix completion), -0.31 (interactive fixed effects), -0.27 (two-way fixed effects), and -0.29 (crime-specific trends).

As in the main text, the incidence series here is estimated, so I propagate its sampling uncertainty into the treatment effect with a bootstrap of 1,000 replications that combines two sources. To reflect uncertainty in the estimated incidence, each replication resamples the underlying sex crime complaints with replacement, re-estimates the duration model, and recomputes the implied incidence series; the calibrated share of never-reporters is held fixed across replications, and Online Appendix Figure G.10 reports the estimate for alternative values. To reflect uncertainty in the counterfactual, each replication resamples the control crime categories with replacement. For each replication, I re-estimate the treatment effect under each counterfactual model on the resulting panel. The reported standard errors are the standard deviations of the bootstrap distribution, with p-values from the implied t-ratios, and the confidence intervals in the event-study figures are the 2.5th and 97.5th percentiles of the bootstrap distribution in each quarter.

I conduct a series of robustness exercises. First, as presented in Table G.1, my results are robust to alternative counterfactual models. This includes a two-way fixed effects (TWFE) model and an interactive fixed effects (IFE) model (Xu, 2017), both of which display an excellent fit in the pre-treatment period (see Figure G.8). Second, the sign, magnitude, and significance of the estimates are not driven by any single crime category in the control group: the effect remains negative and significant at the 1% level under every leave-one-out control set, even with standard errors that propagate the first-stage duration-model uncertainty (see Table G.2).

Table G.1: #MeToo Effect on Sex Crime Incidence – Duration Model

	(1)	(2)	(3)	(4)
	Dependent Variable: Estimated Sex Crimes (in logs)			
#MeToo (indicator)	-0.29***	-0.31***	-0.27***	-0.29***
	(0.09)	(0.09)	(0.09)	(0.07)
Estimator	MC	IFE	OLS	OLS
Fixed Effects				
Incident date	✓	✓	✓	
Crime type	✓	✓	✓	✓
Seasons				✓
Time Controls				
Crime-specific linear trends				✓
Common post-#MeToo indicator				✓
Control Groups				
Murders	✓	✓	✓	✓
Assaults	✓	✓	✓	✓
Robberies	✓	✓	✓	✓
Burglaries	✓	✓	✓	✓
Standard Errors	Bootstrap	Bootstrap	Bootstrap	Bootstrap
N Observations	280	280	280	280

Notes: This table presents the Me Too movement's effect on estimated sex crime incidence based on the duration model; all estimates target the average treatment effect on the treated. The dependent variable is estimated sex crime incidence (log scale), obtained by dividing reported sex crimes by the estimated probability of observing a crime (see Equation 7). The panel covers New York City at the quarterly level between 2006 and 2019. Periods are quarter-length windows whose boundaries are shifted two weeks earlier than calendar quarters, so that a period boundary falls on the movement's viral date (October 15, 2017); this yields 56 periods per crime, the first covering only the first two weeks of January 2006. Column 1 uses the Matrix Completion (MC) method (Athey et al., 2021). Column 2 uses an interactive fixed effects (IFE) model (Xu, 2017). Column 3 is a two-way fixed effects OLS regression. Column 4 is an OLS specification that replaces time fixed effects with crime-specific linear trends and crime-specific seasonal fixed effects, and includes a common post-#MeToo indicator shared by all crime types, so the reported coefficient is the additional post-#MeToo shift specific to sex crimes. Standard errors (in parentheses) are bootstrap standard deviations from the procedure described in the text, which resamples the underlying complaints to re-estimate the duration model and resamples the control crime categories (1,000 replications). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

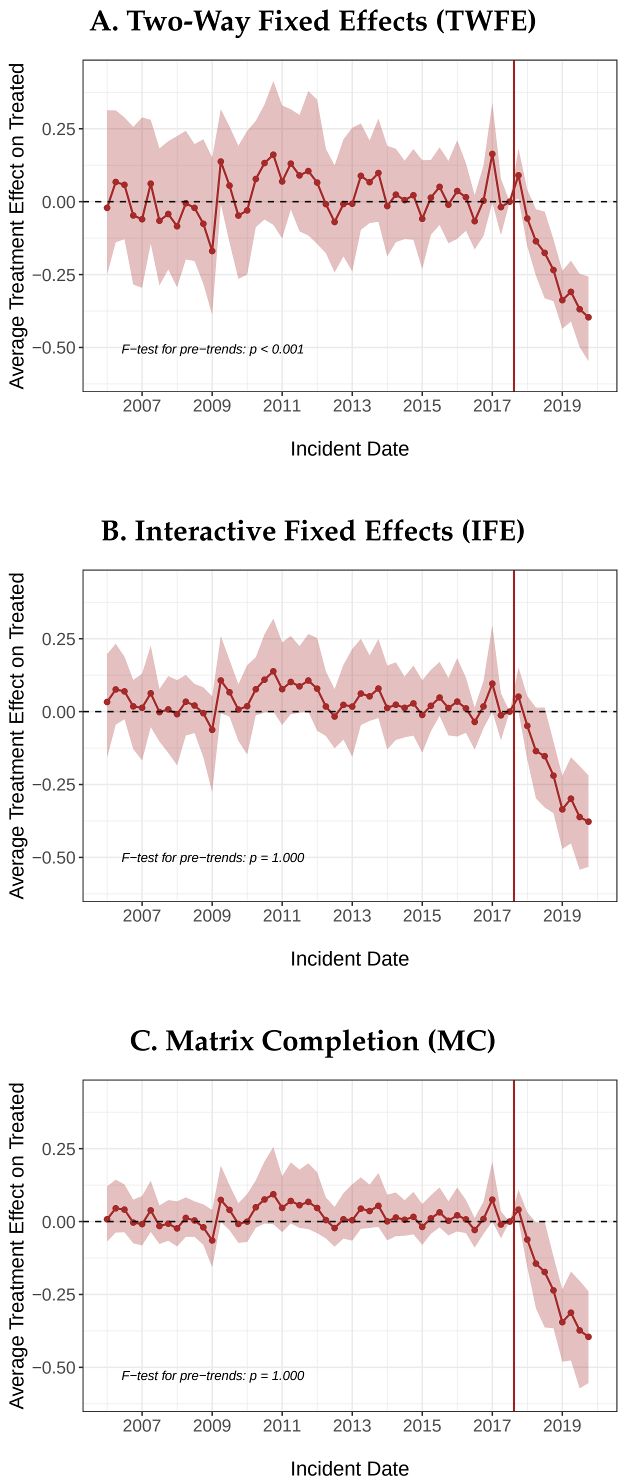


Figure G.8: #MeToo Effect on Sex Crime Incidence – Duration Model

Notes: Quarterly estimates of the average treatment effect on estimated sex crime incidence for three counterfactual models. The main results are presented in Table G.1. 95% confidence intervals are the 2.5th and 97.5th percentiles of the bootstrap distribution described in the text (1,000 replications). The vertical solid red line corresponds to the Me Too movement's mediatization.

G.4.3 Robustness – Alternative Never-Reporter Shares

The duration model requires calibrating the baseline share of never-reporters, set through the intercept α in Equation 11. Figure G.9 assesses the sensitivity of the results to this calibration by re-estimating the model while varying the baseline share of never-reporters from 60% to 90%. While the level of the estimated series shifts mechanically with α, the qualitative conclusions – a steady increase in victim reporting and a large decrease in sex crime incidence over the period – hold across all calibrations.

#MeToo's effects on sex crime incidence seems robust to this calibration. Figure G.10 reproduces the event-study estimates of the #MeToo effect on sex crime incidence for the three counterfactual models – two-way fixed effects, interactive fixed effects, and matrix completion – but replaces the bootstrap confidence bands with the envelope of point estimates obtained as the baseline share of never-reporters varies from 60% to 90%. While the level of the estimated effect shifts mechanically with α, the post-#MeToo decline in sex crime incidence remains negative across the entire range of calibrations in all three specifications.

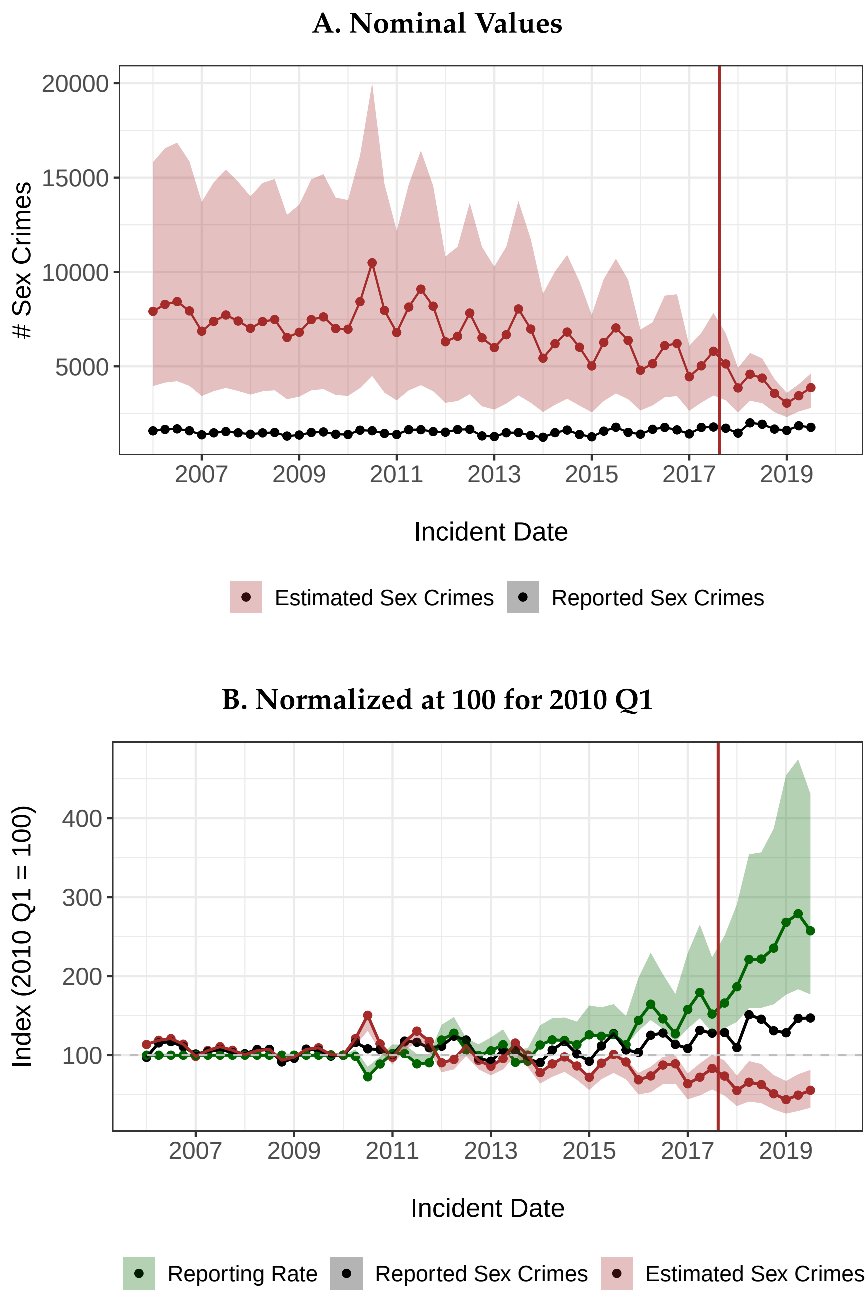


Figure G.9: Robustness of Estimates to Alternative Never-Reporter Shares

Notes: Estimates of sex crime incidence and victim reporting for different values of the baseline never-reporter share. The point estimates assume 80% of never-reporters at baseline as in the rest of this appendix. The shaded bands are built by varying the share of never-reporters at baseline from 60% to 90%. Breaks in the baseline hazard are set after 1, 30, 90, 180, and 365 days. The likelihood appropriately corrects for double-truncation. There is no unobserved heterogeneity. The vertical solid red line corresponds to the Me Too movement's mediatization.

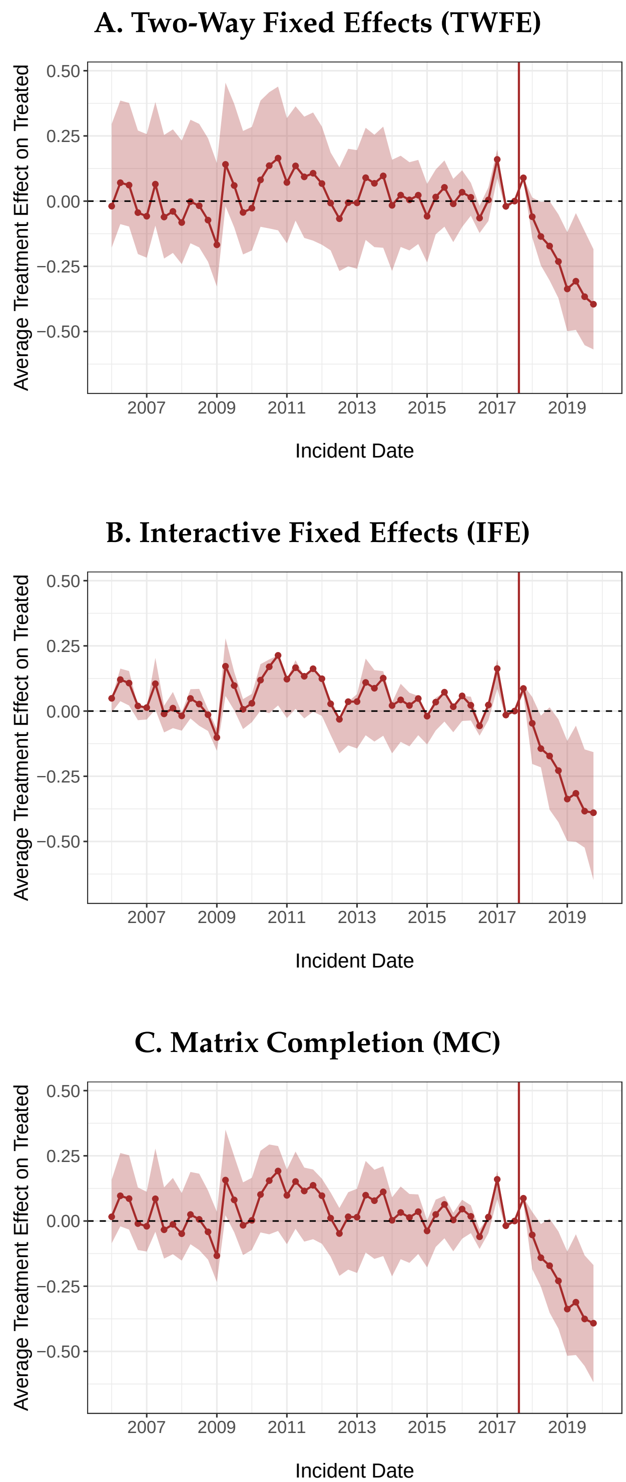


Figure G.10: Deterrent Effect on Sex Crime Incidence – Robustness to the Never-Reporter Share

Notes: Quarterly event-study estimates of the #MeToo effect on estimated sex crime incidence for three counterfactual models. The solid line is the point estimate under the baseline calibration (80% never-reporters in 2006); the shaded band is the envelope of point estimates as the baseline share of never-reporters varies from 60% to 90%. Unlike Figure G.8, the band reflects sensitivity to this calibration rather than sampling uncertainty. The control group includes non-sexual assaults, murders, robberies, and burglaries. Estimates are normalized to the third quarter of 2017. The vertical solid red line corresponds to the Me Too movement's mediatization.

G.4.4 Robustness – Dropping Control Crimes

Table G.2 assesses whether the deterrent effect estimate depends on any particular crime category in the control group. I sequentially exclude each non-sexual crime from the control group and re-estimate the Matrix Completion model. The point estimates are stable and uniformly negative, ranging from -0.18 to -0.34. Every column reports a bootstrap standard error that propagates the first-stage duration-model uncertainty, and the effect remains statistically significant at the 1% level in every specification. The sign, magnitude, and significance of the deterrent effect are therefore robust to the choice of control crimes.

Table G.2: Robustness – Dropping One Control Crime (Duration Model)

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: Estimated Sex Crimes (in logs)				
#MeToo (indicator)	-0.29***	-0.29***	-0.18***	-0.31***	-0.34***
	(0.09)	(0.11)	(0.04)	(0.11)	(0.09)
Counterfactual Model	MC	MC	MC	MC	MC
Control Groups					
Murders	✓		✓	✓	✓
Assaults	✓	✓		✓	✓
Robberies	✓	✓	✓		✓
Burglaries	✓	✓	✓	✓	
Standard Errors	Bootstrap	Bootstrap	Bootstrap	Bootstrap	Bootstrap
N Observations	280	224	224	224	224

Notes: Robustness of the estimated #MeToo effect on sex crime incidence to excluding individual control crimes. All columns use the Matrix Completion counterfactual and the bootstrap standard errors described in the text (duration-model resample plus control resample, 1,000 replications). Columns 2–5 sequentially exclude one control crime. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

G.4.5 Probabilities of Arrest

Finally, I combine NYPD arrest data with the duration model's reporting estimates to compute the probability of arrest conditional on committing a sex crime – the product of the arrest rate (arrests over complaints) and the estimated victim reporting rate. If the surge in reporting after #MeToo had been driven by lower-quality complaints, we would expect this probability to fall. Figure G.11 shows the opposite: the probability of arrest conditional on committing a sex crime stays in the 8–13% range through 2017, rising gradually as victim reporting increases, and then accelerates sharply after #MeToo to reach roughly 30% by 2018. Applying the trend-adjusted calculation of Section 6 to these duration-model estimates yields an arc elasticity of sex crime incidence with respect to the probability of arrest of approximately -0.76 (-0.71 to -0.81 across specifications), combining a duration-model decline in log incidence of about 0.29 with a rise in the log probability of arrest of about 0.38 (roughly 47%) above its pre-2017 trend. This is somewhat more conservative than the main-text estimate of about -1, but points to the same conclusion.

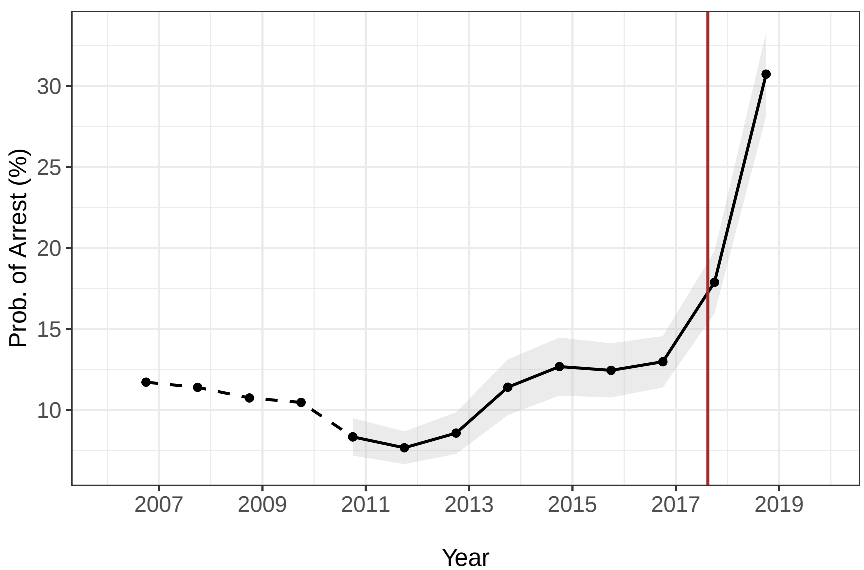


Figure G.11: Probability of Arrest for Sex Offenders – Duration Model

Notes: Yearly estimates of the probability of arrest conditional on committing a sex crime in New York City. The series combines arrest rates (arrests over complaints) with the duration model's estimates of the victim reporting rate. The dashed line shows the pre-estimation period where the reporting rate is assumed constant; the solid line shows the estimated period (2010 onwards). 95% confidence intervals (in grey) are constructed with the duration model's non-parametric bootstrap (1,000 replications). The vertical solid red line corresponds to the Me Too movement's mediatization.

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